

The Pill That Could

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*The improbable making of the first
targeted cancer drug*

MEIRA DAVIDSON

for the jessica family kids — be curious, stay open, be kind.

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The improbable making of the first targeted cancer drug
Meira Davidson

Set in Iowan Old Style and Helvetica Neue.
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PROLOGUE

A Patient in Portland



Brian Druker is thirty-seven years old in the autumn of 1993, in his first months at Oregon Health Sciences University, newly running his own lab, and he is sitting across from a patient he cannot save.

The patient has chronic myeloid leukemia — CML. He has been told this already; that is why he is here. Druker is rarely the first oncologist a CML patient sees. People with rare cancers tend to make their way around the country, looking for someone who knows something the last person didn't, and a few years out of fellowship Druker has begun to acquire that kind of reputation. He has spent his postdoctoral years thinking about this exact disease at the molecular level. If you have CML in 1993 and you read journals, you might end up in his clinic.

What Druker has to offer, though, is the same thing every other oncologist has to offer in 1993, which is not very much.

He has hydroxyurea — a pill that brings the white blood cell count down by interfering with DNA replication. It is not a treatment for CML in any meaningful sense; it is a way to hold the line. The disease keeps progressing under it, just more quietly. He has interferon-alpha, which patients inject several times a week,

which makes them feel as if they have the flu more or less constantly, and which produces a deep and durable response in something like ten or fifteen percent of patients. Most of the rest get the side effects without the benefit. And he has, for the patients who are young enough and lucky enough to have a matched donor, the option of an allogeneic bone marrow transplant — wiping out the bone marrow with chemotherapy and radiation, replacing it with cells from someone else, hoping the new immune system kills any remaining cancer without killing the host.

The transplant works, sometimes. It is also one of the most dangerous treatments in mainstream oncology. Roughly a quarter of patients who attempt it do not survive the procedure itself.

That is what Druker has on offer. A pill that controls counts but does not change the course of the disease. A weekly injection that helps a minority and exhausts almost everyone. A cure that kills one in four.

The patient, like most CML patients in 1993, is in what doctors call the chronic phase. He feels mostly fine. He noticed something — fatigue, maybe, or a fullness on his left side where his enlarged spleen has begun to push against his ribs — and he saw a doctor, and the blood work came back wrong, and now he is here. He looks well. He probably is well, in the immediate sense; the chronic phase can last months or years. But the disease is going to progress. The chronic phase will give way to an accelerated phase, and the accelerated phase will give way to what oncologists call blast crisis — the bone marrow flooded with immature cells, an acute leukemia in everything but name, and untreatable. The median survival from diagnosis is something like four to five years.

This is what Druker has to tell him, and tell him in a way that leaves room for hope without dishonesty, because there is a thing Druker has not yet said and is not sure when to say.

The thing is this. He knows what is wrong with this man's blood.



Not in the way doctors usually mean it. Doctors mean they have a name for the disease, a sense of which cells are involved, a notion of how it tends to behave. Druker means it the other way. He means he knows the specific molecular event that has gone wrong inside one stem cell in this man's bone marrow — the broken switch that is stuck in the on position, telling that cell and all of its daughters to grow and divide and refuse to die when they should.

By 1993 this is not even controversial. It is in the textbooks. Two chromosomes, the ninth and the twenty-second, have swapped pieces. The swap happened once, in one cell, possibly years before the patient noticed anything was wrong. It produced a fused gene that should not exist in nature, made of part of one normal gene and part of another, and that fused gene produces a fused protein, and the fused protein is an enzyme. A kinase — the kind of enzyme that puts a small chemical tag on other proteins to relay signals around the cell. The fusion has broken the regulation that normally keeps this kinase quiet most of the time. The kinase is on. It is always on. The cell receives, continuously and without pause, the molecular equivalent of *grow, divide, survive*. So it does.

The cancer is, in a real sense, a single broken enzyme.



If you are a particular kind of person — and Druker is — a fact like that is not just a fact. It is an invitation. If we know the disease is one broken enzyme, then in principle we know what would fix it. We would need a small molecule that fits the active site of that specific kinase well enough to block it, and that does not fit the active sites of the many other kinases the body needs in order to function. We would need, in the language of pharmacology, a selective inhibitor.

In 1993, “selective kinase inhibitor” is something the field talks about the way an earlier generation of physicists talked about controlled fusion: as a thing that would clearly be wonderful if it existed, and that there are real reasons to suspect cannot. Kinases share a common architecture. The pocket where they bind their fuel — a small molecule called ATP, which every living cell uses to do work — is broadly similar across the human kinome, the set of all human kinases, of which there are over five hundred. The conventional wisdom is that any drug small enough to fit the pocket will fit too many pockets, and that hitting them all will be either useless (because the cell finds a workaround) or lethal (because some of those kinases are doing essential work). The pharmaceutical industry, in 1993, has been trying to make selective kinase inhibitors for years, with limited success. There is a strong professional opinion that it cannot quite be done. Or that even if it can, the resulting drug will be too toxic, or the cancer will adapt around it, or the development cost will not be recoverable from a disease as uncommon as CML.

Druker does not believe any of this. Or rather, he believes the technical objections are real and he believes the conclusion that follows from them is wrong.

He has reasons. During his postdoctoral years at Dana-Farber Cancer Institute, in the lab of a biochemist named Tom Roberts, Druker had spent his time developing tools to study kinases in living cells — antibodies that could light up exactly the chemical tag a kinase leaves on its substrates, so you could tell when the kinase was active and how active it was. He had used those tools to look at the BCR-ABL kinase, the fused enzyme made by the broken chromosome, and seen how distinctive its signature was and how

cleanly you could read it. Cells that have BCR-ABL look different from cells that don't, in a way that a good antibody can detect at a glance.

He has also been talking, on and off for several years now, with a chemist named Nick Lydon at a Swiss pharmaceutical company called Ciba-Geigy. Lydon's team has been making compounds aimed at one kinase or another, more as research tools than as drugs, and some of the compounds they have made are interesting. One of them, a molecule with a code number rather than a name, is interesting in a particular way: in early tests it inhibits a small handful of kinases and ignores the rest. The selectivity that the field considers impossible is starting to look, in this one molecule, like an engineering problem rather than a law of nature.

The molecule has not been tested against BCR-ABL. There are reasons to think it might work — the kinases it does inhibit are structurally related — but no one has done the experiment.

Druker is going to do the experiment. He has the cells, he has the antibodies to read out the result, and Lydon, who has his own reasons for caring about whether the molecule does anything useful, is willing to send some.



This is the situation, then, in 1993. The patient sitting across from Druker has a disease that is going to kill him in roughly four to five years on the current standard of care, or possibly faster, or possibly with a one-in-four chance of being killed by the cure. The cause of his disease has been worked out at the level of single atoms. A molecule that might address that cause exists, in milligram quantities, in a freezer in Switzerland. And the gap between those

two facts — the distance between knowing what to do and being able to do it for the man in the chair — is the entire subject of this book.

It is a wider gap than it sounds.

The molecule has to be tested in cells. If it works in cells it has to be tested in animals. If it works in animals it has to be made in larger quantities, and then tested in the first humans, and then in more humans, and then in many more humans, and at each stage it can fail, and at each stage someone — usually a committee — has to decide whether to keep going. Most molecules that get this far do not become drugs. Most drug projects that get this far are abandoned. The molecule under discussion has, in 1993, a number of features that the people whose job it is to decide whether to develop drugs find unattractive. CML is uncommon. The class — a kinase inhibitor — has a poor track record. There are, in any committee meeting where the molecule comes up, more reasons to walk away than to keep going.

The patient does not have a few years to spare on committee meetings.



Druker, in the clinic in 1993, does not know what is coming. He does not know that the company he is about to start working with will merge with another company and become an institution called Novartis. He does not know that the molecule will be given a new code number, and then a new code number after that, and finally a name. He does not know that the first phase 1 trial — the first time anyone takes the drug — will not begin until June of 1998, five years from now, three years too late for some of the people in his clinic. He does not know that when the trial does begin it will produce results so striking that the doctors running it will spend the first weeks not

believing what they are seeing. He does not know that the FDA, when it finally sees the data, will approve the drug in roughly ten weeks, faster than any cancer drug had been approved before. He does not know that some of the people in his clinic in 1993 — the ones who survive long enough on the inadequate options he has to offer them — will be alive in 2026, taking a pill every morning, watching their grandchildren grow up.

What he knows is that he has a patient he cannot save, and a molecule he has not yet tested, and a hypothesis about how cancer might be treated that the field largely does not share.

The question, at the end of his clinic day in 1993, is the question this book exists to answer.

It is not, despite how the story is sometimes told later, whether the science was right. The science was right. By 1993 the science had been right for several years, and a small number of people had already begun to act on it.

The question is what it takes — in time, in argument, in stubbornness, in luck, in the unglamorous labor of running an experiment and then running it again and then running it a third way because the second way did not convince anyone — to move a fact about a disease from a journal article to the cabinet of someone who has the disease.

It takes, in the case of this molecule, eight years from the beginning of Druker's collaboration with Lydon to the day the FDA approves the drug. It takes longer than that if you start the clock earlier, in the Philadelphia laboratory where the first piece of the puzzle was found in 1960. It takes a number of careers, a number of arguments inside a number of companies, a small group of patients willing to take a drug no one had ever taken before, and several pieces of luck that could easily have gone the other way.

This is the story of how it happened, and what it meant, and what it did not quite manage to do.

It begins thirty-three years before Druker's clinic, with two men in Philadelphia looking at a chromosome that should not have been there.

CHAPTER 1

1

Philadelphia, 1960



The microscope is a Zeiss, the slide is a smear of cells from a man with leukemia, and the man looking down the eyepiece is David Hungerford. It is the autumn of 1959 or the spring of 1960; the dates are not perfectly clear in the historical record, and Hungerford himself, asked years later, gave slightly different versions. What is clear is what he is looking at.

He is looking at a single cell, frozen in the moment of dividing in two. The cell has been treated with a chemical that arrests division partway through, at the stage where the chromosomes have already condensed but haven't yet been pulled to opposite ends of the cell. They sit in the middle of the field, scattered, like loose grains of rice. There are forty-six of them, which is what there are supposed to be in any normal human cell, and which is also a number that the field has only known for a few years.

Hungerford counts them, the way he has counted them in thousands of cells before. He matches them up. Chromosomes come in pairs — twenty-two matched pairs plus the two sex chromosomes, X and Y in a man, X and X in a woman — and matching them is most of the work. You look at length and at the position of the centromere, the pinched-in waist where the two halves of the duplicated chromosome are held together. You group the pairs from longest to shortest. You write the result down.

In this cell, in this leukemia patient, twenty-one of the autosomal pairs match cleanly. The smallest pair — the chromosomes labeled 22, the shortest of the autosomes, which are the chromosomes that aren't sex chromosomes — should look the same as each other. They don't. One is the right size. The other is too small. A piece of it, a noticeable piece, is missing.

Hungerford makes a note. He goes back to his microscope.



You can find Peter Nowell, in the photographs from this period, looking like a young man trying to look a little older. He is in his early thirties. He is a pathologist by training, on the faculty at the University of Pennsylvania School of Medicine, in West Philadelphia. He is tall and slightly formal in a way pathologists of his generation often were. He is also, by accident more than design, in the middle of a small revolution in how people study human cells.

The accident is this. A few years earlier, Nowell had been trying to get a clean look at human white blood cells under a microscope. White blood cells — leukocytes, which is the source of the word *leukemia*, blood that has gone white in some specific way — are the cells that are diseased in the leukemias, and they are also the cells you can most easily get hold of in a clinical setting, because you can

draw blood from a patient's arm. The trouble is that when you spin down a tube of blood, what you mostly get is red cells. The white cells are buried in there, and getting them out cleanly is fiddly.

There is a substance called phytohemagglutinin — PHA, in everyday lab usage — which is extracted from kidney beans and which clumps red blood cells together, so that they fall to the bottom of a tube and leave the white cells suspended above them. PHA was being used, around 1960, as a way to separate white cells from red. Nowell was using it for that. Then he noticed that something else was happening. Some of the white cells, after exposure to PHA, started to divide.

This was not supposed to happen. White blood cells circulating in the body, generally speaking, don't divide. They are made elsewhere — in the bone marrow, in the lymph nodes, in the thymus — and the cells you draw from a vein are mature cells doing mature-cell things, not dividing. To see chromosomes you need cells that are dividing, because chromosomes are only condensed and visible during division. The rest of the time the genetic material is decondensed into something more like loose thread, and you can't see anything useful.

PHA, in other words, was solving a problem the field hadn't quite figured out it had. Before PHA, if you wanted to look at human chromosomes, you had to get cells from bone marrow (which required a needle in a hip bone) or from a tumor (which required a tumor and a surgeon) or from a cell line grown in culture (which only worked for some kinds of cells). After PHA, you could draw a tube of blood, add the bean extract, wait a couple of days, and have a population of dividing cells that you could then arrest in mid-division and stain and look at. It made human cytogenetics — the study of human chromosomes — practical for ordinary patients in ordinary hospitals.

Nowell published the PHA finding in 1960. He was not the only one to notice the effect, but he was early, and he was thorough, and the technique spread. It is, in retrospect, a finding that probably should have made him famous on its own. It is also what made his next finding possible.



David Hungerford was, in 1960, finishing his PhD at the Institute for Cancer Research, the small institution in Fox Chase, just north of the city, that would later become Fox Chase Cancer Center. He had spent his graduate years learning to read chromosomes. This was, in 1960, a relatively rare skill. The field of human cytogenetics, as a recognizable field, was four years old.

Four years old, because in 1956 a Swedish-Indonesian team — Joe Hin Tjio and Albert Levan — had published a paper showing that humans have forty-six chromosomes. This was a surprise. The textbooks of the time said forty-eight. They had said forty-eight for decades. The number had been counted, by various people using various methods, and they had all gotten forty-eight, and it had become one of those facts that everyone knew and no one rechecked. Tjio and Levan, working with a hypotonic-shock technique that swelled cells and spread their chromosomes apart cleanly, recounted, and got forty-six, and were right. (The earlier counts had been done on cells in which the chromosomes were too clumped to count accurately, and people had simply miscounted, repeatedly, in a direction that subsequent counts then anchored to.)

So as of 1956, the field knew how many chromosomes a human has. As of 1959, in a paper by Jérôme Lejeune in France, the field knew that one extra copy of chromosome 21 produced Down syndrome — the first time a specific chromosomal abnormality had been linked to a specific human condition. As of 1960, the field had

a few methods for getting cells to divide and for spreading and staining their chromosomes. It had a small number of practitioners, mostly young, mostly working in just a few labs around the world.

It did not have, at this point, any clear understanding of what chromosomes had to do with cancer.



Cancer, in 1960, was understood in broad strokes and almost no detail.

The broad strokes were these. Cancer is a disease of the body's own cells. A cell that is supposed to grow only in certain circumstances starts growing in others; a cell that is supposed to die at a certain point doesn't; a population of these cells expands, takes up space, interferes with the work of organs, and eventually kills the person. Different cancers come from different starting cells — leukemia from blood-forming cells, breast cancer from the cells lining the milk ducts, and so on — and behave differently because of it. Some cancers are linked to specific exposures: tobacco smoke and lung cancer; certain chemical dyes and bladder cancer; ionizing radiation and basically everything. Some cancers are linked to viruses, at least in animals. (Peyton Rous had shown in 1911 that a virus could cause a tumor in chickens, a finding so far outside the conventional wisdom that it took the field about fifty years to take seriously.) Some cancers run in families.

What the field did not have, in 1960, was a mechanism. There was no agreement on what was actually happening, at the level of the cell, when a cell turned cancerous. The major hypotheses were several. Maybe cancer was a virus, like Rous's sarcoma virus, that just hadn't been found in most cancers yet. Maybe it was something about cellular metabolism, a permanent shift in how a cell handled energy. Maybe it was a mistake in the body's defenses, an immune

system failing to recognize and kill abnormal cells. Maybe it was something genetic, written into the DNA, although what exactly might be written there and how it would have gotten there were not at all clear.

There was an old hypothesis, sitting in the literature, due to a German biologist named Theodor Boveri. Boveri had proposed in the early 1910s, working with sea urchin eggs, that abnormalities in the chromosomes might cause cells to behave like cancer cells. It was a careful argument, made on the basis of clean observations, and it had aged reasonably well. It was not, however, an argument anyone was actively pursuing in 1960. Boveri had died young; the methods to test his idea in human cells had not existed for most of the intervening half-century; and the consensus, to the extent there was one, was that cancer was probably more complicated than that.

Then, in the late 1950s, a new piece of information started to creep into the literature. People who looked at chromosomes from cancer cells — and there were not many people doing this, but there were some — kept finding that the chromosomes were a mess. The numbers were wrong. There were too many of some, too few of others, fragments that didn't belong, occasional ones that looked deformed. This was a curious thing, but the obvious interpretation was that cancer cells, being already disordered, were probably bad at copying their chromosomes properly, and the chromosomal mess was a *consequence* of being a cancer cell rather than a cause. The chromosomes were a downstream symptom, like the fever in a sick person — a sign of the disease but not its source.

The way you would distinguish between symptom and cause, if you wanted to, would be to find a cancer in which the chromosomal abnormality was the same in every patient. If chromosomes go wrong randomly because the cell is sick, you'd expect a different

mess in every case. If something specific in the chromosomes is causing the disease, you'd expect to see the same specific thing every time.



Nowell and Hungerford had been looking at leukemia samples — the easiest cancer to study, because the cancer cells circulate in the blood and can be drawn through a needle — and they had been looking at the chromosomes from those samples carefully. They had noticed, in patients with one particular kind of leukemia, that one chromosome looked unusually small. Not different sizes in different patients. The same too-small chromosome, in the same place in the karyotype, in patient after patient.

The leukemia in question was chronic myeloid leukemia. CML. The same disease that, thirty-three years later, would have Brian Druker sitting in his Portland clinic with nothing to offer.

CML in 1960 was, as in 1993, a fatal disease. The available treatments — busulfan, an alkylating agent that knocked white cell counts down at the cost of considerable toxicity; splenectomy, removal of the swollen spleen, which provided some symptomatic relief — extended life by months, sometimes a year or two, and did not change the eventual outcome. CML patients in 1960 mostly died of CML, in their fifties or sixties or, often enough, much younger.

There was, however, something useful about CML for what Nowell and Hungerford were trying to do. CML produces a great many abnormal white cells, circulating freely. You could draw a tube of blood from a CML patient and have, almost without effort, a high yield of the very cells you wanted to look at. PHA wasn't even strictly necessary — although it helped — because some of the CML

cells were already actively dividing. The disease was, from the perspective of a young cytogeneticist trying to read cancer chromosomes, an unusually cooperative subject.

So they read them. Patient by patient. They looked at samples from a handful of people with CML. In every one, the same small chromosome. A piece of it, looking at it through the microscope of 1960, appeared to be missing.



The paper appeared in *Science* in the autumn of 1960. It is short. The total length, by modern standards, is barely a long paragraph — about a page of journal text, with a small figure showing a karyotype. Its conclusions are stated carefully, in the conditional language of a paper not wanting to overreach. The authors note that they have observed an unusually small chromosome, present in the cells of patients with chronic granulocytic leukemia (which was the more common name for CML at the time), absent in the same patients' normal cells when those could be obtained, and consistent across the cases they had been able to examine. They suggest that this finding may be relevant to the understanding of the disease.

They do not claim to have found the cause of cancer. They do not claim to have shown that CML is a chromosomal disease. They report what they have seen and let the field draw its own conclusions.

The field's first conclusion is that this is interesting but perhaps not entirely surprising. People have been seeing chromosomal abnormalities in cancer cells for a while; here is another one. The novelty of Nowell and Hungerford's paper, when it is appreciated, is not that the chromosomes are abnormal but that they are *consistently* abnormal — the same abnormality in every patient. This is the

feature that distinguishes a cause from a symptom, at least potentially. It is the first time anyone has shown such a thing in a human cancer.

A few years later, at one of the small conferences where the new field of cytogenetics was working out its conventions, the practitioners agreed on a naming system for newly discovered chromosomal anomalies. The system was simple: name the anomaly after the city in which it was first reported. The small chromosome from the CML patients, having first been described in Philadelphia, became the Philadelphia chromosome. It is, as far as I can tell, the only piece of human anatomy named after the city of Philadelphia.



Now here is the thing about the Philadelphia chromosome.

It was the first specific, reproducible genetic abnormality ever linked to a specific human cancer. In a field that had been arguing for decades about whether cancer had genetic causes at all, it was a clear data point on the side of yes. It is, with the benefit of hindsight, one of the most important findings in twentieth-century cancer biology.

It also did, in the immediate term, almost nothing.

It did almost nothing because nobody could do anything with it. The tools to ask the obvious follow-up questions did not exist. *What does the missing piece do?* — unanswerable; you couldn't isolate it, couldn't read it, couldn't even reliably tell what was missing as opposed to what was abnormal in some other way. *What protein is encoded there?* — unanswerable; the relationship between specific stretches of chromosome and specific proteins, in humans, was known in only a handful of cases, and not this one. *Could you make a drug that fixes the problem?* — laughable; the field had no drugs that

targeted specific genes or specific proteins, didn't really have the concept of doing so, and wouldn't develop the necessary toolkit for two more decades.

So the Philadelphia chromosome went into the literature, was cited respectfully, became a piece of standard knowledge about CML, and sat there. Doctors learned to look for it; finding it confirmed the diagnosis of CML in ambiguous cases. Researchers occasionally wrote review articles about it. It was a fact in good standing.

What it wasn't was a useful fact. It told you what was wrong, in some general sense, but not specifically enough to act on. It was as if you had been told that a particular kind of car always broke down because of a problem in the engine, and you opened the hood to find the engine encased in concrete. There was nothing to do with the information.

The interpretation that hardened, in the years after 1960, was that the small chromosome had simply lost a piece. A deletion, in the language of the field. A chunk of chromosome 22 had broken off and been lost, and that loss was somehow involved in the disease — perhaps because something important had been on the missing piece, perhaps because the loss had disrupted some balance. Either way, the missing piece was gone, and there was nothing further to be done.

This interpretation was not exactly wrong, but it was missing something important, and the something important would not be found for thirteen more years, when a researcher in Chicago, on a sabbatical, with a new staining technique that revealed banding patterns inside chromosomes, looked at the Philadelphia chromosome and at the cells it came from and saw something nobody else had noticed.

The missing piece wasn't missing.

PHILADELPHIA, 1960

It was somewhere else.

CHAPTER 2

2

The Translocation



In a small office at the University of Chicago, in the spring of 1973, Janet Rowley is laying out photographs of chromosomes on her desk.

There are forty-six in each photograph. She has cut each one out of a printed micrograph of a single dividing cell, taken from a patient with leukemia, stained with a chemical that makes parts of each chromosome fluoresce under ultraviolet light. The photographs are small and grainy. Each shows a single chromosome, photographed end-on, looking like a pale rod with bands of darker and lighter stripes running across its length.

She is matching them up. The work is patient and slow and looks, from the outside, exactly like solitaire. From the inside it is something else.

The work travels with her. Some of the time she is at her office desk. Some of the time she is at home, on the dining room table, cutouts spread across the polished wood, her four sons either grown or out of the house, her husband — a pathologist at the same university — long used to seeing the chromosomes laid out alongside his evening reading. She has been doing this kind of work, on and off, for most of a decade. She still works part-time. She still has, formally, the part-time appointment she took when her children were small.

The chromosomes she is looking at this spring come from patients with chronic myeloid leukemia. She is looking, specifically, for the Philadelphia chromosome — the small chromosome 22 that Nowell and Hungerford reported thirteen years earlier. She has found it, in patient after patient, and confirmed what is already known: the Philadelphia chromosome is real, and it shows up reliably in CML.

What she has been seeing more recently, though — with a new staining technique she learned during a sabbatical in England — is that the Philadelphia chromosome is not alone in being abnormal.

Chromosome 9, in these same cells, has something extra on it.



Janet Rowley grew up in Chicago, the only child of a librarian mother and a college-instructor father. She finished high school early, started the University of Chicago's accelerated program at fifteen, and finished medical school at twenty-three. She married Donald Rowley, a fellow medical student who would become a pathologist, the day after she got her degree. They had four sons in the next ten years.

Through those ten years, and for some years after, Rowley worked at her medical career in the way it was widely assumed a woman with four children would work at one if she did at all — part-time, fitting it around the home life, in clinics and labs that would have her on those terms. She did not stop working. She also did not become a full-time researcher, in the way a man with the same training would have, until much later in her career.

This is worth noticing, because the science she is doing in 1973 is the kind of slow, careful, attention-intensive work that benefits from a researcher who has spent fifteen years patiently looking at things rather than rushing toward results. Whether or not she would have done it differently in a less constrained career, the career she had was one in which she had learned, very thoroughly, how to see chromosomes.

She had moved into the chromosome work properly in 1962, joining a group at the University of Chicago that was studying cells from patients with developmental abnormalities. By the late 1960s she was a serious cytogeneticist. In 1970 she took her family to England for a sabbatical year — a rarity for a part-time appointment, and a piece of her career she had to push for — and worked in Oxford in the lab of a researcher named Henry Harris. While she was there, the field she worked in changed.



What changed was that someone figured out how to see inside a chromosome.

Up until 1970, a stained chromosome looked like a uniform bar — a pale rod, with the centromere somewhere along its length, and not much else to distinguish it. To tell one chromosome from another you measured length and centromere position and grouped them into a few broad categories, and that was about it. You could

spot major abnormalities — the Philadelphia chromosome was unmistakably small — but anything subtle, anything where two chromosomes might look similar but be different in ways that mattered, was effectively invisible.

Then a Swedish biophysicist named Torbjörn Caspersson, working with a team in Stockholm, tried something unusual. He stained chromosomes with a chemical called quinacrine mustard, which is a derivative of an old antimalarial drug, and which has the property of binding preferentially to certain regions of DNA. Specifically, it binds to regions rich in the bases adenine and thymine — two of the four molecular letters of DNA — and ignores regions that are richer in the other two, guanine and cytosine. Then he illuminated the stained chromosomes with ultraviolet light. The quinacrine fluoresced. The AT-rich regions lit up. The GC-rich regions stayed dark.

What appeared, when he looked through the microscope, was a striped pattern. Each chromosome had a sequence of bright and dark bands along its length, like a barcode. And the pattern was reproducible. The bands on a particular chromosome appeared in the same order, with the same spacing, in cell after cell, in person after person. Each of the twenty-three human chromosomes had its own distinctive pattern. You could now tell them apart not just by size but by their internal structure.

The technique was called Q-banding, for quinacrine. Within a year or two it was joined by G-banding, which used a different stain (Giemsa, a dye that had been used in microscopy for nearly a century) after a brief treatment with an enzyme called trypsin, and which produced a similar barcode in ordinary visible light. By 1971, at a conference in Paris, the cytogenetics community had agreed on a standardized nomenclature. Each chromosome was divided into

bands. Each band had a number. You could now describe a region of a chromosome — a specific stretch of a specific chromosome — with the same precision a surveyor would describe a piece of land.

This was the technique Rowley brought back from her sabbatical year. It was being adopted across the field, but slowly, because it took practice, and because the published photographs in textbooks all still showed the unstained version. She had spent her year in Oxford getting good at it. She came home to Chicago in 1972 with the ability to read the barcode of a chromosome in a way that, in the United States, very few people could yet match.



Before going further, it is worth pausing on what a chromosome actually contains, because the rest of this story depends on it.

Inside a chromosome is DNA. Inside the DNA is a sequence of letters — four chemical bases, A, T, G, and C, repeated in long strings, tens of millions of letters in each chromosome. In humans, the total length of the DNA across all forty-six chromosomes is something like six billion letters.

Stretches of those letters spell out what are called genes. A gene is a section of DNA that contains the instructions for making a particular protein. Proteins are the molecules that do most of the work of being alive — the enzymes that catalyze chemical reactions, the structural fibers that hold tissues together, the receptors that sit in cell membranes and listen for signals, the transporters that move things in and out of cells. There are perhaps twenty thousand protein-coding genes in the human genome (the field will not have a confident count until the early 2000s, but the rough order of magnitude is settled).

The flow of information goes one way. The DNA is read, by a series of molecular machines, to produce RNA — a closely related molecule that serves as a working copy. The RNA is read, by a different molecular machine, to produce protein. The protein folds into a particular shape and does its particular job. DNA to RNA to protein. This is sometimes called the central dogma of molecular biology, a name proposed by Francis Crick in the late 1950s, slightly tongue-in-cheek (he meant it more as a working assumption than a theological commitment), and the name has stuck.

If a gene is broken — if its DNA sequence is altered, or if it is moved to a region of the chromosome where it gets switched on at the wrong time, or if it is fused to a piece of another gene — the protein it makes will be different. Sometimes the difference is harmless. Sometimes the protein is no longer made at all. Sometimes the protein is made in larger amounts than usual, or smaller, or in the wrong cells, or at the wrong moments. Sometimes a new protein is created, made of pieces of two genes that should not be near each other, with properties neither of the originals had.

This last possibility — a fusion protein, made by accidentally joining two genes — is one of the things that can happen when a chromosome breaks and is repaired wrong. The cell has machinery for repairing broken chromosomes, but the machinery is not perfect. If a break occurs in the middle of a gene, and the chromosome rejoins to the wrong partner, the result can be a gene whose first half comes from one chromosome and whose second half comes from another. The protein made from such a gene is a hybrid. Whether it does anything useful, anything dangerous, or nothing at all depends entirely on which two genes were involved.

In 1973, all of this is known in principle but not in detail. Genes have been identified in bacteria, in yeast, in fruit flies. In humans they are mostly inferred. The molecular tools to read a human gene

letter by letter — the techniques that will become known as DNA sequencing — are still a few years away. The field is in a curious position: it knows what genes are, it knows what proteins are, it knows how DNA codes for protein, and it knows almost nothing about which specific human genes do which specific things. The map is mostly blank.

What you can do, in 1973, is look at chromosomes. And with banding, you can look at them well enough to see when one of them has been rearranged.



Rowley, looking at her CML cells with Q-banding, was seeing two things at once.

The first was the Philadelphia chromosome itself, the small chromosome 22, with a noticeably shorter long arm than its partner. This was familiar; this was what Nowell and Hungerford had described in 1960. With banding, Rowley could now see exactly which bands of chromosome 22 were missing. The break was on the long arm, in a band the new nomenclature labeled q11 (the q referring to the long arm, the 11 to the position of the band). The piece beyond the break — the distal portion of the long arm — was gone.

The second thing was on chromosome 9.

Chromosome 9 is, in a normal cell, recognizable by its banding pattern. The long arm has its own characteristic sequence of light and dark stripes, and the very end of the long arm is, by Q-banding, a region that fluoresces brightly. In the CML cells, chromosome 9 looked almost normal — but with something extra at the end of the long arm. A pale band, beyond what should have been the terminal bright band. A piece of something attached.

Rowley measured. The extra material on chromosome 9 was about the same size as the piece that was missing from chromosome 22. The banding pattern of the extra material — once she had it isolated cleanly enough to read — matched the banding pattern of the missing tip of 22.

The piece that was supposed to be on chromosome 22 was on chromosome 9.

It had not been deleted. It had been moved.



This kind of move is called a translocation. A piece of one chromosome detaches and reattaches somewhere else.

Translocations come in two varieties. In a non-reciprocal translocation, a piece of one chromosome moves to another, and the donor chromosome simply loses what it gave away. In a reciprocal translocation, two chromosomes exchange pieces — a piece of A goes to B, and a piece of B goes to A — and both chromosomes are altered. The cell now has the same total amount of DNA it had before; it has just shuffled some of it.

Rowley's reading of the Philadelphia situation was that it was reciprocal. The piece missing from chromosome 22 was on chromosome 9, but chromosome 9, looking carefully, also had something extra at the tip of its long arm — and the small chromosome 22 had received, in exchange, only a tiny piece of chromosome 9, which is why the chromosome ended up shorter overall. The two chromosomes had swapped pieces. 9 and 22 had each broken on their long arms; the broken-off pieces had stuck to the wrong partner; the cell had carried on with the reshuffled set.

In the language that would become standard, the Philadelphia chromosome was a t(9;22) — a translocation between chromosomes 9 and 22 — with breakpoints in specific bands (q34 on chromosome 9, q11 on chromosome 22, in the eventual nomenclature).

This was a reinterpretation of a well-known abnormality, not a discovery of a new one. But the reinterpretation mattered. It changed the question from “what does it mean for a piece of chromosome 22 to be lost?” — a question with no obvious experimental approach in 1973 — to “what does it mean for a piece of chromosome 22 and a piece of chromosome 9 to be joined?” — a question with a much sharper edge, because joining two pieces of DNA can create something specific, namely a junction. At the junction, two sequences that should never have been near each other are now adjacent. If a gene happens to span the junction, you get a fusion gene. If the gene downstream of the junction is normally regulated by the sequences it now finds itself next to, it will be regulated differently. Either way, the junction is the kind of thing you can, in principle, investigate.

You couldn’t investigate it in 1973. The tools weren’t there. But you could now state, clearly and correctly, what kind of investigation would be needed.



Rowley’s paper appeared in *Nature* in June 1973. Like the Nowell and Hungerford paper before it, it was short — a one-page letter, with a small figure showing the banded chromosomes from a CML patient and an arrow pointing to the extra material on chromosome 9. The paper’s title was careful: “A New Consistent Chromosomal Abnormality in Chronic Myelogenous Leukaemia identified by

Quinacrine Fluorescence and Giemsa Staining.” The word “new” referred to the previously unrecognized abnormality on chromosome 9, not to the Philadelphia chromosome itself.

The paper’s reception was mixed. Some readers — the ones already thinking about cancer in chromosomal terms — saw immediately what it meant. Others were skeptical that translocations could really be reproducible features of cancer; the standard view was that cancer cells were chromosomally unstable in a generic way, and that a specific translocation appearing in patient after patient was probably a coincidence or an artifact. Rowley herself encountered some of that skepticism firsthand. An earlier paper of hers, on a different translocation in a different leukemia, had had a difficult time finding a home before being published.

What changed the field’s mind, over the next several years, was that Rowley kept finding translocations. So did other people. The list grew.

Rowley had already identified, slightly earlier, a t(8;21) translocation in a subset of patients with acute myeloid leukemia — a different leukemia from CML, with a different translocation, but also reproducible. By the mid-1970s, the picture was filling in: Burkitt lymphoma was associated with a t(8;14); follicular lymphoma with a t(14;18); a particular subtype of acute promyelocytic leukemia, identified later, with a t(15;17). Each cancer had its own characteristic chromosomal swap. The swaps were not random. They appeared in the same patients, the same diseases, the same breakpoints, over and over.

It became increasingly difficult to argue that the translocations were incidental. The simpler explanation — the only explanation that fit the accumulating data — was that the translocations were causal. Something specific happened at the breakpoint that turned a normal cell into a cancer cell. Different breakpoints in different

cancers meant different specific events. The genetics of cancer, which had been an abstract question in 1960 and an open question in 1970, was starting to look, by the late 1970s, like a tractable problem with specific cases.



What this meant, by the end of the 1970s, was that the field had a new question.

The old question, the one that had been frozen in place since 1960, had been: what does the Philadelphia chromosome do? The new question, made possible by Rowley's work and the cascade of similar findings in other cancers, was: what gene is at the breakpoint?

The two questions are not the same. The old question treated the chromosome as the unit of analysis — a chunk of stuff missing or rearranged, in some general way. The new question treated the gene as the unit of analysis — a specific piece of DNA, with a specific function, broken or fused or moved in a specific way. The new question pointed at experiments. If you knew where the breakpoint was on chromosome 22, you could try to find the gene at that location. If you knew where it was on chromosome 9, you could try to find the gene there. If both were genes, and the translocation was joining them, you might be able to identify the fusion gene and learn what protein it made.

This was a research program. It was not a research program that anyone could pursue in 1973 with any realistic hope of success — the tools were too crude — but it was a research program that the field could see, even from where it was standing. The technology to investigate genes at the molecular level was being developed in parallel, in other labs, for other reasons. Recombinant DNA techniques — cutting DNA at specific sequences, copying it, moving

it from one organism to another — were being worked out in the early 1970s. DNA sequencing, the technique of reading the letters of DNA one by one, was still primitive but was getting better. The tools that would eventually let you go from “the breakpoint is in band q11 of chromosome 22” to “the breakpoint is in the middle of this specific gene, and here is the sequence of the resulting fusion protein” were, in the early 1970s, not yet in working order. They would be, soon.

Rowley herself would continue to work on cancer cytogenetics for the rest of her life. She would identify or co-identify a number of additional cancer-associated translocations over the next two decades. She would receive, eventually, the National Medal of Science, the Lasker Award, and the Presidential Medal of Freedom — most of the major scientific honors short of the Nobel, which she did not get. She would die in 2013, at eighty-eight, having lived to see the small chromosome from her photographs become the molecular target of a drug that worked.

The drug was a long way off in 1973.



Here, then, is where the story stands at the end of the 1970s.

Cancer is genetic, in a real and specific sense. The genetics can be read, with the right tools, in the chromosomes of cancer cells. Specific cancers have specific chromosomal rearrangements, reproducible from patient to patient. In CML, the rearrangement is a swap between chromosomes 9 and 22 — the swap that creates the Philadelphia chromosome, and that has, somewhere in the swap, joined together two pieces of DNA that should not be joined.

What is at the joint?

What protein, if any, does it make?

What does the protein do, that it would turn a stem cell in a patient's bone marrow into a stem cell that produces a slow-rising flood of abnormal white blood cells, and eventually kills the patient?

The next chapter is about people figuring out the answers to those questions. It takes place in the 1980s, in laboratories on three continents, using techniques that did not exist in 1973. By the end of it the field will know, in the precise language of biochemistry, what the Philadelphia chromosome does.

Knowing what it does, of course, is not the same as being able to fix it. But it is the necessary first step.

CHAPTER 3

3

BCR-ABL



In a laboratory at Erasmus University in Rotterdam, in the autumn of 1982, Annelies de Klein is holding an X-ray film up to a lightbox.

The film shows a column of dark bands floating on a gray background. Each band is the answer to a small question: does this fragment of DNA, taken from this kind of cell and cut in this particular way, contain the gene she is hunting for? The dark bands say yes. The blank spaces say no.

She has been doing this experiment, on and off, for months. The technique is called Southern blotting, after Edwin Southern, the British biologist who developed it in the mid-1970s. It is, in 1982, the standard way to ask whether a specific gene is present in a sample of DNA. You take the DNA out of the cells, you cut it into fragments using an enzyme that recognizes a specific stretch of letters, you separate the fragments by size in a slab of gel, you transfer them

onto a sheet of nitrocellulose, you wash the sheet with a small radioactive piece of the gene you're looking for, and you put the whole thing against an X-ray film overnight. If the gene is there, the radioactive piece sticks to it. The film, when you develop it the next morning, shows a dark band.

The gene de Klein is hunting for is called ABL.

What she is finding is that ABL is in the wrong place.



To make sense of why this matters, we need to back up several years and follow three threads. They start in different places in the 1970s and come together, by 1982, into a question that one experiment can answer.

The first thread is a virus that causes leukemia in mice. The second is a discovery about how cancer-causing genes get to be cancer-causing. The third is a piece of unexpected biochemistry — a chemical reaction nobody knew cells were doing, found in the wrong place by accident.

By 1982 these threads have braided together enough that a careful scientist in Rotterdam can ask whether a particular known gene sits at the breakpoint of a particular known translocation. By 1985 she'll have the answer to a sharper version of the same question. By 1990 the field will have shown, in mice, that a single fused gene at that breakpoint is sufficient — by itself, with no other help — to cause a disease that looks like CML.

The threads are worth taking one at a time.



The virus first.

In 1969, a researcher named Herbert Abelson, working at the National Institutes of Health, was studying viruses that cause leukemia in mice. He had a strain of one such virus that, when injected into young mice, caused an unusually fast and aggressive disease — a lymphoma involving immature B-cell precursors, killing the animal within weeks. The virus was eventually named the Abelson murine leukemia virus, or A-MuLV, and the gene inside it that was responsible for the cancer-causing activity was given the short name *v-abl* — the *v* for viral, the *abl* for Abelson.

This was, in 1969, a particular kind of finding. The field had known for decades that some viruses could cause cancer — Peyton Rous's chicken sarcoma virus, the one that took the field fifty years to take seriously, was the original example. By the late 1960s, several such viruses were known: the Rous sarcoma virus in chickens, several murine leukemia viruses including A-MuLV, an avian leukosis virus complex, a feline leukemia virus. Each one had a small genome — just a few genes — and each one carried, somewhere on that small genome, a single gene that was responsible for the cancer-causing activity. These genes were called oncogenes, from the Greek *onkos*, a swelling or mass.

v-abl was the oncogene of A-MuLV. *v-src*, the cancer-causing gene of the Rous sarcoma virus, was the most-studied. There were, by the early 1970s, perhaps a dozen viral oncogenes catalogued. Each was named after the virus that carried it. None of them, at this point, had been connected to anything in the cancers of humans.

The question that hung over the field in 1970 was: where do these genes come from?



That is the second thread.

In 1976, two researchers at the University of California, San Francisco — a virologist named Harold Varmus and a biochemist named J. Michael Bishop — published a paper in *Nature* that answered the question, at least for one virus, in a way that nobody quite expected.

They had been working with v-src, the chicken-virus oncogene. They made a small radioactive piece of v-src — essentially the same kind of probe de Klein would later use, made a few years later — and used it to ask whether anything in normal chicken DNA bound to it. If v-src was a uniquely viral creation, nothing in chicken cells should match. If it had been picked up from a cell at some point in the virus's evolutionary past, something should.

Something did. Normal, healthy, non-cancerous chicken cells contained a gene that was almost identical to v-src. Then they tried it in other species. Quail had a similar gene. So did mice. So did fish. So did humans. The gene that the Rous sarcoma virus had been carrying around in its tiny viral genome was, in some only slightly altered form, present in essentially all vertebrates. It had a normal function in normal cells. The virus had picked it up at some point — perhaps tens of millions of years ago, perhaps more recently — and kept a slightly modified version, and that slightly modified version was what caused cancer.

The implications spread quickly. The cellular version of v-src — called c-src, with the c for cellular — was not, in itself, a cancer gene. It was a regular gene doing a regular job. The viral version differed from it in a few details, and those details were what made it cause cancer. Cancer-causing genes, in other words, were not foreign agents introduced from outside. They were ordinary cellular genes, present in every healthy cell, that became cancer-causing when something specific went wrong with them.

Bishop and Varmus would receive the Nobel Prize for this in 1989. By then, the same logic had been applied to many other viral oncogenes. Each of them — v-myc, v-ras, v-fms, v-erb, and so on, including v-abl — had been shown to have a normal cellular cousin. The cellular cousins were given the same names with a c- prefix. The category of “cellular oncogenes” came into being, although the more careful word, used by people who wanted to distinguish the cancer-causing version from the normal version, was “proto-oncogene” — a normal gene that, when altered, becomes capable of driving cancer.

This was the conceptual move that the Philadelphia chromosome story needed. If the Philadelphia translocation had broken in the middle of a gene, and joined that gene to a piece of another gene, the result might be doing something analogous to what the Rous virus had done: creating a slightly altered version of a normal cellular gene, with new and dangerous properties.

The hypothesis sharpened. If you wanted to find the cancer-causing event in CML, you could look for proto-oncogenes near the breakpoints on chromosomes 9 and 22.



The third thread is shorter, and weirder, and closer to a punchline.

In 1979, a young researcher named Tony Hunter at the Salk Institute in La Jolla, California, was trying to study the protein made by an oncogene from a different tumor virus, the polyoma virus. The protein was being phosphorylated — chemical phosphate groups were being attached to it by some kind of enzyme — and Hunter was trying to find out where on the protein the phosphates went.

A brief side note on phosphorylation, because the rest of the chapter depends on it.

A phosphate group is a small chemical tag, a single phosphorus atom surrounded by oxygens, that cells use as a kind of universal switch. Enzymes called kinases attach phosphate groups onto specific spots on specific proteins; other enzymes, called phosphatases, take them off again. The presence or absence of a phosphate at a given spot can change the shape of the protein, change its activity, change which other proteins it sticks to, change where in the cell it goes. It is the cell's main way of relaying signals — of saying *things have changed, do something different now* — quickly and reversibly. Most of the proteins in any cell at any given moment have phosphates being added and removed continuously, in elaborate patterns that encode what the cell is up to.

Phosphates can be attached to several different kinds of amino acid — amino acids being the small building-block molecules that proteins are made of — but in 1979, the consensus was that essentially all biological phosphorylation happened on two amino acids in particular: serine and threonine. There was a small literature on a third amino acid, tyrosine, but it was quiet, and most working biochemists assumed that if any phosphorylation happened on tyrosine, it was rare and probably unimportant.

Hunter, working with the polyoma protein, ran his samples on a special two-dimensional plate that separates phosphorylated amino acids by their chemical properties. He had done this many times. The serine and threonine spots were where they always were. But there was an extra spot, in a place where, by his understanding, no spot should be. He went back and ran the experiment again, more carefully this time, with fresh buffers. The same extra spot appeared.

It turned out to be phosphotyrosine. Hunter had found, by what was widely understood to be partly accident and partly very careful technique, that the polyoma oncoprotein was phosphorylating tyrosines.

He went back and looked at the famous v-Src protein, which had been studied for years. v-Src had been known to have kinase activity since 1977; the assumption had been that it phosphorylated serines or threonines, like every other kinase known at the time. With the right method, Hunter saw that v-Src was phosphorylating tyrosines too.

Within a year or two, the same was shown for v-Abl. The Abelson virus protein, the cancer-causing gene from the leukemia virus that had given ABL its name, was a tyrosine kinase. So was its cellular counterpart, ABL itself. So were a growing number of other oncoproteins.

It was striking. Of the many things an oncogene could in principle have turned out to be, a disproportionate number of them were turning out to be tyrosine kinases. The class — a relatively rare type of enzyme that hardly anyone had been thinking about a few years earlier — was suddenly central to cancer biology. Whatever was special about tyrosine phosphorylation, it seemed to be the kind of signal that, if it got stuck in the on position, could push a cell toward becoming a tumor.



Now de Klein in Rotterdam.

In 1982, the situation around CML looked like this. The Philadelphia chromosome had been described twenty-two years earlier. Rowley had reinterpreted it as a t(9;22) translocation nine years earlier. The breakpoints had been mapped to specific bands — q34 on chromosome 9, q11 on chromosome 22 — with the new banding techniques. ABL, the cellular cousin of the Abelson viral oncogene, had recently been mapped to chromosome 9, in roughly the right region. It was a tyrosine kinase. It was a proto-oncogene. And the question that suggested itself — obvious in retrospect, less

obvious at the time, the kind of question that requires several different threads to have already been pulled before anyone can think to ask it — was whether the Philadelphia translocation broke chromosome 9 inside the ABL gene.

This was the question de Klein was asking. She was working in the laboratory of Gerard Grosveld, a molecular biologist at Erasmus University, with collaborators including Ad Geurts van Kessel and others, and in close communication with a group at the National Cancer Institute in Frederick, Maryland — John Groffen and Nora Heisterkamp, working in Stephen O'Brien's lab — who were doing complementary experiments.

The technique de Klein used was somatic cell hybridization. You take a human cell and a mouse cell and you fuse them together. The resulting hybrid cell, in the early days of growth, has all the chromosomes from both parents. As it divides, it tends, for reasons that are a little arcane, to lose human chromosomes more or less at random while keeping the mouse ones. After many divisions you have a population of hybrid cells, each of which retains only some of the original human chromosomes. You can sort the cells, find ones that have, say, human chromosome 22 but not human chromosome 9, or human chromosome 9 but not 22, and ask: does the human ABL gene track with chromosome 9, or with chromosome 22?

In normal human cells, you would expect ABL to track with chromosome 9, because that is where it lives. In cells from a CML patient, with the Philadelphia translocation, the answer might be different. If the translocation broke the chromosome at a point distal to the ABL gene — farther out toward the tip than ABL — then ABL would still be on chromosome 9, where it had always been. If the translocation broke the chromosome at a point closer to the centromere than ABL, then the piece carrying ABL would have moved with the rest of the distal long arm to chromosome 22.

De Klein's blot showed ABL on chromosome 22.

In the hybrid cells from CML patients, the ABL gene tracked with the small Philadelphia chromosome — the one that should not have it — not with the normal-looking chromosome 9. The translocation had broken chromosome 9 on the centromere side of ABL, picking up the ABL gene along with the rest of the distal long arm and depositing all of it onto chromosome 22.

This was the gene at the breakpoint. The Philadelphia translocation, in 1982, finally had a known piece of cellular machinery on at least one of its sides.



The paper appeared in *Nature* in November 1982, with seven authors and a careful, narrow title: "A cellular oncogene is translocated to the Philadelphia chromosome in chronic myelocytic leukaemia." It was a clean result, and the field absorbed it quickly. The cellular cousin of v-Abl was on the Philadelphia chromosome. ABL, a tyrosine kinase that was already on the field's list of suspects in cancer biology, had been moved by this specific translocation in this specific disease.

The next question was the partner. ABL was now sitting on chromosome 22, but where exactly? The translocation breakpoint on chromosome 22, in band q11, must have happened in or near some other gene. What was that gene?

Heisterkamp and Groffen, with collaborators that included the same Rotterdam group, took up this question over the next two years. They cloned the breakpoint region from chromosome 22, working outward from where ABL had landed. They found something striking: the breakpoints from different CML patients all

clustered in a relatively small stretch of DNA on chromosome 22, no more than a few thousand letters long. Different patients had breaks in slightly different places, but always within this one tight region.

They named the region the breakpoint cluster region, abbreviated bcr. The gene that contained this region was named, with the same letters, BCR. By 1984 the picture had filled in: the t(9;22) translocation broke the ABL gene on chromosome 9 in one of two introns (an intron being a non-coding stretch within a gene that gets spliced out before the gene's protein is made), and broke the BCR gene on chromosome 22, also typically in an intron, and joined them. The resulting fused gene, which sat on the small Philadelphia chromosome, was BCR on its 5' end (the upstream end, the part that gets read first) and ABL on its 3' end (the downstream end). When the cell read this fused gene and made a protein from it, the protein was a hybrid: BCR up front, ABL behind.

It was a fusion gene. Just as the new questions, in the late 1970s, had suggested it might be.



Now what did it do?

The first answers came from Owen Witte's lab. Witte had trained as a postdoctoral fellow with David Baltimore at MIT in the late 1970s, working on the Abelson virus protein, and had been part of the work showing that v-Abl was a tyrosine kinase. By the early 1980s he had moved to UCLA and set up his own lab, where he and a graduate student named John Konopka turned their attention to the BCR-ABL fusion protein.

They wanted to know two things. First, did the fusion protein have kinase activity, the way both v-Abl and normal ABL did? Second, if it did, was the activity regulated normally, or was it stuck in the on position, the way v-Abl's activity was?

The answer to the first question was yes. The fusion protein, which they showed was a single polypeptide of about 210 kilodaltons (a kilodalton being a unit of protein mass; 210 kilodaltons makes BCR-ABL a fairly large protein, but well within the normal range), retained the kinase domain of ABL and could phosphorylate substrates in the test tube. The ABL part of the fusion was still doing its enzymatic job.

The answer to the second question was also yes — in the bad direction. The fusion protein was a more active kinase than normal ABL was. The BCR portion at the front of the fusion was somehow disrupting the regulation that normally kept ABL quiet. (We now know, in some detail, how: the BCR sequences cause the fusion protein to dimerize — to pair up with another copy of itself — and the dimerization triggers the ABL kinase domain to activate the way it would normally activate only in response to the right cellular signal. In 1985 the molecular details weren't yet that clear, but the functional answer was.) BCR-ABL was a tyrosine kinase whose activity was no longer being properly turned off. It was, in the language we now use without thinking, constitutively active — stuck in the on position, phosphorylating its substrates whether or not the cell wanted it to.

This was, as a finding, exactly what the field had been hoping it would be.



There was one more question, and it was the question that mattered most for whether any of this would translate into anything useful for patients. Was BCR-ABL, by itself, sufficient to cause CML? Or was it just one piece of a more complicated puzzle — a contributor to the disease, but not the whole story?

The question is harder to answer than it sounds. CML, in a person, develops over years, in the context of all the other things that happen in a complex organism. You can show that a fusion gene exists in CML cells. You can show that it makes a fused protein with abnormal activity. But to show that the fusion is *the cause* — that it's enough to make CML happen, with nothing else required — you need to put just the fusion into a cell that doesn't already have it, and watch what happens.

In 1990, a paper appeared in *Science* that did this. The first author was George Daley, a graduate student in David Baltimore's lab at MIT (Baltimore by this point being one of the central figures in molecular biology, having shared the Nobel Prize fifteen years earlier for the discovery of reverse transcriptase). The other lead author was Richard Van Etten, a postdoc in the same lab. Their experiment was, in concept, straightforward.

They built a retrovirus carrying the BCR-ABL gene. (A retrovirus is a virus that, when it infects a cell, inserts its genetic material into the cell's DNA; the Abelson virus from the start of this chapter is a retrovirus, and so is HIV. Researchers had learned to take advantage of this property by removing the disease-causing parts of certain retroviruses and replacing them with whatever gene they wanted to deliver.) They took bone marrow from healthy mice. They infected the bone marrow cells with the BCR-ABL retrovirus, so that some fraction of the marrow cells now had the BCR-ABL gene inserted in their genome. They transplanted the infected marrow into mice whose own marrow had been wiped out with radiation — mice that, without the transplant, would have died of marrow failure within days.

The transplanted mice survived the marrow failure, because the donor cells took over and started making blood. And then most of them got sick. The disease they developed looked, in clinical and

pathological detail, like CML: a high white blood cell count, a swollen spleen, infiltration of the bone marrow with too many immature myeloid cells, and — in mice that lived long enough — progression to something resembling blast crisis. The mice were dying of a disease that BCR-ABL alone, expressed in normal mouse marrow cells, was producing.

The case was as close to closed as biology gets. CML, the disease, was caused by BCR-ABL, the fusion gene. One translocation, one fusion, one constitutively active kinase. Everything that went wrong in a CML patient — the slow rise in white cell counts, the enlarged spleen, the eventual transformation into something acutely fatal — was downstream of one stuck-on enzyme.



This is the picture as the 1980s ended.

A reciprocal translocation between chromosomes 9 and 22, occurring in a single cell at a single moment in a single hematopoietic stem cell in a person's bone marrow, fuses the 5' part of the BCR gene to the 3' part of the ABL gene. The fused gene makes a fused protein, p210 BCR-ABL, in which the BCR portion forces the ABL portion into a permanently active state. The protein is a tyrosine kinase — an enzyme that attaches phosphate groups to tyrosine residues on its substrates. The substrates are proteins involved in cellular pathways that drive proliferation, suppress programmed cell death, and modify how the cell interacts with the bone marrow environment. Because the kinase is always on, those pathways are always active. The cell carrying the fusion grows when it shouldn't grow, divides when it shouldn't divide, fails to die when it should die. Its descendants — a whole lineage of myeloid cells, all carrying the same fusion — do the same. Over months and years,

they accumulate. They expand into the bone marrow. They spill into the blood. They eventually transform into something more aggressive and kill the patient.

All of this is downstream of one enzyme.

If you could turn the enzyme off, you might be able to turn the disease off.



Brian Druker is, in 1990, finishing his postdoctoral fellowship in Tom Roberts's laboratory at the Dana-Farber Cancer Institute in Boston. He has been there since 1984. He was trained, originally, as a medical oncologist; he came to Dana-Farber because he wanted to understand cancer at the molecular level rather than just treat it, and Roberts's lab was a place where you could do that. He has spent his fellowship developing antibodies that recognize phosphotyrosine — antibodies that, in essence, light up tyrosine-phosphorylated proteins so you can see them and measure them. The antibodies are good. Other labs have started asking him for them.

He has also been reading the BCR-ABL literature.

The picture, by 1990, is the one this chapter has just laid out. Druker can see it as clearly as anyone in the field. He can also see, more clearly than most, what it implies. The disease is one enzyme. The enzyme is in a class — tyrosine kinases — that does its work through a particular chemical reaction. Chemical reactions can, in principle, be inhibited by small molecules that fit into the active site of the enzyme. If you could find a small molecule that fit into the active site of BCR-ABL, and didn't fit into the active sites of the kinases that the body needs in order to function, you might have a drug that turned off CML without killing the patient.

This is the obvious idea. It is also, in 1990, an idea that essentially nobody at any major pharmaceutical company is willing to take seriously, for reasons we'll come to in the next chapter.

There is, however, one chemist Druker has heard of, working at a Swiss company called Ciba-Geigy, who is making compounds that look as if they might be selective inhibitors of particular kinases. The compounds are intended as research tools, not as drugs. The chemist's name is Nick Lydon. The compounds are sitting on a shelf in Basel.

Druker writes him a letter.

CHAPTER 4

4

The Compound



In a small laboratory at Oregon Health Sciences University, on a winter morning in 1995, Brian Druker is counting colonies in petri dishes.

There are two stacks of dishes on the bench in front of him. Each dish contains a thin layer of soft agar — a clear, jelly-like medium — with cells suspended in it. The cells, when they're healthy and dividing, multiply over the course of a week or so into small, dense balls of cells called colonies. You can see the colonies under a low-power microscope. You can count them. The number of colonies in a dish tells you how many of the cells you started with were able to grow.

The dishes in the first stack contain cells from a patient with CML. The dishes in the second stack contain bone marrow cells from a healthy donor — normal cells, the controls. Each stack is split into two groups: dishes treated with a small molecule that arrived

several months earlier in a glass vial from Switzerland, and dishes treated with the same volume of the same buffer but no small molecule.

Druker counts the untreated CML dishes. Lots of colonies, as expected; the leukemia cells grow well in soft agar. He counts the treated CML dishes. The colonies are gone. There are a few stragglers, but the difference is unmissable: the dishes that received the small molecule have lost essentially all of their CML colonies.

He counts the untreated normal dishes. A reasonable number of colonies. He counts the treated normal dishes. Almost the same number of colonies. The drug, whatever it is, is not bothering normal bone marrow cells.

He counts everything again, slowly, because you don't get to see this kind of result very often, and when you do, you check it.



Before going any further, it's worth being clear about what kind of object this small molecule is, and why making it had taken seven years and a small company's worth of effort.

A kinase, as we established in the last chapter, is an enzyme that attaches a phosphate group onto a specific spot on a target protein. A kinase inhibitor is a small molecule that prevents the kinase from doing this. The way most kinase inhibitors work is by occupying the spot where the kinase normally puts the phosphate — not the spot on the target protein, but the spot on the kinase itself, where the kinase grabs the phosphate from in the first place.

Where does the kinase get the phosphate from? It gets it from a small molecule called ATP — adenosine triphosphate, the universal energy currency of every cell in every organism on earth. ATP is, structurally, a sort of three-part molecule: a base, a sugar, and a chain of three phosphate groups attached to the sugar in a row. The

bond between the second and third phosphate is somewhat unstable; breaking it releases a useful amount of energy, which the cell can apply to whatever job it needs done. Most of what cells do, in the way of work, is powered by breaking that bond. A kinase, in the act of phosphorylating its target protein, breaks the bond by transferring the third phosphate from ATP onto the target. The kinase is, in effect, a tiny machine that takes ATP, takes a target protein, and combines them so that the phosphate ends up where it should.

To do this the kinase has, somewhere in its three-dimensional structure, a pocket shaped to hold ATP. The pocket is roughly the same shape from kinase to kinase, because ATP is the same shape in every kinase. The pocket has a few specific contact points where particular parts of the ATP molecule fit snugly. Detailed-features-wise, the pocket varies a little: a particular kinase has a slightly different lining, in places, from the next kinase. But the overall geometry is conserved, because they all have to bind the same fuel.

A kinase inhibitor that works by occupying the ATP pocket is, in essence, a small molecule shaped well enough like ATP that the pocket accepts it, but unable to do ATP's job once it's in there. With the inhibitor stuck in the pocket, the kinase can't load real ATP, and so can't transfer a phosphate. The enzyme is, for as long as the inhibitor is bound, switched off.

This is, in principle, an attractive way to drug a kinase. ATP is small — small enough that the molecules that compete with it are small too, and small molecules tend to be the kind of compounds that can be made into pills, get absorbed through the gut, distribute around the body, and reach their targets. The trouble — the reason this approach had been failing for years before Druker's package arrived from Switzerland — is selectivity. There are, as the prologue mentioned, more than five hundred kinases in the human genome.

They all have an ATP pocket. The pockets vary, but not by a lot. A molecule that fits the ATP pocket of one kinase very often fits the ATP pockets of many others. And since the body needs many of those kinases doing their jobs in order to keep functioning, a molecule that hits all of them is not a drug, it's a poison.

The challenge, in 1990, was making a molecule that exploited the small differences between kinase ATP pockets to bind one kinase tightly while leaving the others alone. Drug chemists had been trying for years. The general view was that selectivity at the level required — hitting a single kinase out of five hundred, with no significant effect on the other 499 — was probably not achievable. The kinases were just too similar. Every time someone made a compound that worked nicely on one kinase, follow-up testing would reveal that it also worked on a dozen others. You could *kind of* hit a kinase. You couldn't selectively hit one.

This was the consensus that the small group of chemists in Basel, in the late 1980s, was about to start arguing with.



Ciba-Geigy in 1988 was a Swiss pharmaceutical company of moderate ambition and considerable history. It was the result of a 1970 merger between two older Basel firms, Ciba and J. R. Geigy, both of which had begun in the nineteenth century making industrial dyes and had, like several other Basel chemical companies, eventually moved into pharmaceuticals on the back of the chemistry expertise. The company had a respectable but not spectacular drug portfolio. Its oncology pipeline, by the standards of the larger players, was thin.

Inside the company there was a small research effort, run by a physician-scientist named Alex Matter, whose remit was to develop new cancer drugs. Matter had come to Ciba-Geigy from Schering-

Plough; he was a clinically trained oncologist who had moved into the research side, with strong opinions about what kinds of drugs would actually help patients and a willingness to make bets on hypotheses that other people thought were premature. He had decided, around the time he took the job, that targeting kinases looked like the most interesting frontier in cancer drug discovery, and that Ciba-Geigy should be working on it whether or not the rest of the industry was.

He needed people. He hired Nick Lydon, a Scottish biochemist who had been working at Schering-Plough on tyrosine kinases, and brought him to Basel as the head of a small biology effort. He also brought on a chemist named Jürg Zimmermann, who would do the actual work of designing and making the compounds. Both were young; both had reputations as careful, patient workers; both believed, as Matter did, that the selectivity problem was an engineering problem and not a fundamental obstacle.

Their first target was not, in fact, BCR-ABL. It was protein kinase C — PKC — a serine/threonine kinase (one of the more conventional kinds, phosphorylating serine and threonine rather than tyrosine) that had been linked to several diseases, including some cancers. The early work was a screening campaign: Zimmermann's chemists made or purchased a library of small molecules, the biology team tested each one against PKC in a test-tube assay, and the molecules that inhibited the enzyme were identified as starting points.

Several scaffolds came out of the screen. One of them was a class of compounds called 2-phenylaminopyrimidines — a chemical structure consisting of a six-membered ring with two nitrogens (the pyrimidine), with another ring (the phenyl) attached through a

nitrogen-containing linker (the amino bridge). It was a flexible scaffold; you could decorate it with various chemical groups in various positions, and Zimmermann began making variations.

Some of these variations inhibited PKC, as intended. Some of them, on follow-up, also turned out to inhibit other kinases. This was, in the standard interpretation, a problem — a ‘dirty’ compound that hits more than its intended target is harder to develop into a drug, because it’s harder to predict what it will do in a body. The conventional response was to optimize toward selectivity for the intended target.

Lydon had a different idea. If the compounds were hitting other kinases, that information was worth keeping. Tyrosine kinases were emerging as cancer-relevant targets across the board, and a series that hit several of them was, from the right angle, a starting point for several different programs rather than a contamination problem in one. In particular, some of Zimmermann’s 2-phenylaminopyrimidines were inhibiting tyrosine kinases that Lydon was personally interested in: PDGFR, the platelet-derived growth factor receptor, which was thought to be involved in glioblastoma and certain other tumors; and the closely related kinase c-KIT, which would later turn out to drive a different family of cancers.

And ABL.

Lydon had been a tyrosine kinase biochemist at Schering-Plough. He had read the same BCR-ABL literature that Druker had read. He knew exactly what it would mean if you had a selective inhibitor of ABL.



The optimization was incremental. Zimmermann would make a new variation, the biology team would test it against a panel of kinases, the data would come back, and Zimmermann would make another variation. The goal was to find the modification that pushed potency higher against ABL while pushing potency lower against everything else.

There were many variations. The 2-phenylaminopyrimidine series was eventually elaborated into hundreds of compounds, each with slightly different chemical decorations. Most of them were not impressive. Some of them were a little better than the last batch. Occasionally a single change — a methyl group added in this position, an amine swapped for that one — would produce a compound noticeably more selective than its predecessors.

Around 1992 or 1993 the team converged on a molecule that, in their assays, inhibited ABL with respectable potency and inhibited only a small handful of other kinases meaningfully. It was given a Ciba-Geigy code number: CGP 57148. (The salt form actually used in animal and clinical work was the methanesulfonate salt, sometimes written CGP 57148B. For our purposes, the code number is the molecule.) Its known activity was: ABL, BCR-ABL, PDGFR, c-KIT. Its known non-activity was: most of the other kinases the team had been able to test. The off-target hits were, by the standards of the field, an unusually short list.

By the standards of pharmaceutical drug discovery, however, CGP 57148 was a problematic compound. It was not, in 1993, considered a viable drug candidate by most of the people inside Ciba-Geigy who were paid to make those calls. The reasons were partly chemical — its physical properties, in some respects, were not ideal — and partly strategic. Even if it worked, the company asked,

what would it work on? CML was uncommon. The class — selective kinase inhibitor — had no approved precedent. The compound looked more like a research tool than a marketable medicine.

This was the assessment of CGP 57148 inside Ciba-Geigy in early 1993. It was the kind of compound that made a useful chemical biology paper. It was not the kind of compound that any normal pharmaceutical project would advance toward the clinic.



Brian Druker, in late 1992, was a few months from leaving the Dana-Farber Cancer Institute to take a faculty position at Oregon Health Sciences University. He had been corresponding with Nick Lydon for some time — since the late 1980s, intermittently, on the strength of overlapping interests in tyrosine kinases. Lydon had sent Druker compounds before; some of the early Ciba-Geigy molecules had been useful in Druker's antibody work, as research tools to manipulate kinase activity in cells. The relationship was collegial and informal. Lydon was a chemist working on inhibitors; Druker was a clinician-scientist with the antibodies needed to read out what those inhibitors did to cells. They were natural collaborators.

Druker, by 1993, knew about the 2-phenylaminopyrimidine series. He knew, in general terms, that some of those compounds inhibited ABL. He had not yet had access to the more recent compounds, the ones that were more selective; those were, internally at Ciba-Geigy, being held closely while the chemistry was still being optimized. But the trajectory was visible. He wrote to Lydon, around the time he was starting at Oregon, and asked whether he could test a specific compound — the most ABL-selective one Lydon had — against BCR-ABL in CML cells.

Lydon agreed. Internally at Ciba-Geigy, sending compounds to outside investigators required navigating some bureaucracy — you had to make sure they wouldn't be used in ways the company didn't approve of, that intellectual property was protected, that the compound wasn't being shared with competitors. Lydon worked through it. By 1993 or 1994, Druker had a vial of CGP 57148 in his lab in Oregon.

His test plan was straightforward, and it was straightforward because he had spent his postdoctoral years developing exactly the right tools. He had antibodies that could detect tyrosine phosphorylation in cells. He could expose CML cells to the compound, harvest them, and ask whether BCR-ABL was still active inside the treated cells — whether the kinase, exposed to the compound, was still adding phosphates to its substrates the way it normally would.

It wasn't. The treated CML cells, on the antibody readout, looked like cells with no BCR-ABL activity at all. The compound was reaching its target, fitting into the active site of the kinase, and shutting it down. This was, in itself, encouraging but not yet decisive; many compounds inhibit kinases in cells without doing anything useful for the cells' fate.

The decisive experiment was the colony assay we opened the chapter with. CML cells, plated in soft agar with the compound, didn't form colonies. CML cells without the compound did. Normal bone marrow cells, with or without the compound, formed colonies at about the same rate. The drug was killing CML cells specifically. It was not killing normal cells in any meaningful way.

Druker did the experiment more than once. He did it with cells from multiple CML patients. He did it at multiple drug concentrations. He did it with normal cells from multiple donors.

Every time, the same result: the compound selectively killed BCR-ABL-positive cells while sparing the BCR-ABL-negative cells alongside them.



The reason this is an unusually clean result, the reason Druker spent his evenings in the OHSU laboratory in 1995 counting colonies several times to make sure he had not mis-pipetted somewhere, is that this is what a targeted cancer drug is supposed to look like, and almost no cancer drug had ever looked like this before.

The standard cytotoxic chemotherapies — the drugs that had been the backbone of cancer treatment for decades — work by killing rapidly dividing cells. They damage DNA, or interfere with cell division, or block enzymes that proliferating cells need disproportionately. Cancer cells, dividing more than most cells, are caught preferentially. But normal cells that also divide — the cells of the bone marrow, the lining of the gut, the hair follicles — are also caught. This is why chemotherapy makes patients lose their hair, get nauseous, and get dangerously low blood counts. The drugs aren't selective. They simply happen to hurt cancer slightly more than they hurt the patient.

A drug that kills CML cells while leaving normal bone marrow cells alone is doing something else entirely. It is not exploiting a generic feature of dividing cells. It is exploiting a specific molecular difference — the presence of BCR-ABL — between the cancer cells and the normal cells. Without BCR-ABL, the drug has nothing to do, which is why the normal cells don't notice it. With BCR-ABL, the drug shuts down the one enzyme the cancer cells absolutely depend on, and the cancer cells die.

This is what people had meant, all those years, by the phrase ‘targeted therapy.’ The phrase was older than the achievement; the dream of a magic bullet that picked out diseased cells from healthy ones, leaving the second alone, went back to Paul Ehrlich and the early twentieth century. The dream had survived, mostly intact and mostly unrealized, through ninety years of cancer drug discovery. There were a few partial cases — antibodies against specific cell-surface markers, certain hormones — but not really, in 1995, an oral small molecule that did exactly what targeted therapy was supposed to do. The colonies in Druker’s dishes were, in retrospect, the first clear demonstration in human cells that this was possible.

This was not yet obvious. Druker himself was cautious. The cells were one piece of evidence; you would also need to show that the drug worked in animals, that it could be given safely, that it would do the same thing in a body that it did in a dish. There was a great deal of work between this colony assay and a treatment for a real patient.

But the result, in the lab, was unmistakable. He had a compound that killed CML cells and spared normal cells. It came from a chemist who could send him more of it. The molecule existed, in a freezer, in Oregon and in Switzerland.



There was now a problem that none of the science had prepared anyone to solve.

Druker wanted to take the compound forward. He wanted to test it in animals, then in patients. He wanted Ciba-Geigy, which owned the molecule, to develop it as a drug — to commit the resources required to manufacture it at scale, run the toxicity studies that regulators required, write the regulatory documents, sponsor the

clinical trials. None of this was something an academic clinician at OHSU could do alone. The molecule belonged to the company. The decision to develop it was the company's.

And the company, in 1995, was not enthusiastic.

From inside Ciba-Geigy, the case for development looked weak. CML was a disease of perhaps five thousand new patients a year in the United States, similar numbers in Europe. The market, in pharmaceutical terms, was small. The compound's class — selective kinase inhibitors — had no approved precedent and a long history of disappointing expectations. The compound itself had real chemistry issues that would need to be solved before it could be made at scale. There were fears that any clinical benefit might be brief, with cancers evolving resistance the way they did to other targeted approaches that had been tried. There were fears that the toxicology would surprise everyone unpleasantly. There were fears about a more general principle: a class of drugs that took years to develop and might be quickly outpaced by resistance was not the kind of bet a publicly-traded pharmaceutical company was set up to make. There were, internally, more reasons to walk away than to keep going.

From inside the small group that wanted to keep going — Lydon, Matter, and a few allies — the case looked different. The compound did exactly what a targeted therapy was supposed to do. It came out of a careful, principled research program that had identified a specific target and a specific molecule with all the right properties. Druker's data in CML cells were as clean as anything anyone had seen. CML, while uncommon, was uniformly fatal under the existing standard of care; the unmet need was real and obvious. And if the approach worked here, it would set a template for a generation

of similar drugs. Letting CGP 57148 sit on a shelf because the market was small was, in this view, a profound failure of strategic imagination.

Both views were defensible. Both views were held by people who were trying, in their respective frames, to do good work. The disagreement was not a disagreement between heroes and villains. It was a disagreement between people who saw an unusual opportunity and people whose job it was to make sure their company didn't pursue every unusual opportunity it saw, because most of them, on examination, turn out not to be the opportunity they looked like.

The next chapter is about how that disagreement got resolved.

It took five years.

CHAPTER 5

5

The Long Argument



In Brian Druker's clinic at Oregon Health Sciences University, in the autumn of 1996, a woman is sitting in the chair across from him. She has CML. She was diagnosed eighteen months earlier. She is forty-seven. She has two children in college. She would like to know, given everything Dr. Druker has been able to read in journals and hear at conferences, whether there is something coming.

Druker has, in May of this year, published a paper in a journal called *Nature Medicine*. It is titled "Effects of a selective inhibitor of the Abl tyrosine kinase on the growth of Bcr-Abl positive cells." The paper has eight authors. Druker is the first; Nick Lydon, his collaborator at the Swiss pharmaceutical company that supplied the molecule, is the senior author; the others are mostly lab members from the two sides. The paper shows what Druker has been seeing in his lab for two years — a small molecule, called CGP 57148 in the paper, kills BCR-ABL-positive cells and spares normal cells, in cell culture and in mice. The data are clean. The implications are clear.

Druker has, on the bench down the hall, in the freezer, a few hundred milligrams of the molecule.

He cannot give any of it to the patient in the chair. The molecule is owned by the Swiss company. The Swiss company has not yet decided to develop it as a drug. Without that decision, there is no manufacturing at scale, no formal toxicology studies, no regulatory dossier, no clinical trial, and no way for any patient anywhere to take the compound.

Druker does not, in 1996, have a confident answer to give the patient about when this might change.



It is worth being clear, before going further, about what actually has to happen between a clean result in a dish and a pill in a patient's hand.

There is, first, formal toxicology. A drug candidate has to be tested in animals, generally two species — one rodent (typically rats or mice), one non-rodent (typically dogs, sometimes primates). The animals are given the drug at a range of doses, including doses much higher than what would ever be given to a human, and the people running the studies look at everything they can think to look at: weight, behavior, blood chemistry, organ histology after the animals are sacrificed, the appearance of every tissue under a microscope. The point is to find out what the drug does at high doses to a body, and at what dose problems start to appear. The lower edge of that range — the dose at which the first sign of trouble shows up — sets the upper edge of what regulators will allow you to test in humans, with a built-in safety margin.

There is, second, manufacturing. A clinical trial requires the drug to be made in large enough quantity, with consistent enough purity, that every dose given to every patient is essentially identical. This is

harder than it sounds. A molecule that an academic chemist can synthesize a hundred milligrams of, in glassware, with reasonable yield, is not the same problem as a molecule that has to be made by the kilogram, under controlled conditions, with documented batch-to-batch reproducibility. The chemistry has to be optimized, scaled up, run in larger reactors, and validated. The crystalline form of the final molecule — the way the molecules pack together as a solid — has to be selected and locked down, because different crystalline forms of the same compound can dissolve at different rates and produce different drug levels in the bloodstream. The formulation has to be developed: is the drug a tablet, a capsule, a liquid, and what is in there with it? The formulation has to be tested for stability: does the drug still work after it has been sitting in a bottle for six months at room temperature?

There is, third, the regulatory dossier. To start a phase 1 trial in the United States you have to file an Investigational New Drug application — an IND — with the FDA. The IND collects everything: the chemistry, the manufacturing, the animal data, the proposed clinical protocol. It runs to thousands of pages. Reviewers at the FDA read it and either let the trial go forward (typically thirty days after filing, if no concerns are raised) or put it on hold while questions are answered.

There is, fourth, the trial itself. Sites have to be selected. Investigators have to be trained. Institutional review boards — the local ethics committees — have to approve the protocol. Patients have to be identified, enrolled, consented. The drug has to be shipped, stored, dispensed. The data have to be collected and audited. All of this takes money, infrastructure, and people who do this kind of work for a living, which is to say, the resources of a pharmaceutical company.

An academic researcher with a clean result in a dish cannot make any of this happen on their own. They cannot manufacture a clinical-grade kilogram of compound. They cannot file an IND for a molecule they don't own. They cannot, in any practical sense, push a drug forward without the company that holds the molecule deciding to push it forward too.

What an academic researcher with a clean result can do is everything that comes before the company's decision and everything that helps the company change its mind.



Ciba-Geigy in 1995, when Druker first started lobbying for the compound to be advanced, was distracted.

It had begun to merge with another Swiss pharmaceutical company, Sandoz. The two had been rivals in Basel for most of a century. The deal had been announced in March 1996; it was at the time the largest corporate merger in history, in dollar terms, although it would be quickly surpassed by other late-1990s consolidations. The combined company was called Novartis — a name chosen, in the way these names are chosen, to evoke novelty without committing to anything in particular. The merger was finalized in late 1996. Several thousand jobs were eliminated. Two complete pipelines of research compounds, with their associated programs and people, had to be rationalized into one.

Inside Ciba-Geigy's oncology research, the kinase inhibitor program was small enough that it could plausibly be ended without anyone outside the company noticing. Alex Matter — the head of the program, the one who had hired Lydon and Zimmermann — had been pushing for the compound for years and had kept it alive in the budgets, but he was one voice among many, and the merger threatened to bring in a whole new set of priorities and a whole new

layer of management to clear them with. Lydon, the biochemist whose hand had been on the program from the start, was navigating his own future at the merged company.

Druker, on the outside, was watching this and growing more impatient.



It is worth saying, at this point, that the case against developing CGP 57148 was not stupid.

It was, in fact, the case that any responsible pharmaceutical executive would have made about most molecules in most companies most of the time. Compounds like CGP 57148 — promising in academic studies, owned by the company, advocated by enthusiasts, lacking obvious commercial scale — are common. Most of them, statistically, are not destined to become useful drugs. They fail in animals, or they fail in humans, or they make it through trials but the side effects are unacceptable, or they make it through trials and approval but no one will pay for them, or they cost more to develop than they ever earn back. Pharmaceutical companies that pursue every promising-looking compound from every advocate go bankrupt. Pharmaceutical companies that pursue compounds with care, that demand a clear commercial case before committing development resources, are doing what the system rewards them for doing.

CML was uncommon. The compound was a member of a class — selective kinase inhibitor — that had no approved precedent. The chemistry had real issues. There was a genuine concern, raised internally and at the time entirely reasonable, that even if the drug worked it might work briefly, with cancers evolving around it and patients relapsing within months. There were fears about toxicity. There were fears about resistance. There were fears about the time

and money it would take to develop a drug that would, in the end, be used by relatively few people. None of these fears was crazy. Some of them, as we'll see in chapters nine and ten, were even partly right.

The case for developing CGP 57148, in 1996 and 1997, rested on the narrowness and cleanness of what Druker had shown. The molecule did exactly one thing, very well, in cells. The disease was caused by exactly one thing. There was an unusually direct line, by the standards of cancer drug discovery, between what the drug did and what the patient needed. CML, while uncommon, was uniformly fatal; the unmet need was as clean as the molecular case. And if the approach worked, it would establish a paradigm — not just a drug, but a way of finding drugs — that could be applied across cancers.

Both cases were defensible. They were also, when stated by their advocates inside Novartis, mutually unconvincing.



The new CEO of Novartis, the Swiss-American physician Daniel Vasella, took office in 1996, when the merger was completed. He was forty-three. He had trained as a medical doctor, briefly practiced medicine, and then moved into the pharmaceutical industry through Sandoz, where he had risen rapidly. He had a reputation for being unusually direct and unusually willing to make decisions quickly. He also had, as it would later turn out to matter, the perspective of a doctor; he was not unfamiliar with the experience of having a patient in front of him whom he could not help.

Vasella did not, in 1996 or 1997, have a strong opinion about CGP 57148. He had inherited a pipeline of dozens of compounds, each at different stages of development, each with its own internal

champion and its own internal skeptic, each demanding attention. The kinase inhibitor program was a small part of that pipeline. It was being kept alive, but it was not being prioritized.

What Druker was trying to do, from Oregon, was raise the priority.

He did this in several ways. He gave talks. The Nature Medicine paper had been the first formal publication; it was followed by oral presentations at scientific meetings, the most important being the American Society of Hematology annual meeting in December of each year, where the entire blood-cancer research community gathers to share new data. Druker presented his preclinical results to packed rooms. Hematologists understood instantly what they were seeing. Many of them began calling Novartis directly, asking when the trial would start. Some of them wrote letters. The pressure was diffuse, but it accumulated.

He also worked the academic publication channel relentlessly. The 1996 Nature Medicine paper was followed by additional papers — in *Blood*, in *Clinical Cancer Research*, in other venues — each one extending the preclinical case. The compound worked against primary cells from CML patients. It worked against cells from patients with the more aggressive accelerated and blast-phase forms of the disease. It worked against a related leukemia, Philadelphia-positive acute lymphoblastic leukemia, that shared the same fusion gene. It worked in mouse models. By 1998, the volume of preclinical data on CGP 57148 was greater than what most early-stage compounds carried into the clinic, and most of it had been generated in academic labs by people whose paychecks did not depend on Novartis's decision.

He also kept up direct contact with the people inside the company who were making the decisions. He flew to Basel. He hosted Novartis people in Oregon. He sent updates and reprints. He

made it difficult for the program to quietly die, because every time someone inside the company tried to scale it back, there was an outside investigator with new data calling to ask why.

Lydon was doing the same thing from inside. He had, in 1997, left Novartis for a small biotechnology company in California, but he stayed in close contact with the program and its leadership. The departure removed a senior internal advocate, but Matter remained, and a small number of others.



There was, around 1996 or 1997, a problem with the dog tox studies.

The dose-finding work in dogs — the standard non-rodent species — turned up signs of liver toxicity at high doses. This was, depending on whom you talked to, either a deal-breaker or a nuisance. The deal-breaker reading went: a drug that hurts dog livers may hurt human livers, and a liver-toxic drug for a small disease is a non-starter. The nuisance reading went: the toxicity appeared at doses well above the therapeutic range, the safety margin was acceptable by ordinary pharmaceutical standards, and the relevant question was whether the drug would be safe at the doses where it actually worked. Both readings were consistent with the data. The two camps inside the company read the same numbers and disagreed about what they meant.

The argument took months. It was eventually resolved in favor of moving forward, on the grounds that the toxicity in dogs occurred at exposure levels well above what was projected to be needed in humans, that the molecular target was specific enough that off-target effects in human liver should be limited, and that the disease was severe enough to warrant proceeding into a phase 1 trial where careful monitoring could detect any problems early. The toxicology team signed off. The IND application could go forward.

But the dog issue had cost time. It had given the skeptical reading inside the company more ammunition; for several months it looked, from outside, as if the program might be terminated. Druker, hearing about the worries through Lydon and through his own contacts, was alarmed. He spent the period writing letters and making phone calls and giving talks.

From the perspective of someone with a patient sitting in their clinic, the question “is the dose at which dog livers show damage acceptable” and the question “is my patient going to live” are the same question. From the perspective of someone running a pharmaceutical company, they are not.



What seems to have shifted the situation, more than any single factor, was the accumulation. By 1998, the case for at least running a phase 1 trial of CGP 57148 had become very hard to argue against without sounding like the company was simply unwilling to develop the compound at all.

The preclinical data were exceptionally clean. The molecular target was thoroughly validated. The unmet medical need was evident. The toxicology, after months of argument, was acceptable. The chemistry could be made at the scale a phase 1 trial required. The investigators were in place; Druker at OHSU, Charles Sawyers at UCLA, Moshe Talpaz at MD Anderson Cancer Center in Houston — three sites, three established academic hematologists, all of whom understood the science and were prepared to recruit and treat patients carefully. The protocol had been worked out in detail.

Vasella, by some accounts, made the call to advance the program in late 1997 or early 1998. He has said in subsequent interviews that he saw the case as scientifically compelling and was willing to make the bet. The bet was, in his framing, about whether a careful and

selective drug for a specific disease, if it worked, would justify the development cost — a question whose answer, he has said, he was willing to wait for the trial data to settle. The internal politics of how this decision actually got made, what conversations Vasella had with whom, who pushed back and who relented, are not fully captured in any public account, and probably never will be.

What is captured is that the program moved from “being kept alive” to “being actively developed” over the course of about a year. The IND was filed in early 1998. The FDA cleared it in due course. The phase 1 protocol — a dose-escalation study in CML patients who had failed interferon, the standard first-line therapy — was finalized. The first patient was scheduled to be enrolled in June.

The molecule, by this point, had been renamed. It was no longer CGP 57148. Internally at Novartis it had been given a new code: STI-571. The letters stood for “signal transduction inhibitor,” the broad class of drug Novartis was now describing it as. It was the same molecule. It had a slightly more marketable acronym.



There is a tendency, in the way this kind of story is told later, to compress everything that happened between 1993 and 1998 into a few sentences about delay and turn the company into the villain of the piece.

It is worth resisting that tendency, because it is wrong, and because it makes the actual story less interesting.

Novartis was not the villain. Novartis was a large, publicly-traded pharmaceutical company doing its best, by its own internal standards, to make rational decisions about which compounds to develop. CGP 57148, throughout the period in question, was a compound with real questions hanging over it: a small market, an unprecedented mechanism, real toxicology concerns at high doses,

the fear of resistance. Reasonable people inside the company, looking at the same data Druker was looking at, came to a different conclusion about how to weight the evidence. They were not corrupt and they were not stupid. They were operating in a system that selects for, and that has historically been right to select for, a high bar of evidence before committing the resources required to develop a drug.

What Druker had, that allowed him to push past the bar, was three things. He had unusually clean data. He had unusual persistence — a willingness to spend years writing letters and giving talks and making phone calls and accepting that no single one of these activities would by itself change the outcome. And he had, inside the company, a small group of allies who were willing to argue his case in rooms he was not in. Without any of these three, the molecule probably would have stayed on the shelf.

Even with all three, it took five years from the first cell-line result to the first patient. The patient who walked into Druker's clinic in 1993, the one the prologue described, had — if she was lucky and her chronic phase was on the longer end — about that long before the disease progressed. Some of the patients Druker met in his early years at Oregon were alive when the trial started. Some were not. There is no way to recover the patients who weren't, and no point in pretending that the cost of the delay was zero. But there is also no clean alternative history in which a more decisive company would have produced a usable drug substantially faster, given the actual state of the science and the actual constraints on what the system can do quickly. The five years were too long. They were also, given everything, not crazy.

Druker himself, in interviews after the drug was approved, has been thoughtful about this. He has said that he understands why Novartis was cautious, that he believes the people he worked with

there were trying to do the right thing as they saw it, and that the system that produces drugs is the system that produces drugs. He has also said that he does not regret pushing as hard as he did, and would do it the same way again. Both things can be true.



On a morning in late June of 1998, in Druker's clinic at Oregon Health Sciences University, the first patient enrolled in the phase 1 trial of STI-571 — a man whose CML had progressed past every other available treatment — swallowed the first capsule of a drug no human had ever taken before. The dose was very low. The starting dose for any phase 1 study is, by design, low; the goal of a phase 1 trial is to find the dose at which problems first appear, and you do that by starting well below where you expect problems and working upward in small steps.

At the dose he was given, on the basis of the animal studies, no one expected the drug to do much of anything. The point of the early doses was to demonstrate safety. Efficacy, if there was any, was not expected until the doses got considerably higher.

This is what nearly everyone running the trial believed in June 1998.

It is not what happened.

CHAPTER 6

6

First in Human



The Oregon Health Sciences University Hospital sits on a hill in southwest Portland, in a complex of brick and concrete buildings that look out over the Willamette River and the Cascade Range beyond. The hematology clinic where Brian Druker sees patients is on one of the upper floors. Most days, sitting in the waiting room, you can see Mount Hood through the windows on a clear morning, and on a less clear morning, which is most mornings, you can see the lower edges of the cloud bank that the building sometimes seems to be inside.

On the morning of June 22, 1998, in an examination room down the hall from the waiting area, the first patient enrolled in the phase 1 trial of STI-571 swallowed a single capsule of the drug with a paper cup of water. He had failed interferon. He had refused or was ineligible for transplant. His CML was, by any reasonable measure, going to kill him on the timeline this kind of CML kills people on. The capsule he had just swallowed contained twenty-five milligrams

of the molecule. Twenty-five milligrams was the starting dose, chosen — as starting doses in phase 1 trials always are — to be lower than the dose anyone expected to do anything. The point of the morning was to find out whether anything bad happened. The point of the trial was to find out, by climbing slowly upward in dose, where the bad things would start.

Nobody in the room expected the drug, at this dose, to do anything good.

It was a phase 1 trial. Phase 1 trials are not, in conventional cancer drug development, where benefit shows up. They are where toxicity shows up. The patients in them are typically people who have run out of other options, who consent to taking a drug at doses too low to help them in order to give the field information about whether the drug is dangerous. The benefit, if there is one, comes in later trials at higher doses. This is what everyone running and enrolling in the trial understood.

Druker, watching the patient leave the exam room, did not know what would happen next. Nobody did. The animal data suggested that the drug should be tolerable in humans. The cell data suggested that, at doses considerably higher than twenty-five milligrams, the drug should kill CML cells. Whether either of those would be true in an actual person was the question the trial existed to answer.



Phase 1 trials in cancer follow a standard design. You enroll a small number of patients — typically three — at the lowest planned dose. You observe them for some weeks; you watch their blood counts, their liver function, their kidney function, their general condition. If the dose is well tolerated, you escalate. The next three patients get a higher dose. If that's tolerated, you escalate again. You keep doing this, three patients at a time, until you start seeing dose-limiting

toxicity — side effects severe enough that you wouldn't want to push higher. The dose just below that becomes the maximum tolerated dose, and that is the dose carried forward to phase 2.

The escalation steps in the STI-571 trial were not small. The starting dose was 25 milligrams once a day. The next planned dose was 50. After that, 85; then 140; then 200; then 300; then 400; then 500; then higher if needed. The plan covered a range of more than twentyfold, on the expectation that somewhere in that range you would either find toxicity or find efficacy or both, and on the further expectation — derived from animal data — that interesting drug levels in the bloodstream would not be reached until the upper portion of the range.

The first three patients took 25 milligrams a day. Druker and his team watched them. Nothing alarming happened. After several weeks, with no toxicity to speak of, the trial moved up to 50 milligrams. Three more patients enrolled. Nothing alarming happened. The trial moved to 85.

It is somewhere in here — sometime between 85 and 140 milligrams, sometime in the late summer of 1998 — that something started to happen.



CML is unusual among cancers in that you can monitor it, week by week, with a blood test. The disease is the bone marrow producing too many white blood cells, and the white blood cells circulate in the blood, and you can count them. A normal white blood count is in the range of four to ten thousand cells per microliter. A patient with untreated CML often presents with counts in the hundreds of thousands. The count is the disease, in a fairly direct sense; if the

count comes down, the disease is getting better, and if the count keeps coming down to normal, the disease is in some kind of remission.

There were patients in the trial whose counts started to come down.

Not all of them. The patients on the lowest doses, by and large, didn't respond. Their counts stayed elevated, more or less in proportion to where they had been before they enrolled. But as the dose escalated, more of the patients in each cohort began to show falls in their counts. The drops were slow, over weeks; this was not the dramatic same-day collapse of cell counts you sometimes see with cytotoxic chemotherapy. It was steadier than that. The white cells came down. The platelets, which had been elevated in some patients, came down too. The spleen, in patients who had had enlarged spleens, started to shrink. People reported feeling better.

Charles Sawyers, running the trial at UCLA, was the first to mention to Druker, in a phone call sometime in the autumn of 1998, that his patients were doing strangely well. Druker, comparing notes, said his were too. Moshe Talpaz at MD Anderson, the third site, was seeing the same thing. Three sites, three independent groups of patients, all responding in ways that the design of the trial had not anticipated.

By late autumn, with the dose up around 300 milligrams, something had become unmistakable. Patients who had failed every other available treatment, patients who had been told they were running out of options, were having their CML go quiet. Their white counts were normalizing. Their bone marrows, on biopsy, were showing fewer leukemic cells. Some of them were starting to feel, for the first time in months or years, more or less well.

This was not what phase 1 trials were supposed to do. Phase 1 trials are supposed to demonstrate safety, not efficacy. Efficacy at phase 1 doses, in cancer drug development, is something you might catch a hint of, in a few patients, at the highest tolerable dose. It is not something you watch happen across multiple sites and multiple cohorts as you climb through the dose range.

Druker, in his lab notebooks and in his calls with Sawyers and Talpaz, kept noting the same thing: this is not what is supposed to be happening. He kept checking, the way he had checked the colony assays years earlier, that he was reading the data right. He kept being told, by Sawyers and Talpaz, the same numbers he was seeing himself.



The way doctors talk about response to leukemia therapy involves a small ladder of escalating goodness, and it is worth introducing the rungs of the ladder before going further, because the rungs are how the rest of this chapter, and most of the rest of this book, will measure what the drug does.

The lowest rung is hematologic response. This means the blood is starting to look more normal. The white count comes down toward the normal range. The differential — the breakdown of which kinds of white cell are present, in what proportions — starts to look more like a healthy person's blood than a CML patient's blood. The spleen shrinks. A complete hematologic response is when the blood looks essentially normal: white count under ten thousand, no immature cells circulating, platelets in range, spleen back to normal size. This is good but it is not the deepest kind of response, because the disease can still be present in the bone marrow without showing up

in the peripheral blood. It is the first thing that responds, and it tells you the drug is working at the level of the cells that the body is actively producing.

The next rung is cytogenetic response. This is measured by karyotyping the bone marrow — the same Q-banding and G-banding techniques Janet Rowley had used in 1973, now standard in clinical hematology labs — and counting the proportion of cells that still carry the Philadelphia chromosome. In an untreated CML patient, essentially every dividing cell in the marrow has the Philadelphia chromosome. With effective therapy, you can watch that proportion fall. A partial cytogenetic response is when the proportion of Philadelphia-positive cells drops below about a third. A complete cytogenetic response is when no Philadelphia-positive cells are detectable on karyotyping at all. This is a much harder thing to achieve than a hematologic response, because it means the drug is not just normalizing the output of the marrow but actually clearing the leukemic cells from the marrow itself — reducing the population of malignant cells, not just the population of malignant cells that happen to be circulating today.

The third rung, which would only become measurable later in the story, is molecular response. The Philadelphia chromosome can be detected by karyotyping, which catches it in roughly a percent of cells — you have to count a hundred cells to find one carrier on average. But the BCR-ABL fusion gene can also be detected by a much more sensitive technique, called PCR, which can find the fusion in a tiny fraction of cells — one in a thousand, one in ten thousand, one in a hundred thousand, depending on how carefully you do it. Molecular response is what you measure when even cytogenetic response has gone all the way to zero and you want to know whether the disease is really gone or merely below the limit of the karyotype's resolution. We will come back to this in chapter ten.

What the trial in late 1998 was seeing was complete hematologic responses. Patients on STI-571, at doses in the 300-milligram range, were having their blood counts normalize. By early 1999, with patients holding dose for several months, partial cytogenetic responses were starting to show up too. By mid-1999, complete cytogenetic responses had appeared in some patients.

Patients whose CML had progressed past every other available treatment were having their leukemia, by every measure available to clinical hematology in 1999, disappear.



The trial's findings were not yet published, and the company's view of how widely to discuss them was, in the standard way of these things, conservative. Phase 1 data are preliminary; you do not want to make claims based on a few patients that might not hold up in larger trials. There was, internally at Novartis, a debate through the autumn of 1998 and the spring of 1999 about how to talk about what was happening.

But it was already, by mid-1999, an open secret in the small world of CML research. Investigators at the trial sites were telling colleagues. Patients on the trial, hearing what kind of responses they were having, were telling other patients. Hematologists not on the trial were calling Druker and Sawyers and Talpaz to ask whether they had any open slots and how their patients might be enrolled.

By the summer of 1999 there was, in the CML community, a wave of demand for the drug that the company was struggling to keep up with. Manufacturing had been scaled up to support a phase 1 trial of perhaps fifty patients; it was not anywhere near what would be needed if even a small fraction of the CML patients in the United States wanted access. The company was also, reasonably, hesitant to expand access ahead of trial results that hadn't yet been formally

analyzed. The position was the conventional one for an early-stage compound: trust the trial process, finish phase 1, plan phase 2 carefully, expand access in the standard way.

From the outside, this looked unbearable. Patients with CML, watching their counts rise on inadequate treatment, hearing about a drug that was producing complete responses in people on a trial they couldn't enroll in, started to push back.



There was, in 1999, a CML patient living in Quebec named Suzan McNamara. She was thirty years old. She had been diagnosed two years earlier, had failed interferon, was running out of options, and had heard about the trial. She wanted in. She was, like many CML patients of her generation, an active member of online patient communities, where word of the trial's results had been spreading rapidly. She had, after several attempts to enroll directly, been told that the trial was not currently accepting patients with her particular profile and that she should wait for phase 2.

Wait for phase 2 was, in McNamara's case and in many similar cases, an answer that did not engage with what waiting actually meant. CML in chronic phase, untreated or inadequately treated, progresses. The chronic phase ends. The accelerated phase begins. After that, blast crisis. Once a patient is in blast crisis, the drug, if it ever becomes available, is much less likely to work, because the disease has acquired additional genetic changes that are harder to reverse. From the patient's perspective, every month of waiting is a month closer to the point where the drug, if it were given, might no longer help.

McNamara wrote a letter. The letter was directed to Daniel Vasella, the CEO of Novartis. It explained, in plain and unadorned terms, what waiting meant. It asked him to expand access to the

drug. She also organized, with other patients she had connected with online, a petition asking Novartis to manufacture more of the drug and make it available to patients outside the trial. The petition collected, by various accounts, several thousand signatures.

Vasella has said in subsequent interviews that he read the letter, and that the patient pressure mattered to him. Other Novartis executives have said similar things. Whether it was the deciding factor in any particular decision is hard to know from the outside, but the company did, over the course of 1999 and 2000, dramatically expand its manufacturing of STI-571 and its plans for further trials. Phase 2 trials were designed and launched in patients across the spectrum of CML disease stages: chronic phase, accelerated phase, blast crisis. Production capacity was scaled up to support a level of demand that, even at a year or two earlier, would not have been considered.

Some of this was patient pressure. Some of it was the trial data itself, which was getting harder to ignore as it kept coming in. Some of it was the people inside Novartis who had been pushing for the program all along. The combination, by late 1999 and through 2000, had moved STI-571 from a small-scale phase 1 study into something that looked very much like a serious development effort.



There was a moment in this period — worth noting because it captures something about how the trial felt at the time, not just to investigators but to the patients in it — that became, in retrospect, a small fixture of the story.

Druker had a patient named Bud Romine, an Oregon truck driver who had enrolled in the trial in 1998 with end-stage CML. Romine's CML had progressed, by the time he entered the trial, past chronic phase. He had been told, before the trial, that he probably

had months. He started on STI-571 at one of the higher dose cohorts. His blood counts came down. His bone marrow improved. He felt, after a few months on the drug, well enough to go back to driving. He did. He drove a truck across the western United States for years on STI-571. He brought, periodically, postcards back to Druker's office from places he had been able to go.

Romine was one patient. The trial had dozens. Most of them did not write postcards, but most of them, at the higher doses, were having responses that fell somewhere on the spectrum from very good to remarkable. By the end of 1999, the senior investigators were sitting on a phase 1 data set with a complete hematologic response rate, in chronic-phase CML patients at therapeutic doses, of essentially one hundred percent.

Hematologic response rate of one hundred percent. There is no comparable number for any other cancer drug at any other phase 1 trial. There are very few cancer drugs at any phase of any trial that produce response rates in the high nineties, and most of those drugs are tested in much more selected patient populations than the STI-571 phase 1 was. Druker, Sawyers, and Talpaz had a result that was, by any historical standard, off the chart.



The phase 1 results were presented at the American Society of Hematology annual meeting in December 1999, in New Orleans. The session in which Druker presented was scheduled in a room of the type ASH typically uses for sessions on uncommon diseases or specialized methodology — a room of moderate size, several hundred seats, in a part of the convention center where you would expect a moderate audience of specialists. By the time the session began, the room was overflowing. Hematologists who had heard

rumors about what STI-571 was doing had come from other sessions to see the data. The hallway outside the room was crowded. The session, by the standards of ASH, was a sensation.

Druker presented carefully. The data were presented as preliminary, with the appropriate caveats: the follow-up was short, the patient numbers were modest, the durability of responses was unknown, the safety profile would need to be confirmed in larger studies. None of this kept the audience from understanding what they were seeing. The cytogenetic response rates, in particular — the proportion of patients who were not just responding clinically but actually clearing the Philadelphia chromosome from their bone marrow — were the numbers that the audience kept fixing on. Cytogenetic response had been, in CML therapy, the holy grail of interferon: the durable cytogenetic responses produced by interferon were the ones that translated into long-term survival, and those responses occurred in only a small fraction of patients. STI-571 was producing them, in the early data, in most of the patients at therapeutic doses, in a matter of months.

By the end of the session, several hundred hematologists had taken in the same set of data. Most of them came away with the same conclusion: this was, in the technical sense of a phrase that gets overused but that occasionally turns out to be appropriate, a paradigm shift. Cancer therapy did not normally work this way. It was working this way now.

The 2000 New England Journal of Medicine paper, when it appeared in April of that year with Druker, Talpaz, Sawyers, and others as authors, made the same case in print. It reported the chronic-phase results of the phase 1 trial. The results were what the audience at ASH had heard about: complete hematologic response in essentially every patient at doses of 300 milligrams or more, complete cytogenetic response in a substantial fraction. The drug

was well tolerated. Toxicity was manageable. The maximum tolerated dose had not, in fact, been reached; the trial had escalated past 600 milligrams a day without the kind of dose-limiting toxicity that would normally cap a phase 1 study. The drug was simply working, at every dose at which it was given, with no real ceiling on how much you could give.

A companion paper in the same issue reported on the blast-crisis arm of the trial. Blast crisis is the late, transformed phase of CML, the phase in which the disease has acquired additional genetic changes and become refractory to most therapy. The hope had been, going into the trial, that STI-571 would do something useful in blast crisis even if other therapies could not. In the event, blast-crisis responses were less durable than chronic-phase responses; many patients responded initially and then relapsed within months, with the leukemic cells acquiring resistance to the drug. (Why this happens, and what to do about it, is the subject of chapter nine.) But even in blast crisis, where the prognosis had been measured in weeks, STI-571 was producing meaningful responses in a substantial fraction of patients. The drug was useful at every stage of the disease where it had been tested.

Together, the two New England Journal papers established — in the formal venue where this kind of thing gets established — that something unprecedented had happened in a phase 1 trial.



It is worth pausing here, because the story we are now in is going to keep accelerating, and there is a thing about the phase 1 results that is easy to lose track of in the rush.

What had happened, from June 1998 to April 2000, was that a hypothesis articulated in the late 1980s had been confirmed in human beings. The hypothesis was this: a small molecule that

selectively inhibits BCR-ABL will turn off CML in patients without seriously harming them. There were many ways the hypothesis could have failed. The drug could have been too toxic. It could have failed to reach its target in vivo. The cells could have responded by upregulating other pathways. The disease could have evolved out from under the inhibitor. Patients could have died of side effects unrelated to BCR-ABL that no one had anticipated. The hypothesis had to survive, in turn, every one of these possible failures.

It survived all of them. The drug was well tolerated, reached its target, shut down BCR-ABL signaling in the cancer cells, and produced clinical and cytogenetic responses on a scale that nobody, including the people running the trial, had expected. The forty-year arc from Nowell and Hungerford's small chromosome to Druker's clinic had ended in a drug that did, in patients, exactly what the science said it should do.

This kind of thing is rare in cancer medicine. The history of cancer drug development is largely a history of drugs that work less well than the preclinical data suggested they would, of compounds that failed in the move from animal to human, of mechanisms that turned out to be more complicated in patients than in dishes. The instances in which the molecular logic translated cleanly into clinical benefit, with no major surprises along the way, are few enough to be enumerable. STI-571 in CML was one of the cleanest.

It was also, although nobody quite knew this yet in April 2000, a stress test of the regulatory system. A phase 1 result this striking creates a problem for the FDA. Phase 1 trials are not, by design, the trials on which approval decisions are made; the standard regulatory path requires phase 2 and then phase 3 randomized trials, with proper comparators, before a drug is approved for general use. But STI-571 had effectively skipped past the kind of question phase 2 is meant to answer. There was no doubt that the

drug worked in CML. The question was how to get it into patients' hands as quickly as possible without breaking the framework that protects patients from drugs that don't work.

The next chapter is about what the FDA did about that question.

It is a story about a regulator confronting a drug whose evidence of efficacy was so unambiguous that the usual procedural questions about how much evidence is enough became, briefly, almost beside the point.

It is also a story about how, on May 10, 2001, in a press release that would become a small landmark in oncology, a drug called Gleevec was approved for the treatment of chronic myeloid leukemia, two and a half months after Novartis filed its application.

Two and a half months.

Patients in Druker's clinic, by then, had been on the drug for almost three years.

CHAPTER 7

7

Approval



On a Thursday afternoon in late February 2001, in offices in East Hanover, New Jersey, a team at Novartis Pharmaceuticals Corporation finished assembling the boxes.

There were dozens of them. Each box contained a slab of paper several inches thick, bound in the kind of three-ring binder that pharmaceutical regulatory affairs departments keep in industrial quantities. The boxes contained, between them, the New Drug Application for STI-571 — the formal request to the United States Food and Drug Administration for permission to sell the drug as a treatment for chronic myeloid leukemia.

The application ran to roughly seventy thousand pages. This is, by the standards of New Drug Applications, neither unusually small nor unusually large. The pages contained the chemistry of the molecule, the manufacturing process, the analytical methods used to verify the purity and potency of each batch, the toxicology results

from rats and dogs, the pharmacokinetic data showing how the drug was absorbed and distributed and metabolized in animals and humans, the full statistical analysis of three pivotal phase 2 trials, the clinical case reports for every serious adverse event observed in any patient on the drug, and the proposed labeling — the document that would become the official prescribing information if the drug was approved. There were also, in supplementary appendices, summaries of the phase 1 data, the preclinical pharmacology, and the rationale for the proposed indication.

The application was submitted on a rolling basis through February 2001, with the final submission completed by the twenty-seventh of the month. This was an unusual procedure. Most NDAs are submitted as a single complete package. Rolling submission — in which sections of the application are sent to the FDA as they are completed, rather than waiting until everything is finished — is reserved for drugs the agency has already agreed to evaluate quickly, on the grounds that the indication addresses a serious unmet medical need.

The FDA had agreed. STI-571 had received both fast track designation and priority review status, the latter shortening the agency's stated review clock from the standard ten months to six. Novartis was hoping for faster than that, and so was the FDA, and so were the people the drug was for.

On May 10, 2001, the FDA approved the drug.

Two and a half months had passed between the completion of the application and the approval. This was, at the time, the fastest review the FDA had ever conducted of a cancer drug. It would remain, for some years, near the top of the all-time list.



It is worth saying something about how drug approval normally works in the United States, because the speed of the STI-571 review is only striking if you know what speeds are usual.

Drug approval in the United States is governed by the Food, Drug, and Cosmetic Act, the framework for which dates to 1938 and the substantive elements of which date to 1962. The 1962 amendments — passed in the wake of the thalidomide tragedy in Europe, which had not, by accident more than design, reached the American market because of the skepticism of an FDA reviewer named Frances Kelsey — require that any new drug, before it can be marketed, demonstrate both safety and efficacy. Safety is the older requirement, going back to the original 1938 Act. Efficacy was the new one. The combination has, since 1962, defined what a pharmaceutical company has to prove to bring a drug to market in the United States, and the system that has grown up around the requirement is, by some measures, the most thorough drug-evaluation regime in the world.

The standard path runs in stages. After a drug candidate has been characterized in cells and animals, the company files an Investigational New Drug application — an IND — which gives the FDA the chance to weigh in before any humans are exposed. If the IND clears, phase 1 trials in humans begin: small studies, usually in patients with advanced disease, designed mostly to identify dose-limiting toxicity. Phase 2 follows: larger studies, typically in the range of dozens to a few hundred patients, designed to demonstrate that the drug has biological activity in the disease of interest. Phase 3 follows: large randomized controlled trials, typically several hundred to several thousand patients, comparing the new drug against either placebo or the existing standard of care, designed to demonstrate that the drug actually improves outcomes that matter

— survival, time to progression, freedom from symptoms, quality of life. Only after phase 3 does the company file the New Drug Application that asks the FDA to allow marketing.

The whole process, end to end, has historically taken something like ten to fifteen years from the first lab work to the first prescription. The clinical portion alone, from the start of phase 1 to the filing of the NDA, has averaged six to eight years. Once the NDA is filed, the FDA's standard review clock is ten months for a normal application, six months for a priority application. The clock is not the entire story — the agency can stop the clock to ask for additional information, and for most applications the clock does in fact stop and start — but it is the framework within which the review happens.

Inside this framework, two and a half months from filing to approval is essentially the floor. It happens only when the data are unambiguous and the disease is serious and the agency has been working closely with the sponsor through the development program.

All three of these conditions were unusually well met in the case of STI-571.



The phase 2 program for STI-571, which had launched in 1999 and continued through 2000, was designed to do, on a larger scale, what the phase 1 had done by accident: produce data showing that the drug worked at every clinically relevant stage of CML. There were three pivotal trials. One enrolled patients with chronic-phase CML who had failed interferon. One enrolled patients with accelerated-phase CML. One enrolled patients in blast crisis.

The trials were not randomized. This is, on paper, an unusual choice for a drug seeking approval; the modern standard for cancer drug evaluation is randomized controlled trials in which the new drug is compared head-to-head with the existing standard of care, and the comparison is what tells you whether the new drug is better. The phase 2 program for STI-571 used a different design — single-arm, with historical controls. Every patient on the trial got the drug. The response rates were compared not against a contemporary control group of patients on a different therapy but against historical data on what patients with the same disease, at the same stage, had been observed to do under standard care.

There is a serious objection to this kind of design. Historical controls are not ideal; the patients you compare against may differ from your trial patients in ways that are hard to control for, and the comparison may flatter or unflatter the new drug depending on which historical cohort you pick. Modern oncology, when it can, prefers randomization.

What made historical controls defensible in the case of STI-571, and what the FDA accepted as a basis for approval, was that the response rates the drug was producing were so far outside the historical range that no plausible adjustment for cohort differences could explain them away. CML in chronic phase, on standard therapy, produced complete cytogenetic responses in something like ten percent of patients, on a timescale of years. STI-571, in the phase 2 trial of patients who had failed interferon, was producing complete cytogenetic responses in roughly forty percent of patients, on a timescale of months. There was no historical cohort, no contemporary cohort, no patient subgroup that produced numbers like these on any other treatment. The drug's effect was so large that the question of whether the comparison was fair became, in a practical sense, moot.

This argument was made formally to the FDA, and the FDA accepted it. There is a regulatory path, called accelerated approval, that allows the agency to grant marketing authorization on the basis of a surrogate endpoint — in this case, cytogenetic response, which had not yet been formally validated as predictive of long-term survival but which everyone in the field believed would be — with the requirement that the sponsor complete confirmatory studies after approval. STI-571 was approved under accelerated approval. The confirmatory commitment was a phase 3 trial, not yet run, that would compare the drug head-to-head against interferon plus cytarabine, the standard of care, with overall survival as the endpoint. That trial would launch in 2000 under the name IRIS. We will come back to it.



The naming process for the molecule had been running in parallel with the regulatory process.

STI-571 was an internal Novartis code; it stood, as the previous chapter mentioned, for “signal transduction inhibitor.” It was a useful research label and an awkward consumer-facing name. Drugs that get to market need two names — a generic name, which is the official chemical name used in the medical literature and on the prescription label, and a brand name, which is the trademarked name the company uses to market the drug. The generic name is set by the United States Adopted Names Council, an independent body that follows international conventions for drug nomenclature. The brand name is chosen by the company, subject to FDA review for whether it might confuse patients or doctors with the names of other drugs.

The generic name assigned to STI-571 was imatinib mesylate. ‘Imatinib’ is the active molecule; ‘mesylate’ is the salt form, the methanesulfonate, that is actually formulated into the capsules. The ‘-tinib’ ending was new at the time and would, over the next two decades, become the standard suffix for tyrosine kinase inhibitors as a class — dasatinib, nilotinib, sorafenib, sunitinib, gefitinib, erlotinib, lapatinib, and dozens more would follow. In 2001, imatinib was the first.

The brand name was chosen, after the usual consumer-research process, as Gleevec. (In Europe and most of the rest of the world, the same drug was marketed under the slightly different brand name Glivec, an early lesson for the company about the difficulty of harmonizing brand names across markets with different linguistic preferences.) Both names are derived from the German verb ‘gleichen,’ meaning roughly to make even or to match — a play on the molecule’s mechanism of fitting precisely into the active site of its target. Whether anyone outside the marketing department appreciated the etymology is unclear, but the name had the right syllabic shape — short, easy to say, hard to confuse with anything else.

From the point of approval forward, in this book, the molecule is Gleevec when it’s a drug being sold and prescribed, and imatinib when it’s being discussed as a chemical or as a research compound. CGP 57148 and STI-571 are retired. The molecule itself is the same molecule it has been since Jürg Zimmermann’s chemists made it in Basel a decade earlier.



The press conference announcing the approval was held in Washington on May 10, 2001.

Tommy Thompson, the Secretary of Health and Human Services, announced the approval personally. This was, by the standards of cabinet-level public statements, unusual. Drug approvals are normally announced by the FDA, in a regulatory press release that is read carefully by industry analysts and reporters covering the relevant therapeutic area, and barely noticed elsewhere. Cabinet secretaries do not normally appear at podiums to talk about a single new oncology indication.

What Thompson said, in the prepared remarks of that morning, was that the FDA had approved a drug for a form of leukemia, that the drug had been developed in unusually short time, and that the approval represented — he used a phrase that the next several years of media coverage would repeat — a new era in cancer treatment. The phrase did some work. It was the kind of phrase that gets attached to product launches in industries with marketing budgets, and it was the kind of phrase that, when used about a real new drug for a real disease that really worked, deserved to be taken seriously. Both things were true at once.

Druker, Lydon, Sawyers, Talpaz, Vasella, and several others were present or spoke during the announcement and the surrounding events. The approval was, by any measure, a moment in their careers. It was also, more importantly, a moment in the lives of the patients who had been waiting.

On the day of approval, Gleevec was available by prescription in the United States. In the days after, Novartis began shipping. The phase 1 patients, who had been on the drug for almost three years on a research basis, transitioned to commercial supply. The patients who had been waiting through the petition campaigns of 1999 and 2000 began calling their doctors. Within the first week, several thousand prescriptions had been written. Within the first month, demand had exceeded what the company had projected.



Two months after the approval, on May 28, 2001, *Time* magazine published an issue with Gleevec on the cover.

The cover image was a tall yellow capsule, dominating the frame, with a headline reading “There is new ammunition in the war against CANCER. These are the bullets.” Inside, a long feature by the magazine’s senior science writer, Michael D. Lemonick, traced the history of the drug from the Philadelphia chromosome through the Ciba-Geigy program to the phase 1 trial to the *New England Journal* papers and the FDA approval. The piece was sympathetic, well-reported, and unusually accurate by the standards of mainstream cancer journalism. It also did something that had not been done before, at least in this volume: it told a general American audience, in cover-story language, that a fundamental shift in cancer treatment had occurred.

The framing was the magic-bullet framing — the dream of Paul Ehrlich, finally realized, in a single yellow capsule. The framing was, in some respects, a slight oversimplification of what Gleevec was and what it would turn out to be. The drug worked beautifully in CML, well in some related cancers, and not at all in others; it would not generalize as cleanly to all cancers as the cover suggested it might. CML was an unusually clean target. Most cancers are messier. We will come to this in chapter eleven.

But the cover was also, on its own terms, correct. Something had shifted. Cancer drug development, after Gleevec, was not the same enterprise it had been before Gleevec. The idea of identifying a specific molecular driver and building a specific molecule to inhibit it had moved, over the course of one drug’s development, from a thing that ambitious laboratories did to a thing that the pharmaceutical industry took seriously enough to invest in at scale. The *Time* cover was, in a sense, the moment that change went public.



Then there was the question of price.

Novartis launched Gleevec at a list price, in the United States, of approximately twenty-six thousand dollars per patient per year for the standard 400-milligram daily dose, with the price scaling roughly proportionally for higher doses (chronic-phase patients typically received 400 milligrams a day, accelerated-phase 600, blast-crisis 800). The figure that became attached to the drug in popular coverage was about thirty thousand dollars a year, the rough cost for a representative patient. Either number was, in 2001, considered a high price for a cancer drug. It was not unprecedented; some chemotherapy regimens of the era cost similar amounts on a course basis. But Gleevec was meant to be taken every day, indefinitely, for as long as the patient continued to respond. The lifetime cost, for a patient who responded well and stayed on the drug, would compound year after year.

The pricing decision was, internally at Novartis, the subject of considerable analysis and considerable disagreement. The arguments in favor of the price went something like this. The drug had cost more than a billion dollars to develop. (This is the kind of figure that gets quoted in pharmaceutical contexts; it is also the kind of figure that includes a great deal of allocation and amortization that economists argue about, but it was the figure the company used.) The market — perhaps thirty thousand CML patients in the United States, and similar fractions of similar populations in other developed countries — was small enough that the company would have to charge a meaningful per-patient price to recover the cost. Insurance, in most of the developed world, would cover most of the price for most patients, so the bill would not fall directly on patients in the way the headline number suggested. Comparable cancer therapies cost similar amounts. The company also pointed to its

plan, announced at launch, to provide the drug free of charge to any United States patient whose income was below a threshold and who had no insurance coverage — a patient assistance program that would, in subsequent years, distribute Gleevec to many thousands of patients at no cost.

The arguments against the price went something like this. The drug had been developed on the back of decades of basic science largely funded by the public sector — the work of Nowell, Hungerford, Rowley, Bishop, Varmus, Hunter, the BCR-ABL groups in Rotterdam and elsewhere, all of which had been done in academic laboratories with public money. The pharmaceutical industry's contribution had been important and necessary but it had been the final segment of a long relay; charging the public many times over, through both taxes that funded the research and prices that funded the marketing, was double-billing of a particular kind. The patients who needed the drug — in the United States, with American insurance — might mostly be covered, but patients in lower-income countries would not be, and the price would effectively put the drug out of reach for most of the world's CML patients. And the broader signal, that an effective drug for a small disease justified an annual price north of twenty-five thousand dollars, would set a benchmark that other cancer drugs would race to meet and exceed in the years to come.

Both arguments were correct, in their own frames. Both arguments would echo, in different forms, through the next two decades of debate about cancer drug pricing.

What is worth saying, looking back, is that Gleevec at thirty thousand dollars a year in 2001 looks, in 2026, almost charmingly cheap. By the mid-2010s, list prices for new oncology drugs had climbed past one hundred thousand dollars a year as a matter of routine. Some climbed past two hundred thousand. The number of

cancer drugs entering the market at price points that would have shocked the industry in 2001 is, by the standards of 2001, very large. The Gleevec price was, in retrospect, the lower end of a curve that the industry then climbed for two decades. Novartis itself raised the price of Gleevec several times during the period the drug remained on patent, eventually reaching list prices well above one hundred thousand dollars a year before the patent expired and generic versions entered the market in the mid-2010s, at which point prices for the molecule fell sharply.

We will come back to the price question in chapter eleven, when we look at what targeted therapy did and did not deliver across the broader landscape of cancer. For now it is worth registering: Gleevec was, in May 2001, the first drug of its kind, and the price it was launched at — controversial at the time — became the floor for a generation of similar drugs that would mostly be priced higher.



What did the approval mean for the patient sitting in Druker's clinic in 1993?

The 1993 patient is, in this book, slightly composite — standing in for any number of CML patients Druker saw in his early years at Oregon. But the question is concrete enough. A specific patient, diagnosed with CML in 1993, faced a specific clock: somewhere between three and eight years before the chronic phase ended, give or take. A patient diagnosed in 1993 who lived through 1999 — six years — was at the long end of the range, but not impossibly so. A patient diagnosed in 1993 who was still in chronic phase in 2001 was lucky, but not miraculously lucky.

Such patients existed. They had been on hydroxyurea, on interferon, on whatever combination their oncologists had been able to assemble; their blood counts had been managed, more or less, by

the inadequate tools of the era; their disease had progressed, more or less, on the timeline the disease tended to follow. Some of them had been on the phase 1 trial since 1998 or 1999 and were doing well. Some of them, on May 10, 2001, called their doctors to ask how they could now get the drug.

Many of them did get it. The shipment of Gleevec into clinics through May and June of 2001 was, in CML practice, the largest treatment switch in the history of the disease. Patients who had been managing on the older drugs transitioned to imatinib, often within a few weeks of the approval. Their blood counts came down. Their bone marrows cleared. The Philadelphia chromosome, in many of them, went from being detectable in essentially every dividing cell to being detectable in only a few cells, then in none.

Some of those patients are still alive in 2026. They are in their sixties or seventies or eighties now. They take a pill in the morning. They get blood drawn every several months. They live, by every measure their oncologists can apply, more or less normal lives.

What Druker had not been able to give them in his clinic in 1993 — a treatment that targeted the specific cause of their disease without poisoning them — he could give them in his clinic in 2001. The patients who had survived to 2001 transitioned to the drug. The patients who had been diagnosed since 1998, and especially since 1999, mostly went onto the drug as their first or second therapy and never spent significant time on anything else. The disease had not been cured — we will come to what “cure” means in this context in chapter ten — but it had been, for the patients on the drug, transformed into something that could be managed.



The story this chapter has just told is, in its broad shape, a triumph. A drug worked. A regulator moved quickly. Patients lived who would not have lived otherwise. The cover of Time magazine appeared, and was not wrong.

It is also, the rest of this book is going to argue, only part of the story.

There is, first, the question of how the targeted-therapy paradigm extended beyond CML — whether the success in this one disease was a template that could be applied across cancers, or a beautiful exception that turned out to be hard to repeat. The next chapter looks at a parallel case, in a different cancer, that was developed during the same years and approved a few years before Gleevec: Herceptin in breast cancer, where a different kind of drug aimed at a different kind of target produced its own version of targeted therapy. The two stories together — small molecule for leukemia, antibody for breast cancer — set the shape of what targeted therapy was going to become.

There is, second, the question of resistance. Gleevec worked beautifully in chronic-phase CML. It worked less well, and less durably, in patients with more advanced disease. The reason, when researchers looked carefully, was that some CML cells were evolving — acquiring specific mutations in BCR-ABL that prevented imatinib from binding — and the cells with those mutations were the ones that survived and grew. Resistance was, in 2001, a small cloud on the horizon. By 2005 it was a serious clinical problem. Chapter nine looks at how the field responded, and at what the resistance story tells us about the limits of going after a single molecular target.

There is, third, the question of cure. Gleevec controlled CML, in most patients, very well. It did not, in most patients, eradicate it. The Philadelphia chromosome, when measured at the level of molecular

response — the most sensitive techniques the field had — was almost always still detectable in some small fraction of cells, even in patients who looked completely healthy. Most patients who tried to stop the drug, after years of apparent remission, relapsed. CML had become, for most patients, a chronic disease managed with a daily pill rather than a fatal disease cured outright. The distinction between cure and control turned out to matter, in ways that took the field a decade to fully appreciate. Chapter ten is about that.

There is, fourth, the question of what targeted therapy did and did not do across the larger landscape of cancer. Gleevec worked in CML because CML had the unusual property of being driven by a single molecular event in a way most cancers are not. Many subsequent targeted therapies, aimed at other mutations in other cancers, would produce real benefit; some would be transformative. Some would not. The dream that every cancer would yield a targeted therapy — the dream the Time cover, in retrospect, slightly overpromised — turned out to be more complicated than it looked in 2001. We come to that in chapter eleven.

And there is, fifth, the patients themselves — the ones who entered Druker's clinic in 1993 with three to five years to live, and who are now, twenty-five years after Gleevec's approval, sitting in clinics getting routine blood draws and going home to their grandchildren. Chapter twelve is about them. It is the chapter the book has been moving toward from the start.

But first: a different cancer.

Different decade, different molecular logic, different kind of drug. Same fundamental insight: identify the driver, build the inhibitor, test it carefully, give it to the patients who need it.

Genentech. Breast cancer. Herceptin.

CHAPTER 8

8

A Different Cancer



In a laboratory at the University of California, Los Angeles, in the autumn of 1986, an oncologist named Dennis Slamon is looking at a Southern blot.

It is a familiar piece of laboratory technology by now in this book — the dark bands on a gray film, the result of cutting DNA into fragments and asking which fragments contain a particular gene. The technique is the same one Annelies de Klein used in Rotterdam in 1982 to find ABL on the Philadelphia chromosome. Slamon is using it for a different question, in a different cancer.

He is asking how many copies of a particular gene a tumor has.

The gene is called HER2 — short for human epidermal growth factor receptor 2. The tumors are breast tumors, surgically removed from women with breast cancer, frozen, and shipped to UCLA from a network of hospitals around the country. Slamon has been collecting them for several years. He has, in his freezers, a tissue

bank of several hundred breast cancer samples, each linked, with the patient's consent, to clinical follow-up information — how aggressive was the tumor, how big, how widely had it spread, did the patient relapse, did she survive.

He is looking, in the blot in front of him, at a row of lanes. Each lane is a different patient's tumor. Each lane shows the bands from a Southern blot probed with a small radioactive piece of HER2. In a normal cell, you would expect a single band of a particular intensity — the cell has two copies of HER2, one on each of its two copies of chromosome 17, and the band you see reflects those two copies. In some of the tumor lanes, the band is the expected intensity. In others, it is much darker. The cell, in those tumors, has more than two copies of HER2. Sometimes considerably more. In some of the lanes, the dark band is so much more intense than the normal band that the cell must contain dozens of copies of the gene.

Slamon counts the dark lanes. He is finding that they cluster, in his sample bank, with the worst-prognosis tumors.

Patients whose tumors had extra copies of HER2 — amplified HER2, in the language of the field — had relapsed sooner and died younger than patients whose tumors had normal HER2 levels. The pattern was strong enough, in his initial sample of about a hundred and ninety patients, that he could see it through the noise of the cohort. About a quarter of breast cancers had the amplification. Those breast cancers behaved differently from the others. They were faster, more lethal, and apparently a kind of breast cancer all their own.

Slamon, looking at the blot, is looking at the first piece of evidence that breast cancer is not one disease but several, and that one of those several is driven by a single specific molecular event in a way that, if you wanted to, you might be able to drug.



Breast cancer in 1986 was understood, by oncologists, as roughly one disease.

It was the most common cancer in American women. It killed about forty thousand of them every year. The treatment was surgery, supplemented by radiation, and — for the women whose tumors had been there long enough to spread to nearby lymph nodes, or who had the wrong kind of tumor on pathology — chemotherapy. The chemotherapy regimens were combinations of cytotoxic drugs adapted from the broader oncology pharmacopoeia: cyclophosphamide, methotrexate, fluorouracil; later, doxorubicin and the taxanes. They worked, in the sense that they extended life and increased the proportion of women cured by initial treatment. They did not, in most cases, work spectacularly well, and the side effects were the standard side effects of cytotoxic chemotherapy: hair loss, fatigue, nausea, low blood counts, some risk of long-term cardiac damage and second cancers.

There was, by the mid-1980s, one major exception to the cytotoxic paradigm. A drug called tamoxifen, originally developed in the 1960s as a possible contraceptive (it didn't work as a contraceptive; it had been a near-miss in a different program), turned out to block the estrogen receptor — a protein, found in the nucleus of certain cells, that responds to the female hormone estrogen by turning on a particular set of genes. About two-thirds of breast tumors expressed the estrogen receptor and depended, at least in part, on estrogen signaling for their growth. Tamoxifen, by blocking the receptor, slowed the growth of those tumors. It was given for years, sometimes decades, after surgery, as adjuvant therapy — the term oncologists use for treatment given after the primary tumor has been removed, to reduce the chance of recurrence.

Tamoxifen worked. It still works, in 2026, in essentially the same way, in essentially the same patients. It was, for the period in which it was the only such drug, the closest thing oncology had to a targeted therapy. It was relatively non-toxic compared to chemotherapy. It worked specifically on the tumors that had the right receptor. The tumors that didn't have the receptor were not affected by it, but they also weren't damaged by it, so it did no harm in those patients beyond not helping. The decision to give tamoxifen, beginning in the late 1970s, became one of the first applications of what would later be called precision medicine: you tested the tumor, and you treated based on what the test showed.

But tamoxifen was, in 1986, the only such drug. The other two-thirds of breast cancers — the ones without the estrogen receptor, called ER-negative — were treated with the cytotoxic regimens that had been more or less the same for fifteen years. There was no targeted option for them, because there was no known target.



Dennis Slamon was, in 1986, in his late thirties, an early-career physician-scientist at UCLA. He had trained as an oncologist; he had also trained, on the molecular biology side, in the school of thought that the next generation of cancer drugs would come from understanding cancer at the level of specific genes and specific proteins. He was, by training, well placed to look at breast cancer through the lens of molecular biology rather than the lens of clinical oncology.

What he had been thinking about, since his postdoctoral years, was the question we have already met in chapter three: oncogenes, proto-oncogenes, the Bishop-Varmus discovery that cancer-causing genes were normal cellular genes gone wrong. The list of cellular oncogenes had grown rapidly through the early 1980s. Most of them

were on the list because they had been found as the cancer-causing element of some retrovirus and then traced back to a normal cellular gene. ABL, from chapter three, was one of these. So was MYC, fingered in some of the lymphomas. So was RAS, which would turn out to be the most common single oncogene in human cancer.

And so, by the early 1980s, was a gene called *neu* — named, by the laboratory of Robert Weinberg at MIT, after the rat neuroglioblastoma cells in which it had first been identified. Weinberg's group had isolated *neu* by a clever piece of experimental detective work: they had taken DNA from rat tumor cells, transferred it into normal cells in a dish, and watched which fragments of tumor DNA could turn the normal cells cancerous. The fragment that did so contained the *neu* gene. The protein the gene made was a receptor on the cell surface, related to but not identical with another receptor called the epidermal growth factor receptor. The relationship was close enough that the human equivalent of *neu* was eventually given a name in the same family: HER2 — the second of three closely related receptors, with HER1 being the EGF receptor itself and HER3 being a third member of the family that was identified later.

Weinberg's *neu* had been identified through its capacity to cause cancer in laboratory cells. Slamon's question, beginning around 1984, was whether the human gene was doing anything in the actual cancers people got. He needed to find out whether HER2, in human breast tumors, was altered — mutated, amplified, overexpressed, however — in a way that might matter clinically.

He started building the tissue bank.



It is worth pausing on what amplification means, because it is the molecular event the rest of this chapter rests on.

A normal human cell has two copies of every gene that isn't on a sex chromosome. The two copies are on the two homologous chromosomes — one inherited from each parent. If a cell needs more of a particular protein, it normally turns on the existing copies more vigorously, transcribing more RNA from them, translating more protein from each copy of RNA. The number of copies of the gene itself, in the DNA, stays the same.

Sometimes, however, the number of copies changes. A region of a chromosome can be duplicated — copied, intact, so that the same stretch of DNA appears twice in a row. The duplication can itself be duplicated. The duplications can iterate, sometimes producing a region of chromosome that contains many tandem copies of what used to be a single gene. The cell, with extra copies of the gene, can make more of the corresponding RNA and more of the corresponding protein, simply through having more templates to read.

This is, in normal cells, an unusual event. In cancer cells, it is more common. The genetic instability of cancer cells — the kind of instability that produces translocations, mutations, deletions, and duplications at higher rates than in healthy cells — produces gene amplifications too. And when an oncogene happens to fall in a region that gets amplified, the amplification can drive the cancer. The cell ends up with twenty, fifty, a hundred copies of a gene whose protein product, at normal levels, mildly encourages growth and, at radically elevated levels, drives uncontrolled growth.

HER2, when amplified, was doing something like this. The receptor was a transmembrane protein — a protein with one part outside the cell, one part embedded in the cell membrane, and one part inside the cell. The outside part bound a ligand, a small signaling molecule. (HER2's specific ligand was, for a long time, mysterious; we now know it pairs with other HER family members

rather than binding a dedicated ligand of its own.) The binding caused the receptor to dimerize — to pair up with another receptor in the membrane — and the dimerization activated the inside part, which was a tyrosine kinase. The kinase, once activated, phosphorylated targets inside the cell, setting off the same general kinds of pathways BCR-ABL was setting off in CML cells: pathways that drove proliferation, suppressed cell death, and changed the cell's behavior toward its surroundings.

When HER2 was present at normal levels, this signaling happened only when the cell received the right ligand at the right time. When HER2 was amplified — with fifty copies of the receptor on the cell surface instead of two — the dimerization could happen more or less continuously, just from the sheer density of receptors bumping into each other in the membrane. The cell was getting a constant signal to grow, similar in spirit to what BCR-ABL was doing in CML, although through a different molecular route. The receptor wasn't mutated. There was just too much of it.

About a quarter of breast cancers had this. Slamon had identified, in his blots, the molecular driver of an entire subset of human breast cancer.



The paper appeared in *Science* in January 1987. It had Slamon as the first author and a team of UCLA collaborators. The title was "Human breast cancer: correlation of relapse and survival with amplification of the HER-2/neu oncogene." The figure that everyone remembered was a Kaplan-Meier curve — the standard graph oncologists use to display survival over time — in which the patients with HER2-amplified tumors fell off, sharply, from the patients without the amplification. The amplified group had a roughly threefold higher risk of dying. The amplification was, by

some statistical analyses Slamon's group ran, a stronger predictor of outcome than tumor size or lymph-node status, the two variables that had been the workhorses of breast cancer prognosis up to that point.

The paper was, in retrospect, a marker. It established that breast cancer was molecularly heterogeneous — that there was a subset of breast cancer driven by a specific genetic event, that the subset behaved differently from the rest, and that the responsible gene was one whose protein product was a known kind of receptor with a known kind of activity. Everything about it pointed at a drug.

The drug, however, would not be the same kind of drug that imatinib turned out to be.



There was, in the autumn of 1986, when Slamon's blots were going up on the lightbox in Los Angeles, a small biotechnology company in South San Francisco called Genentech.

Genentech in 1986 was about ten years old. It had been founded in 1976 by a venture capitalist named Robert Swanson and a biochemist named Herbert Boyer, the latter being one of the inventors of the recombinant DNA techniques that had made it possible, in the early 1970s, to take a piece of DNA from one organism and insert it into another — the techniques that, in chapter three, made it possible to express BCR-ABL in mouse marrow cells. Genentech had been founded around the bet that recombinant DNA could be used to make human proteins in bacteria, and that the proteins could then be used as drugs. The company's first product had been recombinant human insulin, which it licensed to Eli Lilly in 1978; the second had been recombinant human growth hormone; subsequent products had included tissue plasminogen activator, used to dissolve blood clots in stroke and heart attack.

By 1986 Genentech had begun to look at cancer. Its strategic bet, in this area, was that monoclonal antibodies could be used as targeted cancer therapies.

A monoclonal antibody is a specific kind of protein. Antibodies are the molecules the immune system uses to recognize foreign invaders — viruses, bacteria, anything the body has decided is not itself. Each antibody binds, with very high specificity, to a particular molecular feature on the surface of its target. The immune system makes a vast variety of antibodies, each with a slightly different binding specificity, so that for almost any molecular feature you can imagine, somewhere in the body there is a B cell making an antibody against it. ‘Monoclonal’ means that the antibody comes from a single clone of B cells — all identical, all making the same antibody — so that what you have is one specific antibody molecule, made in industrial quantity, with one specific binding specificity.

The technology to make monoclonal antibodies had been worked out in 1975 by César Milstein and Georges Köhler at the Medical Research Council laboratory in Cambridge, England. They had figured out how to fuse a B cell making a particular antibody with a tumor cell that grew indefinitely in culture, producing a hybrid — a hybridoma — that combined the B cell’s antibody-making machinery with the tumor cell’s immortality. You could grow hybridomas in tanks. You could harvest the antibody from the supernatant. You could, in principle, make as much of any antibody you wanted, as long as you had the hybridoma.

Throughout the late 1970s and early 1980s, monoclonal antibodies had been deployed as research tools — in fact, the antibodies Brian Druker used to detect phosphotyrosine were monoclonals — with great success. As drugs, however, they had been largely a disappointment. The trouble was that the antibodies were made in mice. A mouse antibody, given to a person, was a

foreign protein. The human immune system recognized it as foreign and made antibodies against it. After a few doses, the patient's body had cleared the mouse antibody from circulation faster than the antibody could do whatever therapeutic work it was supposed to do. The problem was called HAMA — human anti-mouse antibody — response, and it had killed several promising therapeutic-antibody programs in the early 1980s.

Genentech, by 1986, thought it had a way around the HAMA problem. The technique was called humanization. You took a mouse antibody with the binding specificity you wanted; you identified the small portions of the antibody that were doing the actual binding (the complementarity-determining regions, technically); and you grafted those portions onto the framework of a human antibody, using molecular biology techniques. The result was an antibody that bound the same target the mouse antibody had bound but looked, to the human immune system, mostly like a human antibody. The HAMA response was much weaker. The drug could be given for longer.

Genentech's bet was that humanized monoclonal antibodies, against the right targets, would be effective as cancer drugs. The bet required them to find the right targets.

Slamon's paper provided one.



The collaboration between Slamon at UCLA and Genentech in South San Francisco began around 1987 or 1988. The Genentech side included an antibody chemist named Axel Ullrich, who had been working on the HER family of receptors before Slamon's paper appeared and whose group had cloned the human HER2 gene. (Ullrich had, in fact, been the first to publish the cloning of human HER2; the gene appeared in the literature with two names — HER2

from his group, neu from Weinberg's group — which is why the chapter has so far been writing it as HER2 with a parenthetical or hyphenated reference to neu, in the way the early literature did.) Other key Genentech people on the project included an immunologist named Mike Shepard, who would lead the antibody development effort, and eventually a clinical scientist named José Baselga, who would later run much of the clinical program.

The plan was straightforward. Make a humanized monoclonal antibody that bound the HER2 receptor on the surface of breast cancer cells. Test the antibody in the dish for whether it slowed the growth of HER2-amplified breast cancer cells. If it did, test it in animals, then in patients. If it worked in patients, develop it as a drug.

Each step was harder than the previous chapter would suggest. Generating the original mouse antibody against HER2 was a months-long campaign. Identifying which of the many candidate antibodies actually killed HER2-amplified cells in culture took further months. Humanizing the lead antibody — figuring out which mouse residues to graft onto a human framework, then making sure the humanized version still bound its target tightly — was a piece of careful molecular engineering that took, in this case, several years. The eventual humanized antibody was called 4D5 in early publications and would become, in 1990, the molecule that Genentech designated as the lead clinical candidate.

It is worth saying briefly how a monoclonal antibody, given as a drug, can kill cancer cells. There are several mechanisms, and the relative contributions of each in any given case can be hard to disentangle. The antibody can simply block the receptor's function — binding to the outside portion in a way that prevents dimerization or activation. It can cause the receptor to be pulled into the cell and degraded, reducing the number of receptors on the

surface. It can, by virtue of being an antibody, recruit immune cells to the cell it has bound: natural killer cells, in particular, recognize antibody-coated cells and kill them, in a process called antibody-dependent cellular cytotoxicity. It can activate complement, a separate arm of the immune system that punches holes in the cell membrane of antibody-coated targets. The antibody Genentech eventually developed appears to do most of these things to some extent. The relative importance of each is still, decades later, a subject of research.

Imatinib, by contrast, has one mechanism. It fits into the ATP pocket of BCR-ABL, prevents the kinase from doing its job, and the cancer cells die because the cancer is dependent on the kinase being active. The mechanism is clean. The mechanism of an antibody is messier, in the sense that more than one thing is happening at the same time. Both kinds of drug can be effective. They are different kinds of drug, doing different kinds of work, with different practical implications.



The Genentech HER2 program proceeded through the late 1980s and early 1990s with a quiet steadiness that, in retrospect, looks similar in shape to the Ciba-Geigy program from chapters four and five. There were internal champions and internal skeptics. The cancer was uncommon enough as a single subtype — a quarter of all breast cancers, but only the HER2-amplified ones — that the commercial case was not obvious. The technology — humanized monoclonal antibodies as cancer drugs — had no approved precedent. The compound itself had real challenges; making large enough quantities of a humanized antibody to support clinical trials was, in 1990, a non-trivial bioengineering problem.

What sustained the program through this period, more than anything else, was Slamon. He was, throughout, the external advocate — the academic clinician who kept showing up at meetings and conferences with new data on HER2 amplification, who kept calling Genentech to push the project forward, who kept the molecule from being shelved during periods when the company's internal priorities shifted. He was, in his way, the Druker of the Herceptin story.

There was also, on the patient side, a movement that did not exist in CML. Breast cancer, by the early 1990s, had a deep and well-organized advocacy infrastructure. Organizations like the Susan G. Komen Foundation, founded in 1982 by Nancy Brinker after her sister's death, had built constituencies of patients and survivors who were active politically and visible publicly. Some of those advocates, hearing about the HER2 antibody, took up the cause of accelerating its development. Marti Nelson, a California physician with HER2-positive breast cancer who could not get into the clinical trial, became one of the visible faces of the campaign for expanded access. She died in 1994, before the drug was approved. Her name became, for many people involved in the program, a reminder that the cost of moving slowly was paid by particular people in particular timelines that were not always generous.

The phase 1 trial of the humanized antibody began in 1992. The phase 2 trials, including a study testing the antibody in combination with chemotherapy, ran through the mid-1990s. The pivotal phase 3 trial, completed in 1998, compared chemotherapy alone to chemotherapy plus the antibody in women with HER2-amplified metastatic breast cancer. The trial showed that adding the antibody extended progression-free survival and, in subsequent follow-up, overall survival. The combination was approved by the FDA in September 1998.

The drug's generic name was trastuzumab. Its brand name was Herceptin.

It was, by some accounts, the first humanized monoclonal antibody approved for any cancer indication. (Rituxan, a different humanized antibody for non-Hodgkin lymphoma, had been approved in November 1997, about ten months before Herceptin; the priority depends on how you count.) It was also, importantly for our story, the first targeted therapy for breast cancer in the modern molecular sense — a drug aimed at a specific genetic alteration in a specific subset of patients, given on the basis of a molecular test.



The molecular test was a piece of clinical infrastructure that had to be built alongside the drug.

Herceptin's approval was, from the beginning, restricted to patients whose tumors were HER2-amplified. The FDA approval was contingent on the use of a companion diagnostic — a laboratory test that classified each tumor as HER2-positive or HER2-negative, and only HER2-positive patients were supposed to receive the drug. The original test was an immunohistochemistry assay called HercepTest, which scored tumor sections on a 0 to 3+ scale based on how much HER2 protein the tumor cells displayed; tumors scoring 3+ were considered candidates for Herceptin. A more accurate but more expensive test, fluorescence in situ hybridization (FISH), counted the actual number of HER2 gene copies. The two tests gave somewhat different answers in borderline cases; the field gradually settled into a workflow in which immunohistochemistry was done first, with FISH used to clarify the 2+ borderline cases.

The point, for our purposes, is that Herceptin established the model that targeted therapy would follow for the next two decades: every targeted drug came paired with a test that identified which

patients had the alteration the drug was aimed at. You did not give the drug to all breast cancer patients; you gave it to the quarter or so whose tumors were HER2-positive, identified by the test. The patients without the amplification didn't benefit and weren't supposed to receive the drug. The patients with the amplification benefited substantially. The pairing of a specific drug with a specific test for a specific molecular alteration in a specific cancer became the template.

Imatinib, which would be approved three years later, fit the same template, although the test in CML's case was simpler — essentially every CML patient had the Philadelphia chromosome by definition, so the test for the disease was the test for the target. In other cancers that would follow — EGFR-mutant lung cancer, BRAF-mutant melanoma, ALK-rearranged lung cancer, and many more — the same pairing held. A targeted drug came with a companion test. The model was, by the late 2000s, the dominant paradigm in oncology drug development.



It is worth, before closing this chapter, noticing what is similar and what is different between the Gleevec story and the Herceptin story.

What is similar is the structural argument. Both diseases were driven, in a substantial subset of cases, by a single molecular event — the BCR-ABL fusion in CML, the HER2 amplification in HER2-positive breast cancer. Both molecular events produced a specific protein doing too much of a specific kind of signaling — in both cases, signaling driven by tyrosine kinase activity. Both could be addressed by drugs that targeted the responsible protein selectively. Both drugs, when developed, produced clinical benefit so much larger than what the previous standard of care delivered that the comparison was not really close.

What is different is the modality. Imatinib is a small molecule. It is a single chemical compound, made by chemical synthesis, that binds inside the BCR-ABL kinase and turns it off. It is taken as a pill once a day. Trastuzumab is a protein — a humanized antibody — made in cultured cells in large bioreactors, purified through a complicated downstream process, given by intravenous infusion every three weeks. It binds the outside of HER2, on the cell surface, and stops the receptor from doing what it does. The two drugs work on the same general kind of target — a tyrosine kinase — but from opposite sides of the cell membrane and through quite different mechanisms.

The two modalities have different practical implications. Small molecules can typically be given orally and reach essentially all tissues, including the brain, although they often need to be taken every day to maintain effective levels. Antibodies are large; they can't be given orally (they would be digested), they don't cross the blood-brain barrier (which is why antibody therapies often fail to control brain metastases that small-molecule therapies might reach), but they have very long half-lives in the bloodstream, which is why they can be given every few weeks rather than every day. Small molecules are cheaper to manufacture; antibodies are much more expensive. Small molecules tend to have more off-target effects; antibodies tend to be more selective. Each kind of drug has its strengths.

What targeted therapy turned out to mean, in practice, was both kinds of drug, used together where useful, deployed in different cancers based on what the cancer's molecular drivers were and what kinds of drugs would reach those drivers most effectively. Some targets were druggable only with antibodies (most cell-surface receptors). Some were druggable only with small molecules (most kinases inside the cell). Some could be approached with either or

both. The toolkit, by the early 2000s, was bigger than it had been in 1990, and the question for any new molecular target was no longer ‘is there a way to drug this’ but ‘what kind of drug, and which one will get there first.’



Herceptin was approved in September 1998. Gleevec was approved in May 2001. The phase 1 trial of Gleevec started in June 1998, three months before Herceptin’s approval. The pivotal phase 3 trial of Herceptin had completed enrollment by the time Druker dosed his first patient.

If you had been an oncology drug developer in 1998, looking at these two programs side by side, you would have seen a coherent picture taking shape. Two cancers. Two different driver alterations. Two completely different kinds of drug — small molecule and antibody — aimed at the same general class of target. Both programs had been developed on the strength of strong molecular evidence and had produced clinical results so striking that the regulatory path had compressed in front of them. Both established the principle that a drug could be developed for a defined molecular subset of a disease, paired with a diagnostic that identified the right patients, and produce benefit substantially better than the cytotoxic standard of care that had been the backbone of oncology for the previous thirty years.

By 2001, with Gleevec approved, the picture had become even clearer. The targeted-therapy era had not just arrived; it had begun to scale. Other compounds were already in development, aimed at other molecular targets in other cancers. Other companies were retooling their oncology pipelines around the idea that finding the molecular driver was the first step in finding the drug. The 1990s, in

oncology drug development, had been the decade in which a specific kind of bet — the bet on selective targeting — went from being a contrarian hypothesis to being the default frame.

What this looked like, in the years after 2001, is the subject of chapter eleven — what targeted therapy did and did not deliver across the broader landscape of cancer.

But before that, two things have to happen in the CML story itself, and they are the subjects of chapters nine and ten.

Some patients on imatinib started to relapse.

And, for the patients who did not relapse, a question began to take shape that the trials of 2001 had not been long enough to answer: what does it actually mean, in this disease, in the long run, to keep taking the drug?

CHAPTER 9

9

Resistance



Charles Sawyers had been worried about resistance from the beginning.

He had run his half of the phase 1 trial at UCLA, alongside Druker in Oregon and Talpaz in Houston, and he had seen what the rest of the team had seen — the unprecedented response rates, the cytogenetic clearances, the patients who came back for follow-up looking healthier than they had in years. He shared the team's optimism about what the drug was doing. He also kept asking, in his own work and in conversations with his collaborators, what would happen when it stopped.

He asked because he was a cancer biologist as well as a clinician, and cancer biologists have an instinct, born of watching what cancer does, that any treatment that works will eventually fail. The reason is not pessimism. The reason is evolution. A tumor in a patient's body contains, at the time of diagnosis, somewhere between a

hundred million and a hundred billion cancer cells, depending on the cancer and the stage. Each of those cells is a slightly imperfect copy of its parent. Each of them carries, in its genome, a small number of new mutations that the parent didn't have. Most of those mutations are silent or harmful or irrelevant to anything. But across a population of a hundred million cells, every single position in the genome has, statistically, been mutated in at least a few cells. The diversity is staggering. It is also, from the perspective of any drug aimed at a specific molecular target, an opportunity. If even one cell in the tumor has, by accident, a mutation that prevents the drug from binding its target, that one cell will survive treatment, while every cell without the mutation dies. The surviving cell will then divide, and its descendants will be a tumor again, this time made entirely of cells the drug can't touch.

This is, in the broadest sense, how cancers become resistant to treatment. It is also how bacteria become resistant to antibiotics, how viruses become resistant to antivirals, how insects become resistant to insecticides. The same Darwinian logic operates anywhere a population of replicating things meets a selection pressure that kills most of them. The few survivors are the ones that, for whatever reason, were already different from the rest. Their descendants inherit the difference. The population that grows back is, on average, less susceptible to the killing pressure than the population that was there before.

Sawyers, in 2000, was one of the people thinking hardest about whether this would happen with imatinib.



There was a clinical signal, even in the early phase 1 data, that pointed at where to look first.

Patients in chronic-phase CML — the early, slow form of the disease — responded to imatinib durably. They went into hematologic and cytogenetic remission, and most of them stayed there for as long as they kept taking the drug. Some of them, we will see in the next chapter, are still in remission in 2026, on the same daily dose they started in 1999 or 2001.

Patients in blast crisis — the late, transformed form of the disease — responded to imatinib too. The response rates were lower than in chronic phase, but they were not zero; many blast-crisis patients had clear and meaningful responses, with disappearance of leukemic blasts from the bone marrow and improvement in their clinical condition. The trouble was that the responses didn't last. A blast-crisis patient who responded beautifully in the first few months would, very often, relapse within a year. The leukemia that grew back would be, by every measure the trial could apply, less responsive to the drug than the leukemia that had been there to start with. Sometimes substantially less responsive. Sometimes essentially completely refractory.

This pattern — dramatic initial response in advanced disease, followed by relapse — is a classic signature of the kind of evolutionary process this chapter opened with. A treatment that works on most cells but not all of them produces a brief remission, after which the cells that didn't respond grow into a tumor that the treatment can't control. The pattern was familiar in oncology from earlier targeted approaches — from the way some leukemia cells survive cytotoxic chemotherapy, from the way HIV adapts to antiretroviral drugs given as monotherapy. It was the pattern Sawyers had been worried about.

What he wanted to know, by 2000, was what was happening at the molecular level in the relapsed cells. If you took blast cells from a patient who had relapsed on imatinib, and asked what was different about them compared to the cells that had originally responded, what would you find?



The simplest possibility, in principle, is that the cells had stopped depending on BCR-ABL. Cancer can do this. A cell driven by one signaling pathway can sometimes, given time and selection pressure, learn to be driven by a different pathway instead. The kinase that was running the show is bypassed, and the cell continues to grow on the strength of some other signal. Drugs that targeted the original kinase would, in this case, not work, because the kinase no longer mattered. The cancer had moved on.

There was a second possibility, more specific and in some ways more interesting. The cells could still be depending on BCR-ABL but could have changed BCR-ABL itself in a way that prevented the drug from binding. Imatinib's mechanism, as chapter four laid out, is to fit into the ATP pocket of the kinase and block it. The fit depends on the precise three-dimensional geometry of the pocket. A change in even one amino acid, in the right place, could disrupt the fit — not enough to disable the kinase, just enough to make it stop accepting the drug.

If this was happening, you would expect to see, in resistant cells, mutations in the BCR-ABL gene itself — specifically in the part of the gene encoding the kinase domain, and especially in the parts of the kinase domain that contacted the drug.

Sawyers, with a postdoctoral fellow named Mercedes Gorre and other members of his lab, took up the question. They obtained samples of blast cells from a series of CML patients who had been on imatinib and relapsed. They sequenced the BCR-ABL gene in each sample, looking specifically at the kinase domain.

What they found, published in *Science* in 2001, was that most of the resistant patients had mutations.

The mutations were not all the same. Different patients had different mutations, in different places in the kinase domain. But they all clustered in a region of the protein consistent with the drug's binding site — the parts of the kinase that, in the three-dimensional structure, would be in contact with imatinib. One mutation appeared in multiple unrelated patients, suggesting it was a particularly favored escape route: a substitution of an isoleucine residue at position 315 of the ABL protein for a threonine, written in the standard one-letter code as T315I. (The T is the original threonine, the 315 is the position number, the I is the isoleucine that replaced it.) Other mutations, occurring at other residues that contacted the drug, included Y253F, E255K, M351T, and others. Each of them, at the level of the protein, made some small change in the shape of the binding site. Each of them, at the level of the cell, made imatinib bind less well.

The cancer was, in short, doing exactly what evolution would predict. It was acquiring point mutations in the gene that the drug targeted, and the cells with the right mutations were the ones that survived.



It is worth pausing on T315I, because it is going to come back several times in this chapter, and because it is the place where the resistance story turns into a story about chemistry as much as biology.

The T315 residue — the threonine at position 315 of the ABL protein — is in a part of the kinase domain called the gatekeeper position. The name is descriptive. The gatekeeper is the residue that sits at the back of the ATP pocket, helping to determine which molecules can fit deep into the pocket and which can't. In ABL, as in many tyrosine kinases, the gatekeeper is a small amino acid — threonine has a relatively short side chain — which leaves room for an ATP-competitive inhibitor to extend a chemical group past the gatekeeper into a pocket beyond it. Imatinib makes specific use of this geometry. A small projection of the imatinib molecule slips past the threonine gatekeeper and contacts a region of the kinase that, in less selectively targetable kinases, would be inaccessible. The threonine, by virtue of being small, lets the drug in.

The T315I mutation replaces threonine with isoleucine. Isoleucine has a longer, bulkier side chain. Where the small threonine had left room for the drug to extend past it, the bulky isoleucine fills the space. The drug can no longer slip past the gatekeeper. It can no longer bind tightly. The kinase is now invisible to imatinib, in the sense that the drug can't grab hold of it.

This is, at the level of one amino acid, why a single point mutation can make a cell completely resistant to a drug that, in cells without the mutation, works almost perfectly. It is also, at the level of strategy, why T315I is harder to address than other mutations. The other resistance mutations — the ones at Y253, E255, M351, and so on — mostly disrupt imatinib binding through subtler means: shifts in the conformation the kinase prefers, slight changes in the contacts the drug makes elsewhere in the pocket. A second-generation inhibitor that bound the kinase a little differently, or a little more tightly, could often work around them. T315I was different. The gatekeeper substitution closed off a part of the binding site that imatinib actively used. Designing a drug that

worked despite a closed gatekeeper required something fundamentally different from imatinib, not just a better-tuned version.

We will come back to T315I.



What followed Sawyers's 2001 paper was a flurry of work, in academic labs and pharmaceutical companies, to make second-generation BCR-ABL inhibitors that could overcome resistance.

Two compounds emerged most prominently. The first was a Bristol-Myers Squibb compound originally numbered BMS-354825 and eventually named dasatinib. It was a molecule with a different chemical scaffold from imatinib — it inhibited BCR-ABL through a slightly different mode of binding, and it was, by orders of magnitude, more potent than imatinib against the wild-type kinase. It also inhibited most of the resistance mutations that had emerged in imatinib-treated patients. It did not inhibit T315I.

The second was a Novartis compound, AMN107, eventually named nilotinib. Novartis, having developed imatinib, had a strong interest in extending the franchise; nilotinib was, in a sense, an explicit second-generation imatinib. The chemists at Novartis had taken the imatinib structure and modified it carefully, addressing some of the chemical issues that had limited imatinib's potency and improving its activity against most of the common resistance mutations. Nilotinib, like dasatinib, did not inhibit T315I.

Both drugs were approved, in 2006 and 2007 respectively, for CML patients who had become resistant to or intolerant of imatinib. They worked. Most patients with imatinib-resistant CML who were not carrying the T315I mutation could be returned to remission with one of the second-generation drugs. The arsenal of effective CML therapies, which in 2000 had consisted of one drug, expanded to

three by 2007. Patients who relapsed on imatinib could be switched to dasatinib or nilotinib; if they relapsed again, they could often be switched to the other; many of them could be kept in cytogenetic remission through a sequence of these drugs for years.

There was a parallel question, opened up by the second-generation drugs, of whether they should be used not just after imatinib but instead of it, as initial therapy. The argument for using them first was that they produced faster and deeper molecular responses than imatinib did, with the implication that fewer patients would progress to advanced disease and fewer resistance mutations would have time to emerge. The argument for using imatinib first was that imatinib had longer follow-up data, well-characterized side effects, and — by the time the question became live in the late 2000s — the prospect of going generic, which would dramatically lower the cost of first-line therapy. Both nilotinib and dasatinib were approved for first-line use in the early 2010s, but the field's practice settled into a roughly even split: some physicians began with imatinib, some began with one of the second-generation drugs, and the choice depended on patient factors and physician preference more than on a clear superiority of either approach for most patients. We will come back to the cost question in chapter eleven.

But there was still T315I, the gatekeeper mutation, against which neither dasatinib nor nilotinib was effective.



T315I is, by some estimates, present in fifteen to twenty percent of imatinib-resistant patients — not the majority, but not a small minority either. For these patients, dasatinib and nilotinib were as ineffective as imatinib. The cancer had taken the route through the gatekeeper, and the only way around the gatekeeper was a fundamentally different molecule.

Several pharmaceutical and academic teams worked on the problem through the 2000s. The chemistry was difficult. The constraints on a molecule that could bind a closed-gatekeeper kinase were severe; the chemists had to find a way for the drug to engage the kinase without using the part of the binding site the gatekeeper normally left open.

A small biotechnology company called ARIAD Pharmaceuticals, based in Cambridge, Massachusetts, eventually produced one. Their compound, originally numbered AP24534 and later named ponatinib, was designed specifically to inhibit T315I-mutant BCR-ABL. It used a chemical strategy quite different from imatinib's: rather than slipping past the gatekeeper, it bound in a configuration that didn't depend on getting past it at all. It was, in the test tube, potent against essentially every BCR-ABL mutation that had been identified in imatinib-resistant patients, including T315I.

Ponatinib was approved in 2012, for patients with CML and Philadelphia-positive acute lymphoblastic leukemia who carried the T315I mutation or whose disease was resistant to other available kinase inhibitors. It was, for those patients, often the difference between effective therapy and no effective therapy.

It came with a problem the earlier drugs had not had. Ponatinib turned out to cause an unexpectedly high rate of arterial thromboses — blood clots in arteries, leading to heart attacks, strokes, and limb-threatening blockages — in patients taking it. The mechanism was poorly understood; it was not obviously related to BCR-ABL inhibition itself, but rather to off-target effects of the molecule on other targets in the cardiovascular system. The frequency of these events was high enough that the FDA, in 2013, briefly suspended the drug's marketing while the data were reviewed. Ponatinib was eventually returned to the market, but with a black-box warning, with revised dose recommendations (lower doses than originally

approved, with the dose lowered further once a patient achieved a satisfactory response), and with a clear understanding among prescribing oncologists that the drug was for patients whose situation was bad enough to justify accepting the cardiovascular risk.

This is, in itself, a piece of the resistance story worth attending to. The drug that closed the gatekeeper escape route for imatinib was a more dangerous drug than its predecessors. The molecular ingenuity required to build it had come at a cost in tolerability. The arc of CML treatment, which had begun with the cleanest of clean targeted therapies, had bent toward something messier as the cancer's evolutionary repertoire forced the field to use more aggressive molecules to keep up.



There is, in 2026, a fourth drug specifically designed for T315I, called asciminib, that takes a different approach again.

Asciminib was developed by Novartis and approved in 2021. It is what chemists call an allosteric inhibitor: rather than binding the ATP pocket the way imatinib, dasatinib, nilotinib, and ponatinib all do, it binds a different site on the kinase — a site called the myristoyl pocket, which is the pocket that, in the normal regulation of ABL, accepts a small fatty-acid tag and uses the binding to keep the kinase folded in its inactive conformation. BCR-ABL, lacking the regulatory machinery that uses this pocket, has the pocket sitting empty. Asciminib slots into the empty pocket and locks the kinase in the inactive conformation — turning the same regulatory mechanism that keeps normal ABL quiet against BCR-ABL, which had escaped the regulation by being a fusion. Because asciminib doesn't bind in the ATP pocket, it isn't affected by the gatekeeper mutation that defeats the ATP-competitive drugs. Because it engages a regulatory

site that doesn't exist in most other kinases, it is unusually selective — even more so than imatinib was. Its safety profile, in the trials that led to its approval, has been better than ponatinib's.

Asciminib is the kind of drug that becomes possible only after years of structural work on a target. The myristoyl pocket was first appreciated as a regulatory site in ABL in the 1990s. The realization that it could be drugged, and the chemistry to drug it, took the better part of two decades after that. Asciminib's approval is, in 2026, one of the more striking examples in oncology of medicinal chemistry catching up with structural biology — of the ability to design a drug that exploits a feature of a protein nobody knew how to use, twenty years earlier, when the first inhibitor of the same protein was approved.

The CML toolkit, in 2026, has five oral kinase inhibitors approved for the disease: imatinib, nilotinib, dasatinib, bosutinib (a third second-generation drug, slightly different in its profile, that we have not paused on), ponatinib, and asciminib. (Five drugs that are oral and approved as monotherapy; bosutinib makes six if you count it separately, which the field generally does.) For most patients with most patterns of resistance, at least one of these drugs will produce a durable response. The question, in CML practice, has shifted from 'is there a drug for this patient' to 'which drug, in which sequence, at what dose, weighed against what side-effect profile.'

This is a different question from the one Druker faced in his clinic in 1993. It is a better question.



What did the resistance story actually mean, looking back?

The first thing it meant is that the early concerns the skeptics inside Novartis had voiced — the worry that resistance might rapidly outpace any single inhibitor — had not been crazy. They had

been right to be worried. CML cells did evolve resistance to imatinib. The evolution happened reproducibly, through specific mechanisms, in a substantial fraction of patients with advanced disease. It was the predictable thing to happen. It happened on the predictable timeline. It produced, in many patients, the predictable end — a relapse on a drug that no longer worked.

The second thing it meant, however, is that the resistance story was, in CML, manageable in a way it has not always been in other cancers. The mutations that arose in BCR-ABL were not arbitrary; they clustered in particular residues, they could be enumerated, and the field could design drugs that worked around them. The cancer's evolutionary repertoire was finite. Each successive generation of inhibitor blocked some additional set of resistance mutations, and patients could be cycled through the drugs as their disease evolved, and most patients could be kept in remission through the cycle. Resistance, in CML, became a chronic management problem rather than a terminal one. The disease did keep evolving, and the field did keep responding, and the pace of the back-and-forth turned out to be one the field could mostly maintain.

The third thing, harder to summarize, is that the resistance story shaped how the field thought about all subsequent targeted therapies. After CML, every targeted-therapy program assumed resistance would emerge — it was no longer a surprise, it was a planning parameter. Companies developing first-in-class inhibitors began thinking, from the outset, about second-generation versions. Trials were designed to monitor for emerging resistance mutations. The model of treatment shifted from 'give the drug, hope it works forever' to 'give the drug, monitor for resistance, switch when needed.' This was a more realistic model of what targeted therapy

could do. It was also, in some other cancers, a more difficult one to execute, because the resistance landscape in those cancers turned out to be wider and messier than CML's.

We will come back to that in chapter eleven.



There is one more thing about the resistance story worth attending to before this chapter ends, because it is the thing that connects what happens in the cells of any one patient to the larger story of the disease.

When a CML patient is in chronic phase, on imatinib, doing well, with the Philadelphia chromosome essentially undetectable in the blood and bone marrow — a patient in deep cytogenetic and molecular response — it is easy to think of the cancer as gone. The disease is, by every measure available, suppressed. The patient feels well. The blood looks normal. The bone marrow, biopsied, shows a more or less normal-looking population of cells, with no obvious leukemic infiltrate.

But the disease is not gone. It is, in most patients, still there, just below the level of detection, in some small reservoir of cells that the drug has not killed. The drug suppresses the cells; it does not eradicate them. As long as the patient keeps taking the drug, the cells stay suppressed. If the drug is stopped, in most patients, the cells re-emerge and the disease comes back. Resistance mutations, when they appear, often appear in this small reservoir — mutations that were already there at low frequency before treatment began, or that emerged during treatment in cells that the drug was selecting against but not quite eliminating. The reservoir is the source from which both relapse and resistance can grow.

What this means, practically, is that imatinib in CML — even imatinib at its best, in patients in deep remission — is a drug that holds the disease in check without finishing it off. The disease is controlled but, in most patients, not cured. The patient has to keep taking the drug. If the patient stops, the disease comes back. If the disease comes back, it is sometimes still controllable with the same drug, and sometimes it is not, because the cells that emerge in relapse have, sometimes, acquired the resistance mutations that the drug couldn't block.

This is a different concept of what a successful cancer therapy looks like than what most of us, growing up, learned to expect. It is not the cure that the surgical-oncology model trains us to think about — take it out, and if the borders are clear, it's gone. It is something more like the management of a chronic infectious disease, or the management of a chronic metabolic disease: a daily intervention that keeps a process from happening, with the understanding that the process is still trying to happen and would resume immediately if the intervention stopped.

What this means, for the long-term outlook of a CML patient, is the subject of the next chapter.

The chapter is about what cure means in a disease that is well-managed but not eradicated, about what the long-term data show for patients who have been on imatinib for fifteen, twenty, or twenty-five years, and about whether there is anyone in CML — with the disease really, truly gone — or only patients in whom the suppression has been so deep, for so long, that the distinction has become unimportant.

The honest answer, as we will see, is somewhere in between. And the resistance story this chapter has told is part of why the answer is honest rather than triumphant.

10

What Cure Means



Bud Romine, the Oregon truck driver from chapter six, took imatinib for nearly twenty years.

He had enrolled in the phase 1 trial in 1998, in late-stage CML, with what he had been told was perhaps months to live. He started on a higher dose cohort. His blood counts came down. His bone marrow improved. He went back to driving. He sent postcards. Through the early and middle 2000s he was, by every measure his oncologist could apply, doing well — the disease was in deep cytogenetic and then molecular response, his blood counts were stable, his Philadelphia chromosome was undetectable on karyotyping. He continued on imatinib because, by every account he received from his oncologist and from the protocols of the trial he had stayed on, that was what you did with imatinib in CML. You stayed on it. You kept taking it. Your disease, in the language oncologists had begun to use, was being managed.

Romine eventually developed an unrelated cancer in his late seventies. He died of complications related to that cancer in 2017, having outlived by roughly nineteen years the prognosis he had been given when he enrolled in Druker's trial. The CML had not killed him. The CML, on every test that had been done in the years before his death, had remained suppressed.

Was Romine cured?

It is a question that depends on what 'cured' means, and the answer that took the field most of the 2000s and 2010s to arrive at, and that is still being refined in 2026, is: not exactly, but maybe close enough that the distinction is, for him and most patients like him, not the question that matters.



What 'cure' has historically meant in oncology is straightforward in concept and complicated in practice.

The conceptual definition is: the cancer is gone, and it stays gone, and the patient eventually dies of something else. The biological criterion is the absence of any remaining cancer cell anywhere in the body — zero, not few. A cancer that has been removed by surgery and that does not recur after many years of follow-up, in a patient who shows no evidence of disease anywhere on imaging or in blood work, is generally called cured. So is a leukemia in long-term remission after a successful bone marrow transplant. So is, by tradition, a tumor whose response to chemotherapy is so complete and so durable that the cancer never reappears for the rest of the patient's life.

The practical difficulty is that you can never quite prove the absence of cancer. Cancer cells are small; the body is big; the most sensitive tests can detect cancer down to some threshold of frequency, below which the cells become invisible to the test. A

patient declared cured of breast cancer, with clean scans and clean blood, may still have a small population of dormant cancer cells somewhere in her bone marrow that no test has found. Most of the time, in such a patient, those cells will remain dormant for the rest of her life. Sometimes, decades after the original treatment, they will reactivate, and the cancer will return — a recurrence after what had looked, by every available measure, like a complete cure. The biological question ‘are there any cancer cells left’ is harder to answer in any specific patient than the medical question ‘has the cancer recurred,’ which is what oncologists actually track.

The way oncology has historically managed this difficulty is through a combination of follow-up and statistics. Patients are followed; recurrences are counted; survival curves are plotted. A treatment is said to cure when, after enough years, the survival curve flattens out — when the rate of new recurrences drops to roughly the background mortality rate of the population, suggesting that the patients still alive are no more likely to die of the cancer than they are to die of the other things people eventually die of. Cure, on this definition, is something you can establish at the level of a population without ever being entirely sure about any one patient. It is also something that takes a long time to establish, because you need many patients followed for many years to be confident the curve has actually flattened.

CML in the imatinib era poses a different problem. Imatinib doesn’t, in most patients, eliminate the disease the way successful chemotherapy or transplant does in some other leukemias. It suppresses the disease, sometimes very deeply, sometimes to levels too low for any clinical test to detect, but the suppression is contingent on the drug. Stop the drug, in most patients, and the

disease comes back. The traditional definition of cure — cancer gone, stays gone, no further treatment needed — doesn't fit. What does fit is something the field had to develop new vocabulary for.



The new vocabulary built up over the 2000s as the long-term data on imatinib began to accumulate. The most consequential single piece of that data came from a trial called IRIS.

IRIS — the International Randomized Study of Interferon and STI571 — was the phase 3 trial Novartis had committed to running as part of its accelerated-approval agreement. It enrolled, beginning in mid-2000, more than a thousand newly diagnosed chronic-phase CML patients across many sites in many countries, randomized them between imatinib and the previous standard of care (interferon plus low-dose cytarabine), and followed them for outcomes that mattered: cytogenetic response, molecular response, progression to advanced disease, overall survival.

The first IRIS readout, published in the *New England Journal of Medicine* in March 2003, was decisive. At eighteen months of follow-up, the rate of major cytogenetic response — a substantial reduction in Philadelphia-positive cells — was 87 percent in the imatinib arm versus 35 percent in the interferon arm. The rate of complete cytogenetic response was 76 percent versus 14 percent. The rate of progression to accelerated phase or blast crisis was about 3 percent versus 9 percent. By every endpoint the trial was measuring, imatinib was so much better than the previous standard that further accrual to the comparison was, in a strict sense, untenable. The trial's data monitoring committee allowed patients on the interferon arm whose disease had not responded adequately to cross over to imatinib, and most of them did.

IRIS continued to follow patients for years afterward, with updates published periodically. By the five-year readout in 2006, the picture had stabilized: imatinib produced complete cytogenetic responses in roughly four out of five chronic-phase patients, the responses were durable in most patients who achieved them, and progression to advanced disease was rare. The rate of progression in patients who had achieved complete cytogenetic response by twelve months was, by year five, very low — single-digit percentages of all such patients had progressed.

The ten-year IRIS update, published in 2017, extended the picture in the direction the early data had suggested. Among the original imatinib-arm patients, overall survival at ten years was about 84 percent. Survival free of CML, specifically — patients still alive with their disease in chronic phase or in remission — was higher than that. Most of the deaths over the ten years had been from causes unrelated to CML: heart disease, other cancers, the things people in their sixties and seventies die of. CML itself, in the imatinib-treated cohort, had killed only a small fraction of the patients enrolled, the great majority of whom had progressed during periods when the drug couldn't be optimized fully — patients who had developed early resistance, intolerance, or who had not been able to maintain adequate dosing.

These numbers represented a transformation. CML in the pre-imatinib era had killed roughly two-thirds of patients within ten years of diagnosis. CML in the imatinib era was killing roughly one-sixth, and most of that one-sixth was made up of patients whose disease had presented in advanced phase, or who had developed early resistance, or who had been unable to take the drug for one reason or another. For the patients on imatinib doing well, ten-year survival was approaching the survival of an age-matched general population. By the fifteen and twenty-year readouts that have come

since, that approach has continued. The patients on the drug, doing well, are dying at roughly the rate that people of the same age without CML are dying. They are not dying of CML.

This is what happens, in clinical statistics, when a fatal disease becomes a chronic one. The mortality of the disease falls off; the mortality of everything else stays where it was; the survival curve of patients with the disease starts to track the survival curve of the general population. By the long-term IRIS data, the imatinib-treated CML population had effectively merged, in survival terms, with the population that had never had CML.



What this did not establish, however, was that any individual patient was cured.

The IRIS data showed that patients on imatinib were not dying of their CML. It did not show that the CML was gone. The deepest molecular tests — PCR-based assays that could detect the BCR-ABL fusion gene at levels of one in tens of thousands of cells — mostly continued to find some small signal, in most patients, even after years of apparent remission. The disease was suppressed deeply enough to be clinically silent, but it was, at the level of molecular detection, generally still there.

It is worth taking a moment on what the molecular tests actually measure, because the rest of this chapter rests on them.

When chapter six introduced the response ladder — hematologic, cytogenetic, molecular — it deferred a careful explanation of the third rung. We can take it now. Molecular response in CML is measured by quantitative PCR, a technique that uses small DNA primers to specifically amplify and count the BCR-ABL fusion transcript in a blood or marrow sample. The transcript is the messenger RNA produced by the fusion gene, the working copy

that the cell uses to make BCR-ABL protein. PCR can detect this transcript with extreme sensitivity — down to roughly one transcript per ten thousand or hundred thousand normal transcripts of a control gene, depending on the assay's design and the sample's quality. Each result is reported on a logarithmic scale: a major molecular response (MMR) is a thousandfold reduction from a standardized baseline, conventionally written as MR3.0; a deep molecular response is ten or a hundred thousandfold (MR4.0 or MR4.5); a complete molecular response is undetectable BCR-ABL on a sufficiently sensitive assay (MR5.0 or beyond).

On these scales, most patients on imatinib in deep remission sit somewhere between MR3 and MR4 — still detectable, but at very low levels. A subset of patients, particularly those on more potent second-generation drugs or on imatinib for many years, achieve MR4.5 or deeper. Some achieve undetectable disease on the most sensitive assays available. The depth of molecular response has turned out to predict, more accurately than any other clinical variable, what happens if the patient stops the drug.



The question of stopping the drug is the question that, more than any other, has defined what 'cure' has come to mean in CML.

By the mid-2000s, with patients accumulating years of imatinib at deep molecular response, the natural question began to be asked. If the disease is undetectable at the most sensitive level of testing, do you still need to be taking the drug? The drug is not without cost — it is taken every day, indefinitely, with side effects that, while generally mild, are not zero. There is muscle and joint pain; there is fluid retention; there is fatigue; there are skin and gastrointestinal effects. There is the psychological weight of being a chronic-disease patient who takes a medication every morning forever. There is the

financial cost, which we will come to in chapter eleven. If a patient could safely stop, even sometimes, that would be a substantial change in what living with CML actually means.

The first formal trials of stopping imatinib, in patients in deep molecular remission, ran in France in the late 2000s. The largest of them was called STIM — Stop Imatinib — led by a French hematologist named François-Xavier Mahon. STIM enrolled patients who had been on imatinib for at least three years and who had achieved complete molecular response (undetectable BCR-ABL on the available PCR assays) for at least two consecutive years. They stopped the drug. They were monitored monthly, by sensitive PCR, for any reappearance of the BCR-ABL transcript.

What STIM showed, when it reported in 2010, was that about 40 percent of patients who stopped imatinib in deep molecular response remained in remission at a year, with no reappearance of detectable disease. The other 60 percent relapsed — their molecular signal returned, usually within the first six months — and were restarted on the drug. Crucially, essentially all of the relapsed patients responded to imatinib again when they restarted; the relapse, when it happened, did not represent the emergence of resistance. Their disease had simply re-emerged from the reservoir, the way chapter nine described, and the same drug was still able to suppress it.

STIM established something new in the CML field. It established that, in a substantial minority of patients in deep molecular remission, stopping the drug was a feasible thing to do. The patients who stayed in remission off the drug were not, on any reasonable interpretation of the data, suppressing the disease through residual drug effects — imatinib washes out of the body in days. They were keeping the disease at zero, or near enough to zero, through some combination of having so few residual cells to begin with and the

body's own mechanisms keeping those few cells from re-expanding. They were, for as long as the remission held, functionally cured. Whether this could be called cure in the traditional sense was a definitional question. They were certainly off the drug, off therapy, with no detectable disease, going about their lives.

STIM also established that this was a minority phenomenon. Most patients who stopped the drug relapsed. The disease, in those patients, was still there, suppressed below detection but not to zero, and as soon as the drug was removed, it resumed. The reservoir, in chapter nine's language, was still active.

Subsequent trials — EURO-SKI in Europe, A-STIM and others elsewhere — expanded the picture. They confirmed the basic finding, refined the criteria for who could safely attempt a treatment-free remission attempt, and produced clinical guidelines that have, by 2026, become part of standard CML practice. The current rough consensus is that a patient who has been on a TKI (tyrosine kinase inhibitor — imatinib, dasatinib, nilotinib, or one of the others) for at least five years, who has been in deep molecular response (MR4.5 or deeper) for at least two years, who is willing to be monitored frequently, and whose physician judges them to be a candidate, can attempt a treatment-free remission. About 40 to 50 percent of such patients will succeed in remaining off therapy long-term, with the precise number depending on the criteria used and the duration of follow-up. The rest will relapse and need to restart.

This is, by any reasonable historical comparison, an extraordinary state of affairs. CML in 1993 was uniformly fatal. CML in 2026, for patients who reach deep molecular response and choose to attempt treatment-free remission, is a disease from which roughly half of those patients can walk away. The other half remain on a daily pill, with normal life expectancy. Some patients in the second group, with continued treatment and continued deep response, will

become candidates for treatment-free remission attempts later. Some won't. Either way, the disease has become something most patients live with, rather than something they die of.



There is, however, a complication, and it is one of the things that makes 'cure' a more interesting word in CML than it might first appear.

Even patients who achieve treatment-free remission — patients who have been off all therapy for years, with no detectable disease — sometimes have, on the most sensitive testing available, a small number of BCR-ABL-positive cells still present somewhere in their bone marrow. Specialized research assays that look at single cells have, in several published studies, detected rare leukemic stem cells in patients who by all clinical measures are in long-term treatment-free remission. The cells are there. They are not, in these patients, expanding. They appear to be held in check by mechanisms that the field is still working out — some combination of the cells' own dormancy, the bone marrow microenvironment they sit in, the ongoing surveillance of the immune system, and, possibly, factors specific to each patient that no one has fully characterized.

What this means is that even apparent functional cure, in CML, may not be the same as biological cure in the strictest sense. The cancer cell that started everything — the original Philadelphia-chromosome-positive stem cell from which all the other leukemic cells descended — may still be there, dormant, in some fraction of patients who appear cured. The patient is, in clinical terms, indistinguishable from someone who has never had CML. They feel well; their blood is normal; they don't take a drug; they live a

normal life. But the disease, at the deepest level of molecular biology, may be in something like a permanent stalemate rather than a complete eradication.

Whether this distinction matters depends on how you weight different ways of being healthy. If a patient lives a normal life span, with normal blood counts, off all therapy, on the basis of a stalemate in which their immune system is suppressing a small population of leukemic cells they don't know about, are they cured? In the practical sense that matters to them, almost certainly yes. In the strict biological sense that someone like Sawyers might use, perhaps not quite. The semantic gap between these two answers is, in the era of imatinib and its successors, the gap that contains a great many patients.



One thing the long-term data have made clear is that imatinib does not, in general, fail in the way the early skeptics had feared.

Recall the worry from chapter five: a class of drugs that took years to develop, the skeptics had said, and that might be quickly outpaced by resistance, was not the kind of bet a publicly-traded pharmaceutical company should make. The fear was that imatinib would work briefly, the cancer would evolve around it, and the drug would be obsolete within a few years of approval.

What actually happened was different. Resistance did emerge in some patients, especially those who started in advanced disease, and chapter nine described how the field responded with second-generation inhibitors, then ponatinib, then asciminib. But for the bulk of patients — the patients who started in chronic phase, the patients who were the great majority of CML diagnoses going forward — imatinib has continued to work, in many cases for the lifetime of the patient on the drug, with no need to switch to

anything else. The IRIS cohort patients who responded to imatinib in 2001 are, twenty-five years later, mostly the same patients on the same drug. Their disease has not evolved out from under them. The reservoir from which resistance might have emerged has, in their case, remained suppressed.

Why imatinib has worked so durably in chronic-phase CML, when the early models of how cancer evolves under selection pressure suggested it might not, is a question with several partial answers. The number of cells generating new mutations is smaller in chronic phase than in advanced disease. The selective advantage that any single resistance mutation confers is partially offset by the fitness cost of the mutation itself — a mutated kinase is often a less efficient kinase. The deep responses imatinib produces leave only a small reservoir to mutate. The tolerability profile allows patients to remain on consistent dosing for years without the dose interruptions that might create the population dynamics resistance needs to emerge. Some combination of these factors, plus probably some elements not yet fully understood, has made imatinib in chronic-phase CML one of the most durable cancer drugs anyone has ever developed.

This was not what was expected in 2001. It was, in some ways, better than what was expected. The skeptics inside Novartis had been right to worry. They had been wrong about how it would actually go.



What this all looks like, from the perspective of a CML patient diagnosed in 2026, is a sequence of decisions in which traditional cancer biology has been replaced by something closer to chronic disease management.

The patient is diagnosed, typically incidentally, on routine blood work that finds an elevated white count. A bone marrow biopsy and karyotype confirm the Philadelphia chromosome. The patient is started on a TKI — most commonly imatinib, often one of the second-generation drugs depending on physician preference and patient factors. They take it every day. They get blood drawn every few months for the first year, then less often, with PCR monitoring of BCR-ABL transcript levels. Their disease, in the standard course, achieves complete cytogenetic response within twelve months and major or deep molecular response within two to three years. They stay on the drug.

Side effects, for most patients, are tolerable: muscle cramps, periorbital edema, occasional rashes, low-grade fatigue. Some patients tolerate one drug better than another and switch within the class to find one that suits them. Most patients on imatinib who have achieved deep molecular response after several years remain on the drug indefinitely; some, meeting the criteria, attempt a treatment-free remission and become candidates for life off therapy. They live, in either case, with the underlying expectation that their CML, as it now stands, is not going to be the thing that kills them.

Compare this to the patient sitting across from Druker in 1993, the one the prologue described. The 1993 patient had a few years to live. The 2026 patient has a normal life expectancy. The 1993 patient's options were inadequate. The 2026 patient's options are, by the standards of any other oncology indication, embarrassingly good. The 1993 patient lived in a world where CML was lethal and where the science was right but the drug did not yet exist. The 2026 patient lives in a world where the same molecular fact — the Philadelphia chromosome, the BCR-ABL fusion, the constitutively

active kinase — has been turned, by chemistry and clinical trials and several decades of cumulative work, into a piece of clinical management.

Whether the 2026 patient is ‘cured’ is a question whose answer depends on definitions. By the practical definition that matters to her — will the disease shorten her life, will she suffer from it, will it limit what she can do — the answer is mostly no. By the strict biological definition — are there zero leukemic cells anywhere in her body — the answer is, in most cases, also no. The disease is, in the language the field has settled into using, controlled. It is not the same as cured. It is also not, in any meaningful sense, the disease that killed people in 1993.



There is a way of thinking about all of this that the field has, slowly, become comfortable with, and that is worth articulating before the chapter closes.

Cancer, on the surgical-oncology model that was still the field’s dominant frame in the late twentieth century, is something you cure or fail to cure. You take it out, or you eradicate it with treatment, or you don’t, and the person dies. There is a binary built into the way the older oncology talked about its work. Many of the dramatic stories the field tells about itself — about chemotherapy, about transplant, about radiation — are stories in which a patient is either rescued completely or is not rescued at all. The intermediate state, in which the cancer is held in check by an ongoing intervention but not eliminated, did not have an obvious place in the older frame. It was treated, at best, as a temporary holding pattern between cure and failure.

What the imatinib era, with its long-term data and its molecular monitoring and its treatment-free remission attempts, has done is to make that intermediate state legitimate. A disease that is held in check indefinitely, with a daily pill, with normal life expectancy and normal quality of life, is not a failure of treatment. It is a form of treatment that wasn't really possible before, in cancer at any rate, and that has turned out to be enough for an enormous number of patients. It is what most CML patients now experience. It is what most patients with several other targeted-therapy-amenable cancers also now experience. It is, in some respects, the analog of what diabetes treatment looked like fifty years after insulin and sixty years after oral antidiabetics: a chronic disease, well-controlled with daily medication, in patients whose disease is not gone but whose lives are not being abbreviated by it.

This is a different kind of victory than the one the *Time* cover in 2001 implied. It is a more complicated victory. It is also, for the patients living it, a real one.



Bud Romine took imatinib for nineteen years before dying of an unrelated cancer in 2017. He never stopped taking it. He could have, by the time he died, met some criteria for attempting treatment-free remission, but he did not try — by his own account, he was content to take the morning pill and not change a thing. His family has said, in interviews after his death, that he understood the drug had given him the years he had with his children and grandchildren and that he saw no reason to risk the loss of any of those years for the sake of finding out whether he could be off the drug.

Some patients in his cohort did try, and stayed off. Some tried and relapsed and went back on. Some chose, like him, not to try. The data the field accumulated on each of these paths — the patients

who stopped successfully, the patients who relapsed and restarted, the patients who never tried — became the basis for the guidelines that govern CML practice today.

What none of these data have produced, in CML, is a path that ends with the cancer eliminated as cleanly and as definitively as a tumor cured by surgery. There is, for most patients, always the small reservoir somewhere, the dormant cells, the molecular signal at the limit of detection. The disease has been transformed but, for most patients, not eliminated. What has been transformed is the patient's life, which is, in the end, the thing the field was trying to transform in the first place.

The next chapter is about how this story — a single targeted molecule, a single transformed disease — generalized, and didn't generalize, to the cancers that came after. CML was an unusually clean target. Many cancers are not. The dream the *Time* cover suggested in 2001 — that targeted therapy would work for cancer the way imatinib worked for CML — turned out to be more complicated when the field tried to apply it to the rest of oncology.

It is the chapter where the book takes seriously what targeted therapy did and did not do across the larger landscape of cancer.

It is also the chapter where the answer is more mixed than the success of the imatinib story might suggest, and where the things that did work — including a story we have so far given less space than it deserves, of a humanized antibody and a kind of breast cancer that, after 2005, became a different disease — deserve their own honest accounting.

The Generalization Problem



Imatinib worked in CML for a reason. The reason is the subject of this chapter, and the reason is also why imatinib's success has not been quite as easy to repeat as the celebratory coverage of 2001 implied.

The reason is that CML is, by the standards of human cancer, almost embarrassingly simple. There is one driver. The driver is genetic. The driver is the same in essentially every patient. The driver is a kinase, and kinases are druggable. The cancer cells depend on the driver in a way they cannot escape, except by acquiring specific point mutations in the driver itself, which the field has, with some effort, been able to address through additional drugs. The cancer's natural history is slow enough — a chronic phase that lasts years, an accelerated phase that lasts months, a blast crisis that ends the disease — that drugging the chronic phase, where the cancer is most genetically homogeneous, can be done

early enough to keep most patients out of the more lethal phases. CML is, in short, an unusually clean target for an unusually clean drug.

Most cancers are not like this.

Most cancers are driven by multiple genetic changes, accumulated over years, in shifting combinations across the population of cells in any individual tumor. Most cancers do not have a single dominant kinase that, if inhibited, brings the whole enterprise down. Most cancers, when you look at them at the molecular level, look less like an engine encased in concrete with one obvious wrench in the works, and more like a malfunctioning city in which several different systems have failed at different times in different parts.

What targeted therapy could do, after CML, depended on which cancers turned out to look more like CML and which turned out to look more like the general case.

Some did. Some didn't. The picture, twenty-five years after Gleevec's approval, is genuinely mixed in a way the 2001 coverage did not anticipate.

This chapter is about that mixture. It will look at where targeted therapy delivered something close to what was promised. It will look at where it did not. It will ask why — in a way the molecular logic the book has been laying out can begin to answer.



Start with the cancers where the targeted-therapy paradigm has worked best.

Several other hematologic cancers turned out, like CML, to be driven by single recurrent genetic events that could be drugged. Acute promyelocytic leukemia, a subtype of acute myeloid

leukemia, is driven by the t(15;17) translocation Janet Rowley helped catalogue — a translocation that fuses two genes, PML and RAR-alpha, into a fusion that does to a different cell lineage what BCR-ABL does to myeloid stem cells. APL had been, until the late 1980s, one of the most lethal acute leukemias, with rapid bleeding complications often killing patients in the first weeks of presentation. The discovery that the disease could be treated, sometimes cured, with a combination of all-trans retinoic acid (a vitamin A derivative that could differentiate the leukemic cells) and arsenic trioxide (yes, the poison — in the right dose, against this specific kind of leukemia, an effective therapy) transformed APL into one of the most curable adult leukemias. The targeted approach in APL came first, chronologically, before imatinib; in some accounts of targeted therapy's history APL gets pride of place over CML. The cancer's molecular logic was similar: one fusion, one driver, one drug class that addressed it.

Gastrointestinal stromal tumors, or GISTs, are a rare type of soft-tissue sarcoma that, before 2001, had been one of the most stubbornly untreatable cancers. Conventional chemotherapy and radiation did essentially nothing. Surgery worked when the tumor was localized but most patients eventually had recurrence. In 1998, a group at Brigham and Women's Hospital in Boston — Jonathan Fletcher, Christopher Fletcher, and others — showed that GISTs were driven, in most cases, by activating mutations in a tyrosine kinase called c-KIT — the same c-KIT that, by happy coincidence, imatinib also inhibited as one of its handful of off-target activities. A clinician named George Demetri, also in Boston, took up the case for using imatinib in GIST. The first patient was treated in 2000. The response was dramatic. By 2002 — a year after Gleevec's approval in CML — the FDA had granted a second indication for the same drug in GIST, on the basis of a single small trial showing response rates

that had been unimaginable in this cancer a few years before. Patients with metastatic GIST who would have been dead within a year went into durable remission on a daily pill. The drug had found a second cancer, almost by accident, that fit its mechanism.

Several lung cancers turned out, in the 2000s, to be driven by druggable kinase events in subsets of patients. The clearest case was EGFR-mutant non-small cell lung cancer, identified in 2004 by groups led by Bill Sellers, Pasi Jänne, and Bruce Johnson at Dana-Farber and Daniel Haber at Massachusetts General Hospital. About fifteen percent of lung adenocarcinomas in Western populations — and a higher fraction in East Asian populations — carried activating mutations in EGFR, the epidermal growth factor receptor. (EGFR, recall from chapter eight, is the first member of the HER family.) Inhibitors of EGFR — gefitinib, erlotinib, and later osimertinib — produced response rates in EGFR-mutant lung cancer of seventy or eighty percent, with substantial extensions of survival compared to chemotherapy. The first generation of these drugs had been developed before anyone knew which patients would benefit; what made them succeed was identifying the molecular subset, then giving the drug specifically to those patients.

ALK-rearranged lung cancer, identified in 2007, accounts for a smaller fraction of lung adenocarcinomas — perhaps three to five percent — but is driven by a fusion gene that is, structurally, more like BCR-ABL than like the EGFR mutations: a chromosomal rearrangement creating a fusion protein with constitutively active tyrosine kinase activity. The first ALK inhibitor, crizotinib, was approved in 2011, with subsequent generations — alectinib, lorlatinib — producing progressively deeper and more durable responses. Patients with ALK-rearranged lung cancer, on the most

current drugs, can have median survival measured in many years rather than months. This is, again, a cancer that turned out to look molecularly more like CML than like generic lung cancer.

BRAF-mutant melanoma, identified in 2002 in work led by Mike Stratton's group at the Sanger Institute, is driven in about half of melanomas by a single recurrent mutation — V600E, a substitution of valine by glutamate at position 600 of the BRAF kinase. BRAF inhibitors, beginning with vemurafenib (approved 2011), produced rapid and dramatic responses in BRAF-mutant melanoma, although the responses were less durable than the kinase-inhibitor responses in CML or ALK lung cancer. The combination of a BRAF inhibitor with a MEK inhibitor (a drug aimed at the next kinase down in the same signaling pathway) extended the responses, and dual BRAF/MEK therapy is, in 2026, standard of care for advanced BRAF-mutant melanoma. Five-year survival rates that were essentially zero in the early 2000s are now in the thirties.

These cancers, and a growing list of others — PARP inhibitors in BRCA-mutant ovarian and breast cancer, FLT3 inhibitors in FLT3-mutant AML, IDH inhibitors in IDH-mutant cancers, KRAS-G12C inhibitors in lung cancers carrying that specific mutation — collectively form the part of the targeted-therapy story that lived up to the 2001 hype. They are not all imatinibs. The responses are sometimes shorter, the drugs are sometimes harder to live with, the cancers sometimes evolve out from under the drugs faster than CML did. But each represents a real clinical benefit, in a defined patient subset identified by a molecular test, of a kind that did not exist before targeted therapy as an idea took hold.



There is one targeted therapy whose success story has been, in some ways, undertold relative to its impact, and it is worth pausing on that one in particular before this chapter moves to the cancers where the paradigm did not work as well.

Herceptin in metastatic breast cancer, approved in 1998 and described in chapter eight, was the first major non-CML success of the targeted paradigm. It extended survival in HER2-amplified metastatic disease by a meaningful margin and established the small-molecule-vs-antibody dichotomy that has shaped much of oncology since. But Herceptin's bigger story, in some ways, came after the 2001 Time cover, in a different setting.

The setting was adjuvant therapy — treatment given right after surgery for localized breast cancer, in patients whose tumors had been completely removed but whose disease was at meaningful risk of recurrence elsewhere in the body. The standard of care, before 2005, was adjuvant chemotherapy and (for hormone-receptor-positive disease) tamoxifen or an aromatase inhibitor. Patients with HER2-positive disease did less well on the standard regimens than patients with HER2-negative disease; the HER2 amplification, as Slamon's 1987 paper had originally established, predicted worse outcomes.

Beginning around 2000, several large randomized trials launched to ask whether adding Herceptin to adjuvant chemotherapy in HER2-positive early breast cancer would reduce recurrence and extend survival. The trials were enormous — the HERA trial enrolled five thousand patients, the joint analysis of two North American trials (NSABP B-31 and NCCTG N9831) covered another three thousand. They took years to mature. Results began appearing in 2005.

What the trials showed was, in the language of breast oncology, a paradigm shift. Adding one year of Herceptin to adjuvant chemotherapy reduced the risk of recurrence by about half. The reduction translated, on follow-up, into substantial improvements in overall survival — absolute differences of about ten percentage points at ten years between the chemotherapy-alone and chemotherapy-plus-Herceptin arms. Patients with HER2-positive early breast cancer, who had previously been the breast cancer subset with the worst prognosis, were now, with adjuvant trastuzumab, doing as well as or better than HER2-negative patients. The prognostic stratification that had defined breast cancer for nearly twenty years — HER2 amplification as a marker of more aggressive disease — had been not just neutralized but, in some respects, inverted.

The breast cancer community absorbed the result quickly. Adjuvant Herceptin was, by 2006, standard of care for any HER2-positive breast cancer with sufficient risk of recurrence. The treatment is, in 2026, given to roughly fifty thousand women a year in the United States and many times that worldwide. The number of HER2-positive breast cancer recurrences prevented, and the number of breast cancer deaths prevented, runs into the hundreds of thousands cumulatively over the past two decades. It is, by any measure of total clinical impact, one of the largest contributions of targeted therapy to oncology so far.

The story did not end with Herceptin. A second-generation HER2 inhibitor, pertuzumab, was approved in 2012 and is now used in combination with trastuzumab for both metastatic and high-risk early disease. An antibody-drug conjugate — trastuzumab emtansine, sold as Kadcyla — was approved in 2013, attaching a cytotoxic chemotherapy molecule to the trastuzumab antibody so that the antibody delivers the chemotherapy specifically to HER2-

positive cells. A more recent antibody-drug conjugate, trastuzumab deruxtecan (Enhertu), approved in 2019, has produced even larger response rates, and has, in a particularly striking 2022 result, been shown to benefit not just HER2-amplified tumors but tumors with merely modest HER2 expression — a much larger fraction of breast cancers — effectively expanding the HER2-targeted franchise from a quarter of breast cancers to perhaps two-thirds of them.

What the adjuvant Herceptin story shows, in the larger frame of this chapter, is that targeted therapy's biggest contribution in some cancers has come not in the headline-grabbing metastatic setting but in the quieter, longer-tail setting of preventing recurrence. The patient who never has a metastatic recurrence does not show up in dramatic before-and-after photographs. She shows up in the survival curves, decades after her surgery, alive and well, having taken a year of an antibody she would not have received in 1998. The adjuvant Herceptin trials saved more lives, in absolute terms, than any individual development in CML did. They are also, partly because there is no single dramatic moment of cure, less famous than they should be.

There are several other cancers where adjuvant or maintenance targeted therapy has produced effects of a similar shape — not the dramatic shrinkage of metastatic disease that makes for a good newspaper article, but a steady reduction in the rate at which a cancer comes back, integrated over years and across populations. PARP inhibitors as maintenance therapy in BRCA-mutant ovarian cancer have done something analogous. Adjuvant osimertinib in EGFR-mutant lung cancer, established by the ADAURA trial in 2020, is now standard. The pattern is consistent: where targeted therapy can be given to patients before the cancer has had a chance to advance, the long-tail benefits compound.



Now the cancers where the paradigm has not worked as well.

Most solid tumors are not driven by a single, dominant, druggable kinase or by any single, simple, druggable event. Most solid tumors, when they are sequenced, contain dozens of mutations across dozens of genes, with substantial diversity within any one tumor between cells in different regions, and substantial diversity between different patients with the same diagnosis. This is not a small problem for the targeted approach. It is the central problem.

Pancreatic ductal adenocarcinoma, the most common form of pancreatic cancer, is the locus classicus of this difficulty. It is one of the deadliest cancers — five-year survival, as of 2026, remains in the low double digits, and most diagnoses are still effectively terminal. Roughly ninety percent of pancreatic cancers are driven, in part, by a mutation in a gene called KRAS — a small GTPase, a kind of molecular switch involved in the same general signaling pathway as many of the kinases the rest of this chapter has discussed. KRAS would seem, on the face of it, to be the perfect target. It is mutated in nearly every patient. It drives the cancer. It has been the subject of intense drug-discovery effort for forty years.

It also turned out, for most of those forty years, to be undruggable. KRAS is small. Its surface is smooth in a structural sense — there are no obvious deep pockets where a small molecule could bind tightly. The active site, where it cycles between active (GTP-bound) and inactive (GDP-bound) states, holds onto its native substrate so avidly that no competitor molecule could realistically displace it. Multiple generations of medicinal chemists, working in multiple laboratories, attempted KRAS inhibitors and found that the protein simply did not provide them with the kind of binding surface a drug needs. KRAS sat, for forty years, on the list of validated cancer targets that nobody could drug.

This changed, partially, in the late 2010s. A group at the University of California, San Francisco, led by Kevan Shokat, identified a particular subtype of KRAS mutation — KRAS-G12C, in which a glycine at position 12 is replaced by cysteine — that introduces, into the otherwise smooth protein surface, a reactive cysteine residue that a covalently-binding drug could grab onto. The covalent KRAS-G12C inhibitors that emerged from this work — sotorasib, approved 2021, and adagrasib, approved 2022 — were the first KRAS-targeted drugs ever approved. They worked, in patients carrying the specific G12C mutation, in cancers including non-small cell lung cancer.

But here is the catch. KRAS-G12C accounts for only about thirteen percent of KRAS mutations. The other eighty-seven percent — G12D, G12V, G12R, G13D, and others — do not have the same reactive cysteine and cannot be targeted the same way. In pancreatic cancer specifically, G12D and G12V are the most common KRAS mutations; G12C is rare. The drugs that exist, as of 2026, for KRAS-mutant cancer are useful for a slim fraction of patients, and the broader population of KRAS-mutant cancers — including the great majority of pancreatic cancers — remains, in a meaningful sense, undruggable. Newer compounds aimed at G12D and the broader KRAS family are in clinical development, with some early results in late 2024 and 2025 looking promising; whether they will match what G12C inhibitors deliver, or whether they will hit a wall the field has not yet fully understood, remains an open question in 2026.

Pancreatic cancer, in short, has the molecular driver and does not yet have the drug. The targeted-therapy paradigm has not yet rescued it. Whether it ever will is uncertain.

Glioblastoma, the most aggressive primary brain tumor, is another cancer that has resisted the targeted approach. Its driver landscape is heterogeneous within any single tumor — different

regions of the same tumor have different mutational profiles — and the blood-brain barrier, the cellular barrier that protects the brain from circulating molecules, makes drug delivery to glioblastoma cells unusually difficult. Targeted drugs that work in other cancers often cannot reach effective concentrations in the brain. Glioblastoma's median survival, despite forty years of incremental improvement in surgery and radiation and chemotherapy, has remained in the range of fifteen to eighteen months. The targeted-therapy revolution has, so far, largely passed it by.

Triple-negative breast cancer — breast cancer that is negative for estrogen receptor, progesterone receptor, and HER2 amplification — is the breast cancer subset for which targeted therapy has had the least to offer. About fifteen percent of breast cancers are triple-negative; they tend to be more aggressive than ER-positive or HER2-positive disease, and they affect younger women disproportionately. The molecular drivers of triple-negative breast cancer are heterogeneous and largely not druggable with current approaches. Treatment has remained, in many patients, conventional chemotherapy. Recent developments — PARP inhibitors in BRCA-mutant subsets, immunotherapy combinations, antibody-drug conjugates targeting other surface markers — have begun to make a dent. But triple-negative disease has, in 2026, no equivalent of Herceptin or imatinib; no clean molecular driver, no transformative single drug.

These are not isolated cases. Many of the most common solid tumors — most colorectal cancers, most stomach cancers, most ovarian cancers (outside the BRCA-mutant subset), the majority of lung cancers (those without an actionable kinase mutation) — have not been transformed by the targeted-therapy paradigm in the way

CML and HER2-positive breast cancer were. They have benefited, sometimes substantially, from improvements at the margins. They have not been remade.



Why some cancers have yielded and others have not is a question with a partial answer the field has, by 2026, become reasonably comfortable with.

The cancers that yield to targeted therapy tend to share several features. They are typically driven by a single dominant genetic event, present in essentially every cell of the tumor. The driver is typically a kinase or a comparable signaling protein with a druggable structural feature — an ATP pocket, an allosteric site, an extracellular domain accessible to an antibody. The cancer cells are typically dependent on the driver in a way that they cannot easily replace by activating some other pathway. The tumor's mutational burden — the total number of additional mutations beyond the driver — is moderate enough that the cells have not already accumulated alternative ways to grow.

The cancers that resist targeted therapy tend to share the opposite features. They are typically driven by combinations of changes, with no single dominant event. Their drivers are often not kinases but transcription factors, structural proteins, or proteins with shapes that don't accept conventional small-molecule inhibitors well. The cells have multiple redundant pathways for growth and survival, so blocking any one of them gets compensated for by another. The mutational burden is high, the tumor's genetic heterogeneity is high, and within any one patient there are usually many subpopulations of cells with different vulnerabilities, so a single drug can't catch them all.

There is also, layered on top of this, the question of stage. Targeted therapy generally works better in earlier-stage disease, where the cell population is smaller and more genetically homogeneous, than in advanced disease, where the cells have had more time and more opportunity to evolve away from any single intervention. CML in chronic phase responded beautifully to imatinib; CML in blast crisis responded less well. EGFR-mutant lung cancer in early-stage disease, on adjuvant osimertinib, has dramatic recurrence-prevention benefits; EGFR-mutant lung cancer in advanced disease eventually evolves resistance, in most patients, to even the best current drugs. The cancers that have generalized targeted-therapy success best are the ones where the molecular insight could be deployed early, against a smaller and more genetically uniform target population, before the cancer had a chance to become as heterogeneous as advanced cancers tend to be.

There are, finally, the cancers where targeted therapy has been tried, has worked partially, and has hit a ceiling. HER2-positive breast cancer is, in some ways, the prototype of this. The drug works, the responses are durable in many patients, but a substantial fraction of metastatic patients eventually progress, and the resistance mechanisms are heterogeneous and harder to address with second-generation drugs than the analogous mutations in CML were. The newer antibody-drug conjugates have extended the franchise, but they have not closed it. HER2-positive breast cancer in 2026 remains a treatable but, in advanced settings, often ultimately fatal disease.



It is worth being clear about the larger frame this chapter has been moving toward.

The 2001 *Time* cover suggested that targeted therapy was a new paradigm that would transform cancer treatment. Twenty-five years later, this is true and not true. Targeted therapy has transformed treatment for a real and meaningful subset of cancers — the chronic myeloid leukemia patients on imatinib, the GIST patients on imatinib, the EGFR-mutant lung cancer patients on osimertinib, the BRAF-mutant melanoma patients on combination BRAF/MEK therapy, the HER2-positive breast cancer patients on trastuzumab and its successors, the BRCA-mutant ovarian cancer patients on PARP inhibitors, and a growing list of others. For these patients, the change has been real and large. Some of them have effectively had their cancer converted from a fatal disease to a chronic one, or from a chronic one to a curable one. Some of them are alive, in 2026, who would not have been alive in 1995, and the difference is the molecular logic and the drugs that came out of it.

Targeted therapy has not transformed treatment for most cancers, in the sense that it has not delivered, for most patients with most cancers, the kind of clean before-and-after benefit it delivered for CML. The pancreatic cancer patient diagnosed in 2026 has, as a function of pancreatic cancer therapy specifically, a slightly better median survival than the patient diagnosed in 2001, but only slightly. The glioblastoma patient has not benefited materially. The triple-negative breast cancer patient has benefited modestly. The ovarian cancer patient outside the BRCA subset has benefited modestly. For the bulk of solid-tumor cancer, targeted therapy has been an additive, incremental improvement, not a paradigm shift.

What the field has done, in response, is broaden its attention. Targeted therapy as the dominant frame of cancer drug development — the frame that the imatinib story established and the *Time* cover propagated — has, since the mid-2010s, shared the stage with several other approaches. Immunotherapy, particularly

the class of drugs called immune checkpoint inhibitors, has produced transformative results in some cancers (melanoma, certain lung cancers, certain kidney cancers, Hodgkin lymphoma) and modest results in others. Antibody-drug conjugates — antibodies that deliver cytotoxic payloads selectively to cells expressing a particular surface marker — have expanded the targetable space substantially, particularly in HER2-positive disease, and are now in clinical development for many other markers. CAR-T cell therapies — patient T cells engineered to recognize and kill tumor cells — have produced striking results in certain blood cancers and are being adapted for solid tumors with mixed success. Each of these approaches has its own logic, its own successes, and its own limitations. None of them, individually, has been the universal solution that targeted therapy was sometimes thought to be in 2001.

The cancer therapy landscape in 2026, taken as a whole, is dramatically better than the cancer therapy landscape in 1995. The improvements are not uniform across cancers and they are not, in any single drug or approach, the kind of complete transformation that the early imatinib results produced in CML. The improvements are, more accurately, a thousand smaller transformations, each in a particular subset of a particular cancer, each driven by a particular molecular insight, each producing benefit measured in months or years for a particular group of patients. Cumulatively, this has added years of life and quality of life to enormous numbers of people. It has not, on the whole, made cancer go away.



The deepest thing the targeted-therapy era has done, perhaps, is not a particular drug but a particular habit of mind.

Cancer drug development before imatinib was largely empirical. You took compounds with broadly toxic activity, gave them to patients, watched what happened. The drugs that worked, you used. The drugs that didn't, you didn't. The mechanisms by which the drugs worked were often only partly understood, and the question of which patients would benefit from which drug was, in most cases, not well answered before the patient got the drug. Cancer drug development was a slower, more gradual, more accidental enterprise than what came after.

Cancer drug development after imatinib has been, increasingly, a different kind of enterprise. The standard sequence — identify the molecular driver, validate it experimentally, design a drug to inhibit it, identify which patients have the driver, give the drug to those patients — has become the default frame. The imatinib story established that this sequence was possible. The subsequent success stories established that it was reproducible, in the right cancers. The failure stories established that it was not universal, that some cancers were going to require different approaches. The combination of successes and failures has shaped what cancer drug development looks like in 2026 in a way it would not have been shaped without imatinib.

This is a real legacy. It is also a different legacy than the one the 2001 coverage promised. The promise was that targeted therapy would solve cancer. The actual delivery is that targeted therapy solved CML, transformed several other cancers, modestly improved many more, and gave the field a habit of asking molecular questions about cancer drugs that has, slowly and unevenly, raised the floor of what cancer treatment can achieve. The patients who have benefited have benefited substantially. The patients who have not benefited are still waiting.

The patients who have benefited — the ones who entered Druker's clinic in 1993 with three to five years to live, and who are now, in 2026, living more or less normal lives on a daily pill — are the subject of the next chapter, and the last chapter of this book. They are the chapter the book has been moving toward from the start. They are the people for whom the long argument from 1960 to 1998 to 2001 was, in the end, the argument that mattered.

Some of them are still in Druker's clinic.

CHAPTER 12

12

The Patients Who Waited



On a Tuesday morning in the spring of 2026, in the same hematology clinic at Oregon Health Sciences University where this book began, Brian Druker is seeing patients.

He is sixty-nine years old. He stepped down a few years ago from running the Knight Cancer Institute, the cancer center at OHSU that he led for many years and that grew, in some part on the strength of imatinib's success, into one of the major American cancer research institutions. He still sees patients. He still runs a research program. He still writes letters about CML and about other cancers and about the people taking the drugs and about the people developing the drugs that come next.

The clinic on this particular Tuesday is a follow-up clinic. Most of the patients on his schedule are CML patients who have been coming to see him for ten, fifteen, twenty, twenty-five years. Many of them he has known longer than he has known some of his own

colleagues. He has watched them through their disease and through the rest of their lives. He has watched their children grow up. He has gone to a few weddings. He has attended a few funerals — mostly, in recent years, for things other than the disease that brought him into their lives.

The first appointment of the morning is a woman in her early seventies. She was diagnosed with CML in 1995, three years before the phase 1 trial of imatinib opened. She was on hydroxyurea and then interferon for three years before she enrolled in the phase 2 trial in 1999. She has been on imatinib continuously since. Her blood counts are normal. Her latest PCR result, on the chart in front of Druker, shows BCR-ABL transcripts undetectable on the standard assay — deeper than MR4.5. She has been at this level for more than a decade. The conversation lasts about twenty minutes. They talk about her grandchildren. They talk about a recurrent shoulder problem that she is going to get a steroid injection for. They talk briefly about whether she might be a candidate for a treatment-free remission attempt; she says she has thought about it and would rather just keep taking the pill, because the pill works and she sees no reason to fix what isn't broken. Druker says that's reasonable. He writes the prescription renewal. They schedule her next appointment for six months out.

She gathers her things and leaves. The next patient is already in the waiting room.



What this clinic looks like, in 2026, is what the prologue's clinic in 1993 was reaching for, across the gap that this entire book has been about.

In 1993, Druker had patients with a fatal disease and inadequate tools. In 2026, he has patients with a managed disease and tools that work. The clinic visits in 1993 had to be conversations about how to live with a worsening cancer. The clinic visits in 2026 are conversations about pill adherence, blood draws, side effects, and the ordinary medical concerns of people who happen to have once had something serious that has, in the language clinicians now use without embarrassment, become a chronic condition.

Some of the patients in this clinic in 2026 — perhaps a quarter of them, perhaps a third — have been coming to see Druker since the 1990s. Many of them are patients he saw in his early years at OHSU, when there was nothing to give them but the inadequate options of the era. Many of them survived long enough to enroll in the phase 1 trial, or the phase 2 trials that came after, or to receive the drug commercially after May 2001. They are, for the most part, alive. They are, for the most part, well.

There are also patients in his clinic this morning who are newer — diagnosed last year, the year before, five years ago. They were not part of the original cohort that lived through the long argument from 1993 to 1998 to 2001. They came in after the drug was already there. They started imatinib, or one of the second-generation drugs, within weeks of their diagnosis. Their experience of CML has been, from the start, a daily pill and a routine blood draw and a normal life expectancy. They are aware, generally, that this was not always how the disease worked. Few of them know the history in detail. Most of them, gracefully, do not have to.



The patient after the first one is a man Druker has known since 1998.

He was sixty-three when he was diagnosed and is ninety-one now. He was one of the patients in the early phase 1 cohort — not the very first patient, but in the first wave, dosed somewhere in the middle of the 1998 escalation. He had advanced disease at enrollment; he had, in the conversation he and Druker had before the trial, been told that the drug was experimental, that it might not work, that he was very ill, and that the alternatives — by which they meant the lack of alternatives — were what they were. He took the first dose with his wife in the room, the way several of the early patients did. The wife is no longer with him; she died of unrelated causes about six years ago. The man is here this morning by himself. He came in by Lyft. He is, by his report, doing well.

His CML, on every test that has been done, is undetectable. He has been off active monitoring for the deepest molecular response measures for several years now — at his age, the question of whether his MR is at 4.5 or 5.0 is no longer of great clinical interest. He still takes the drug. The drug, after a long discussion several years ago, was reduced from his original dose to a lower one, in deference to his age and his renal function. He tolerates the lower dose well. He has the side effects of being ninety-one, which are extensive. He does not, by any reasonable assessment, have the side effects of CML.

Druker spends about thirty minutes with him. The conversation is what conversations between long-known doctors and long-known patients tend to be. They talk about the man's son, who lives in Eugene. They talk about a book the man has been reading. They talk briefly, in the way that elderly patients sometimes raise such things, about how the man would like things to go when his time comes — not soon, he hopes, but not so far away anymore that the question is

academic. Druker listens. He asks about pain management. He asks whether the man has talked with his son about any of this. He notes a few things in the chart.

When the man stands to leave, he steadies himself on the chair. Druker walks him to the waiting area. They shake hands. The man says ‘thank you, doctor,’ the way the man has said ‘thank you, doctor’ at the end of every appointment for almost thirty years. Druker says ‘thank you for coming in.’

The next patient is already waiting.



If you sat in this clinic for a full day, the picture you would build up would not be a triumphal one. It would be quieter than that.

It would be a picture of normal medical practice, by which I mean the practice of seeing people who have a chronic condition and helping them stay well. The patients are not, by and large, dying. They are not, by and large, in crisis. They have a thing they take a pill for, and they come in to make sure the pill is still working, and the pill is, almost always, still working. Their bloods come back normal. Their bone marrows, when they get one (which is less and less often, since modern PCR-based monitoring has reduced the need for bone marrow biopsy in stable patients), are clean. Their lives are continuing in the way lives continue when they are not being interrupted by a fatal illness.

There are, on any given clinic day, exceptions to this picture. There is sometimes a newer patient who is in the early stages of treatment, working out side effects, getting used to the daily pill. There is sometimes a patient whose disease has done something unexpected — a slow rise in PCR, a question about whether the drug needs to be switched. There is sometimes a patient who has come in because they have developed something else — another

cancer, another medical condition — and is here because Druker has been their oncologist for so long that he is the doctor they want to talk to. There is sometimes a patient who has not done well, whose CML has progressed, whose options through the targeted-therapy ladder have narrowed, and for whom the conversation in clinic is more like the conversations Druker had to have in 1993 — honest, careful, with limits acknowledged. These conversations have not entirely disappeared from the clinic. They are rarer.

This is what the targeted-therapy era has done in CML, in the small specific space of one disease and one clinic. The dominant feature of the practice is the absence of crisis. Patients live with their disease. They come in. They go home. They keep taking the pill. They come in again.



Some of the people from the earlier chapters of this book are no longer alive in 2026, and it is worth noting them honestly.

Janet Rowley died in 2013, at eighty-eight. She had been in the field, more or less continuously, since 1962. She had been a cytogeneticist when cytogenetics was a small and somewhat marginal subspecialty, and she had lived to see her work be the basis for the first targeted cancer therapy in history. She continued to work, in some capacity, almost until the end of her life. The Presidential Medal of Freedom that she had received in 2009 sat in her office; the Lasker Award and National Medal of Science she had received earlier had similar fates. Her colleagues, in the obituaries written after her death, used a phrase that recurred: she had been the kind of scientist who was satisfied to do the work, and to have done it, and not to have made too much fuss about it. She did not get the Nobel Prize. She probably should have. Most of the obituaries said so.

David Hungerford, who had counted the chromosomes in 1959 and 1960 — who had, with Peter Nowell, made the original observation that started this story — died in 1993 at the age of sixty-six, of a stroke. He died in the year Druker first walked into his clinic in Portland. He died several years before the phase 1 trial of the drug that would, eventually, do something with the small chromosome he had described. Peter Nowell, the other author of the 1960 paper, lived longer; he died in 2016 at the age of eighty-eight. He did get to see imatinib's approval, the IRIS data, the second-generation drugs, and a long stretch of the unfolding consequence of the original observation he and Hungerford had made. By the time of his death, the Philadelphia chromosome had been the subject of more research papers than any other chromosomal abnormality in human cancer.

Bud Romine, the Oregon truck driver in chapters six and ten, died of an unrelated cancer in 2017. His CML, by every test he had had in the years before his death, had remained suppressed. His family donated some of his memorabilia from the imatinib years to OHSU; the postcards from his trucking trips are, I am told, in a small collection of patient artifacts that the cancer center keeps.

Marti Nelson, the California physician who had advocated publicly for accelerated access to Herceptin, died in 1994. She was thirty-seven. She had not been able to enroll in the trial that would, in 1998, lead to the drug's approval; her HER2-amplified breast cancer, by the time the trial slots that might have fit her opened up, had advanced past the point where Herceptin could have helped her. The activism around her case continued for some years after her death. The Nelson Foundation, established by her family and friends after she died, was for a long time involved in supporting expanded

access to investigational therapies. She is, in the way these things go, less remembered now than she was twenty-five years ago. She should be remembered.

Some patients from the original phase 1 trial in 1998 — the ones who had advanced disease at enrollment, who had started the drug expecting nothing and gotten years they had not expected — are still alive. Most are not. Many of the ones who responded most dramatically and most durably were already in their fifties or sixties at the time of the trial; they have, since, lived through the natural arc of the rest of their lives, and most of those lives have ended in the ordinary ways that lives end, of the things people in their seventies and eighties and nineties die of. The ones who are still alive are doing well. The ones who are not are, generally, not dead because of the disease the trial was about.



It is worth saying something, before this chapter ends, about the people who developed the drug.

Brian Druker received many awards over the years that followed Gleevec's approval. He shared the 2009 Lasker-DeBakey Clinical Medical Research Award with Nick Lydon and Charles Sawyers. He has been mentioned, periodically, as a Nobel candidate; he has not, as of 2026, received the Nobel Prize. (Whether he will is, like all such questions, the kind of thing the Nobel committee answers in its own time, in ways that often seem unrelated to the relative merits of the candidates being considered.) He has continued to work on CML and on related questions; he has founded a research institute; he has trained many of the people who, in 2026, are running their own programs at other institutions. He is, by the standards of academic medicine, one of the most successful clinician-scientists of his

generation, and is so widely known among CML patients that he receives, by his own account, a steady stream of cards and letters from patients he has never met.

Nick Lydon, the Ciba-Geigy chemist whose lab made the original molecule, has continued to work on kinase inhibitors. After leaving Novartis in 1997, he founded a small biotech company called Kinetix that was eventually acquired by Amgen, then later co-founded Granite Biopharma; he has been involved in several other companies since. He shared the 2009 Lasker with Druker and Sawyers. He has, in the years since, kept a relatively lower public profile than Druker has, in the way that the more inside-the-company half of a successful drug-development story usually does.

Charles Sawyers, who ran the UCLA half of the phase 1 trial and led the work on resistance mutations in chapter nine, has had perhaps the most academically distinguished subsequent career of the three. He moved from UCLA to Memorial Sloan Kettering in 2006, where he founded the Human Oncology and Pathogenesis Program; he has been, in the period since, one of the central figures in the broader field of targeted cancer therapy, with substantial work on resistance mechanisms in prostate cancer and on the development of the second-generation androgen receptor antagonist enzalutamide. He has received the National Medal of Science, the Lasker, several other major awards. He is, in 2026, one of the leading senior figures in American oncology research.

Daniel Vasella stepped down as CEO of Novartis in 2010, having held the role since the 1996 merger that created the company. His tenure included Gleevec's approval and many other drug launches; it also included controversies, particularly around executive compensation, that made him a more politically complicated figure in Switzerland than the Gleevec story alone would suggest. He has continued to write and speak about pharmaceutical innovation in

retirement. His memoir from 2003, written before the long-term Gleevec story had played out, remains one of the few primary accounts of the corporate side of the imatinib development effort.

Alex Matter, the Ciba-Geigy oncology research head who had championed the kinase inhibitor program from inside the company, retired from Novartis in 2005 and went on to lead a translational cancer research initiative in Singapore. He is, in 2026, in his eighties, less publicly visible than he was during his Novartis years but still active in cancer research advocacy. The internal champion of a successful drug, in pharmaceutical histories, often gets less credit than the external advocate. Matter's role in keeping CGP 57148 alive through the lean years of the early and mid-1990s was substantial. He is one of the people without whom this story would not have happened the way it did.

Jürg Zimmermann, the chemist who had actually made the molecule — who had synthesized, optimized, and refined the 2-phenylaminopyrimidine series until it produced the compound that became imatinib — retired from Novartis a few years ago and lives, the last I knew, in the Basel area. He has not been a frequent presence in the public-facing accounts of the imatinib story; the chemist who made the drug is, in a pattern that recurs across pharmaceutical history, often less visible than the clinician who tested it. Zimmermann's work was extraordinary. The drug exists because of it.



What does the whole story add up to, looking back?

It adds up, first, to a single drug for a single disease. CML, a disease that killed two-thirds of patients within ten years in 1990, kills a small fraction of patients within ten years in 2026, and the patients who die of CML in 2026 are mostly the ones whose disease

presented in advanced phase, or who could not tolerate the drug, or whose disease developed mutations that the available drugs could not address. The bulk of CML patients now live normal lives. This is, by the standards of any other oncology indication, an unusual outcome. It is also the central thing the story this book has told actually delivered.

It adds up, second, to a paradigm. The targeted-therapy approach — identify the molecular driver, design a drug to inhibit it, give the drug to patients identified by a molecular test — has, since imatinib, been the dominant frame of cancer drug development. Some of the drugs that have come out of this paradigm have produced benefits comparable to imatinib's; many have produced smaller benefits; some have produced none. Cumulatively, across cancers, the paradigm has saved or extended the lives of an enormous number of people. It has not delivered the universal solution that the 2001 coverage sometimes implied it would. It has delivered an enormous number of partial solutions, in particular cancers, for particular patients, that did not exist before.

It adds up, third, to a way of thinking. Cancer, in 1990, was understood by most physicians as a category of disease defined by the organ of origin — breast cancer, lung cancer, colon cancer, leukemia. Cancer, in 2026, is increasingly understood by most physicians as a category of disease defined by molecular features — EGFR-mutant lung cancer, HER2-positive breast cancer, BRAF-mutant melanoma, KRAS-G12C cancer, BCR-ABL-positive leukemia. The unit of clinical thinking has, slowly and incompletely, shifted. Some breast cancers are more like some leukemias, molecularly, than they are like other breast cancers. Some lung cancers respond to drugs that work in melanomas. The map of cancer has been

redrawn, partially, around the molecular drivers rather than the tissues of origin. This redrawing is unfinished. The continuing redrawing is, in some measure, a continuing legacy of imatinib.

It adds up, fourth, to the patients. The thousands of CML patients who, in 1993, were facing a fatal disease and who, in 2026, are not. The hundreds of thousands of HER2-positive breast cancer patients whose recurrence rate has been cut in half by adjuvant trastuzumab. The smaller numbers of patients with rarer cancers — GIST, ALK-rearranged lung cancer, BRAF-mutant melanoma, BRCA-mutant ovarian cancer, IDH-mutant cancers, KRAS-G12C cancers — whose lives have been extended by drugs that did not exist a generation ago. The story this book has told is, in some ways, an institutional and scientific story, but at every step the institutional and scientific story has been about whether a particular person somewhere would have a few more years with their family. Some of those people are around to be asked. The arithmetic of how many of them is, by 2026, well into six figures and rising.



There is a kind of book this book is not.

It is not a book about how everything will be fine. Cancer is not solved. Pancreatic cancer is still pancreatic cancer. Glioblastoma is still glioblastoma. Triple-negative breast cancer is still mostly treated with the cytotoxic drugs from the 1980s. Many cancers, in 2026, are still essentially the same disease they were thirty years ago. The 1993 patient with pancreatic cancer would have been told they probably had a year. The 2026 patient with pancreatic cancer is told they probably have a year and a half. The improvement is real, but small, and it has come more from incremental refinements in chemotherapy and surgery than from the targeted-therapy

revolution this book has been about. There are, for the patients in those clinics, no equivalents of imatinib. There may someday be. There are not yet.

It is not a book about how heroes did this single-handedly. The drug was developed by hundreds of people, most of whom this book has not been able to name — the chemists who tried earlier compounds in the 2-phenylaminopyrimidine series before Zimmermann landed on the right one; the technicians who ran the cell culture work in Druker's lab and Sawyers's lab and Witte's lab and many others; the pharmacologists who did the dog studies and the clinical pharmacology; the regulatory affairs specialists at Novartis and at the FDA who actually wrote and read the documents that allowed the drug to advance; the doctors at sites we have not had room to mention who enrolled patients into trials; the patients themselves, in their thousands, without whose willingness to take an experimental drug there would be no drug. The book has focused on a small number of people because narrative requires this; the actual story involves vastly more people than any narrative can reasonably name.

It is not a book about a single decisive moment. There are, in the history this book has told, many candidate moments — Hungerford at the Zeiss in 1959 or 1960; Rowley at her dining room table in 1973; de Klein at the lightbox in 1982; Druker counting colonies in 1995; the first phase 1 dose in June 1998; the FDA's approval in May 2001 — each of which is, in some accounts, presented as the moment everything changed. None of them is the moment everything changed. The moment everything changed was, accurately, the cumulation of all of them, plus many smaller moments, plus the slow shift in what the field considered possible. The story is the story of a long argument over many decades. The story does not have a single fulcrum.

It is, finally, not a book about heroes and villains. Novartis was not the villain of chapter five; Druker was not heroic in any sense his own description of himself would accept; the FDA was not benevolent or malevolent in chapter seven, just a regulator doing its job in a particular framework. The story is, in its actual texture, a story about ordinary professionals trying to do work they thought was right, in institutional contexts that pushed back on them in some cases and supported them in others, with occasional individual acts of conviction and stubbornness and luck that, in retrospect, made specific differences. There is no single moment in the story that requires a moralizing frame. There are many moments that require a careful one.



The patient is back in Druker's exam room — a different patient now, the third of the morning, a woman in her late fifties who was diagnosed with CML six months ago and is starting to get used to her daily pill.

She is asking about side effects. She has been having some muscle cramps, particularly at night, and some mild fluid retention around her eyes when she wakes up. Druker explains that these are common and usually manageable, that they often improve over the first six months, that there are simple measures she can try — magnesium for the cramps, hydration, sometimes a slight dose adjustment — if they don't. He asks her how she is doing emotionally. She says she is doing all right, mostly; the diagnosis was hard, and the first weeks on the drug were harder, but she is starting to feel less like a sick person and more like a person who takes a medication. He says that's good. He says that's typical. He says that, if her response continues to track the way it has been

tracking on her early monitoring, she should expect, in a year or two, to feel essentially like herself, with the difference being that she takes a pill in the morning that she did not used to take.

She nods. She asks, with some hesitation, whether this is going to shorten her life. Druker, in the way he has answered this question for several thousand patients over the past quarter-century, says that the honest answer is probably not, that the data on chronic-phase CML on imatinib are now mature enough to support a lifespan close to normal, and that there are good reasons to expect the same will be true for her. He says it might not be true — every individual patient is an individual — but that on the available evidence she has good reason to plan for a long life.

She thanks him. She gathers her things. She walks out into the waiting room, where her husband is reading something on his phone.

Druker writes a few notes in the chart. He glances at the next name on his list. He goes to the door of his exam room, opens it, and calls the next patient back.

Outside, on the hill above the Willamette River, it is raining lightly. It is mid-morning. There are several more patients to see before lunch.